

SECTION PLATE t=0 | Year 2000 | D=8 N=26 pairs=28

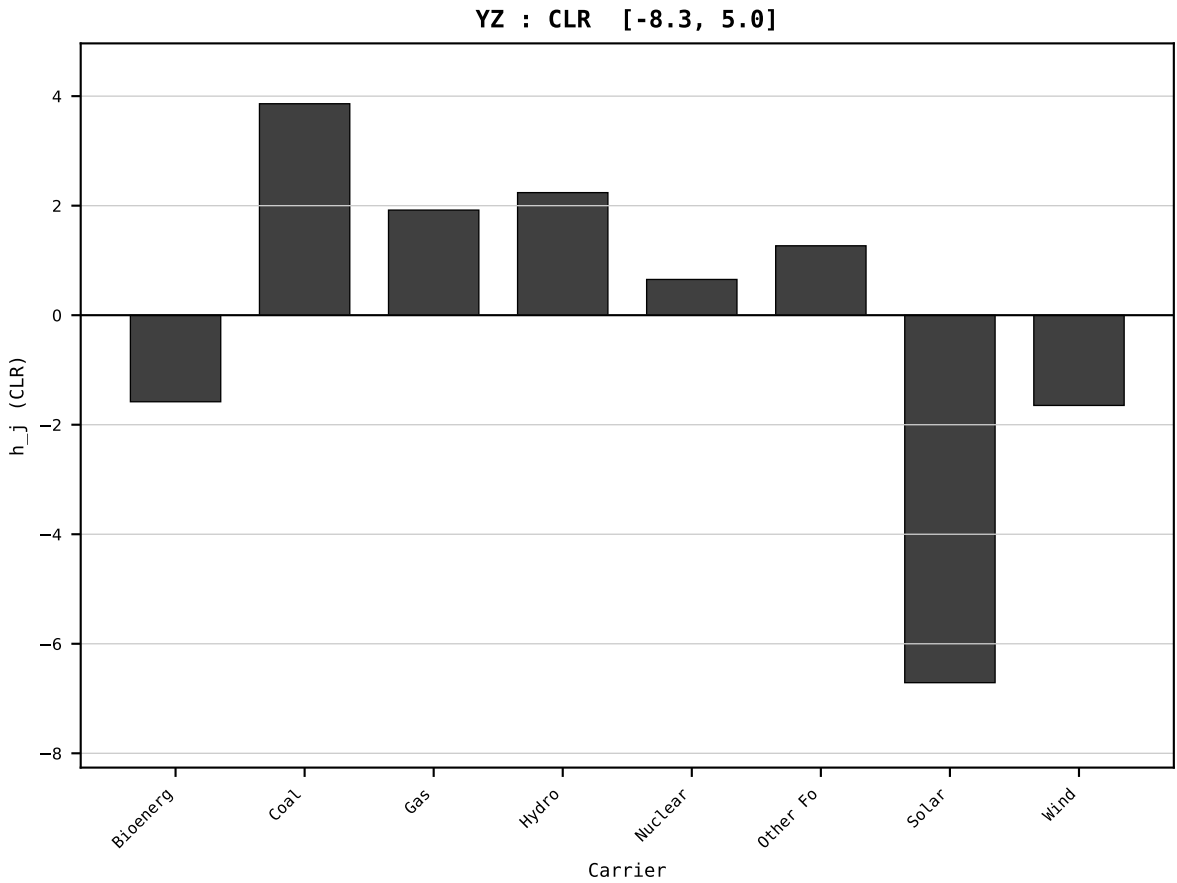
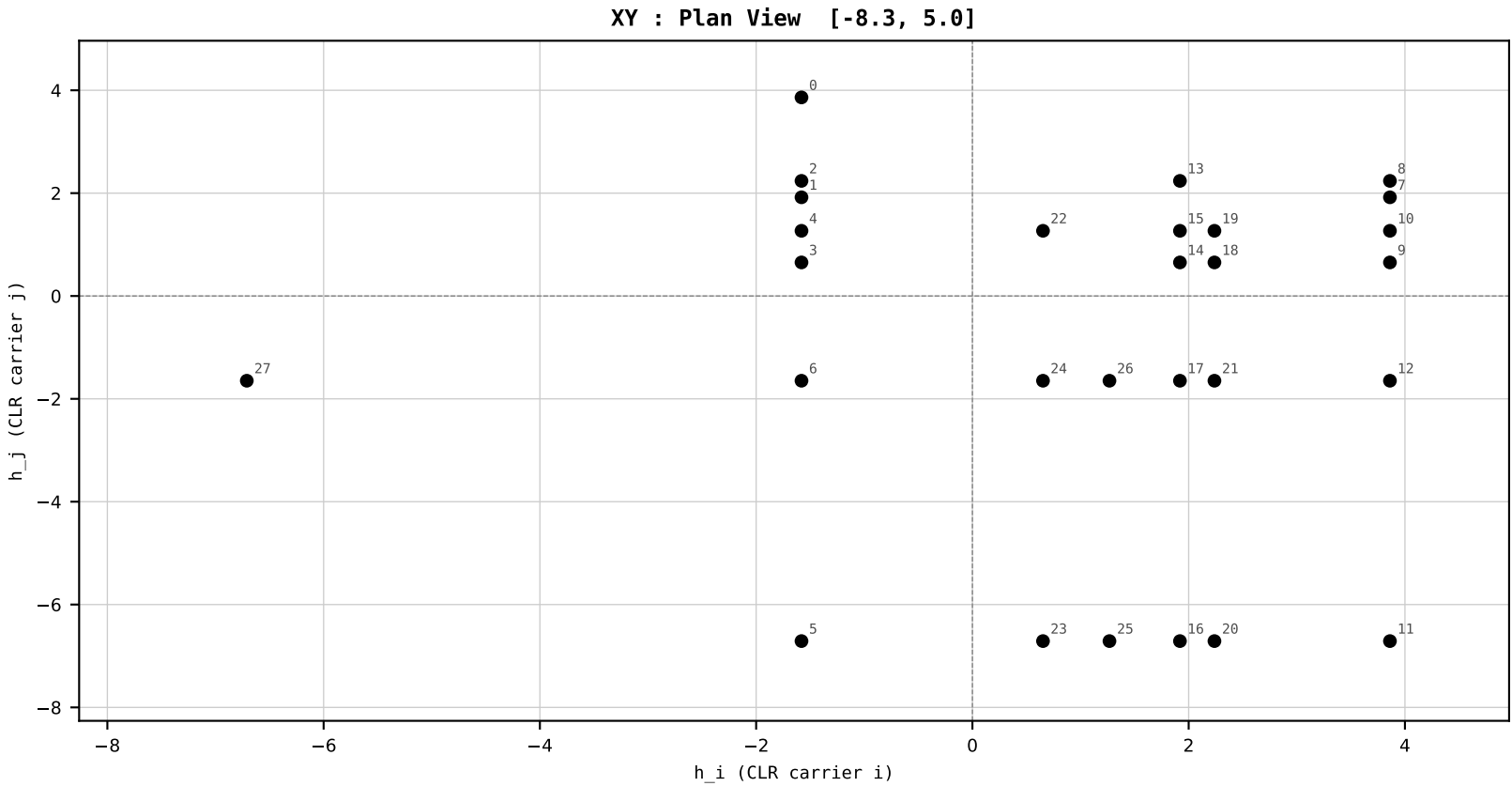
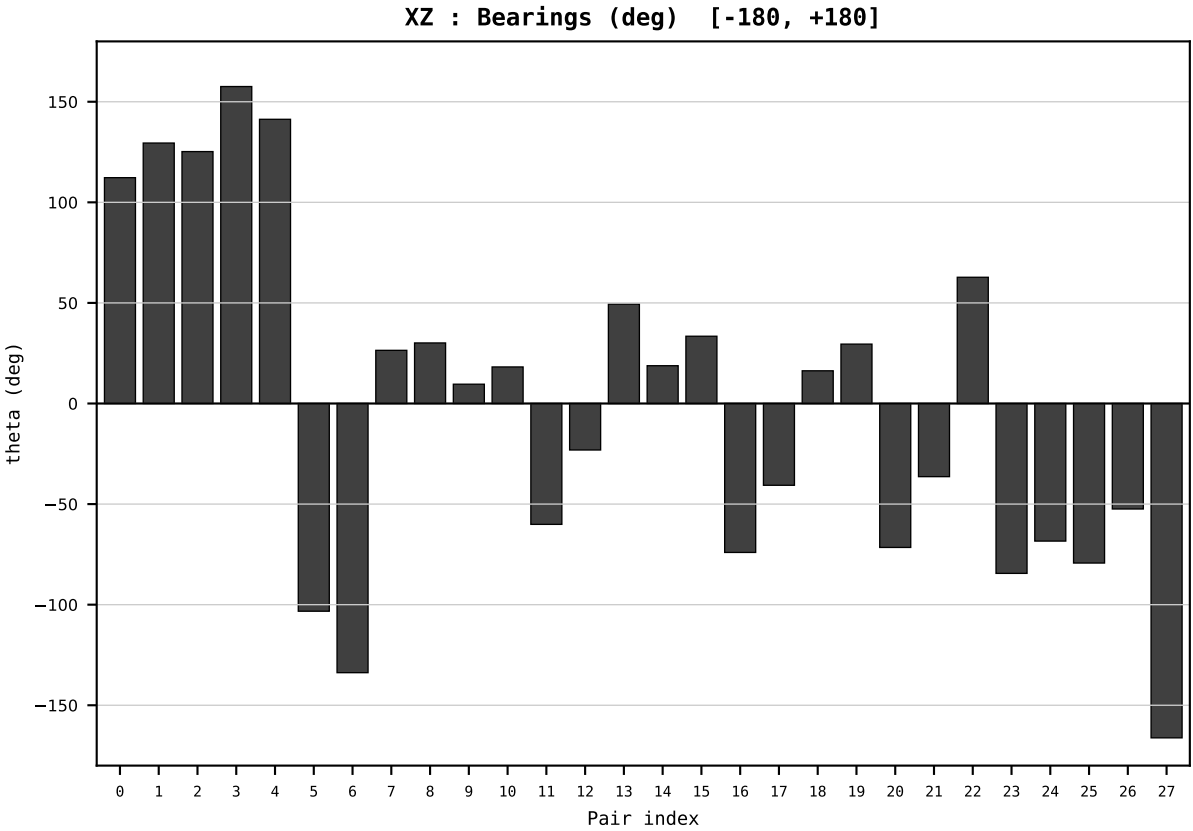
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2000
t : 0 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.498576
Ring = Hs-3
E_metric= 75.8817
kappa_HS= 39023.00
omega = 0.0000 deg
d_A = 0.000000
Helm = ---
Helm d = ---
DR = 10.572 HLR
DR ratio= 39023.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=1 | Year 2001 | D=8 N=26 pairs=28

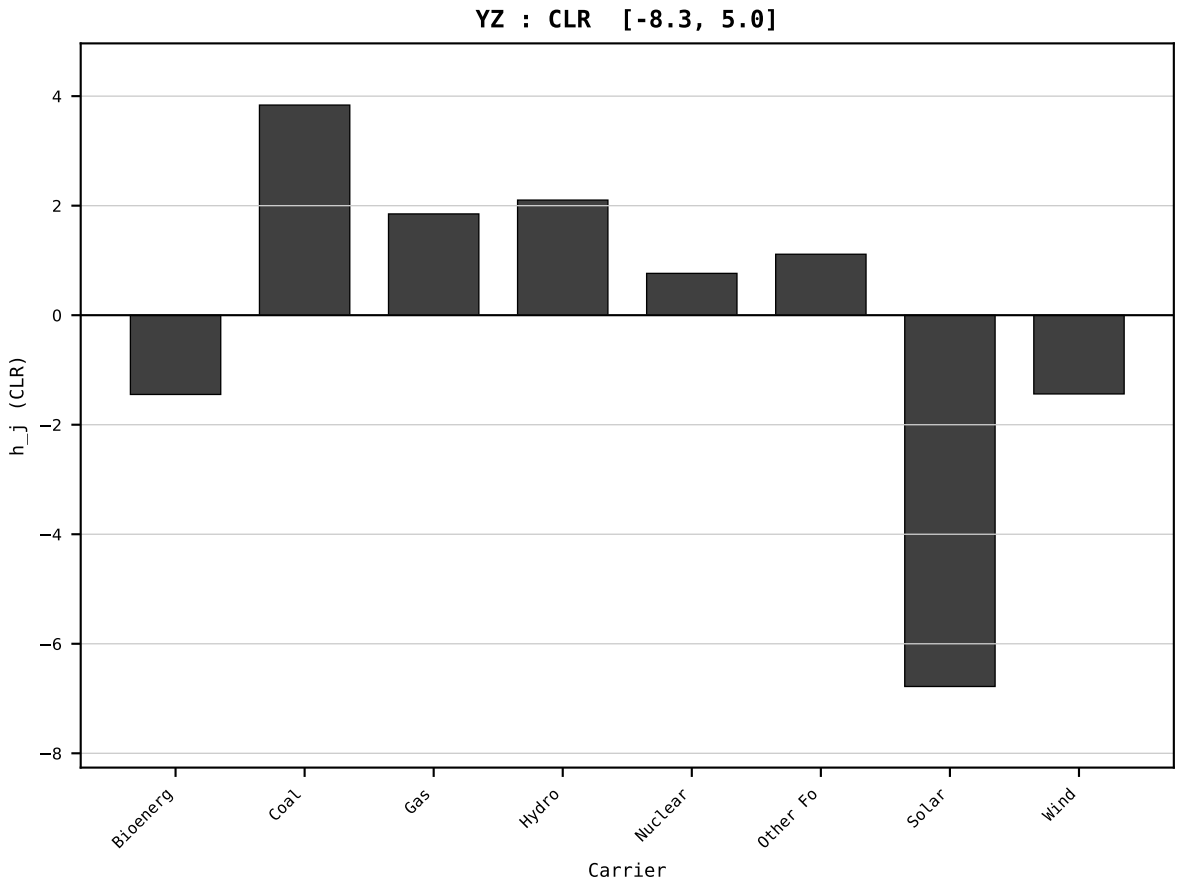
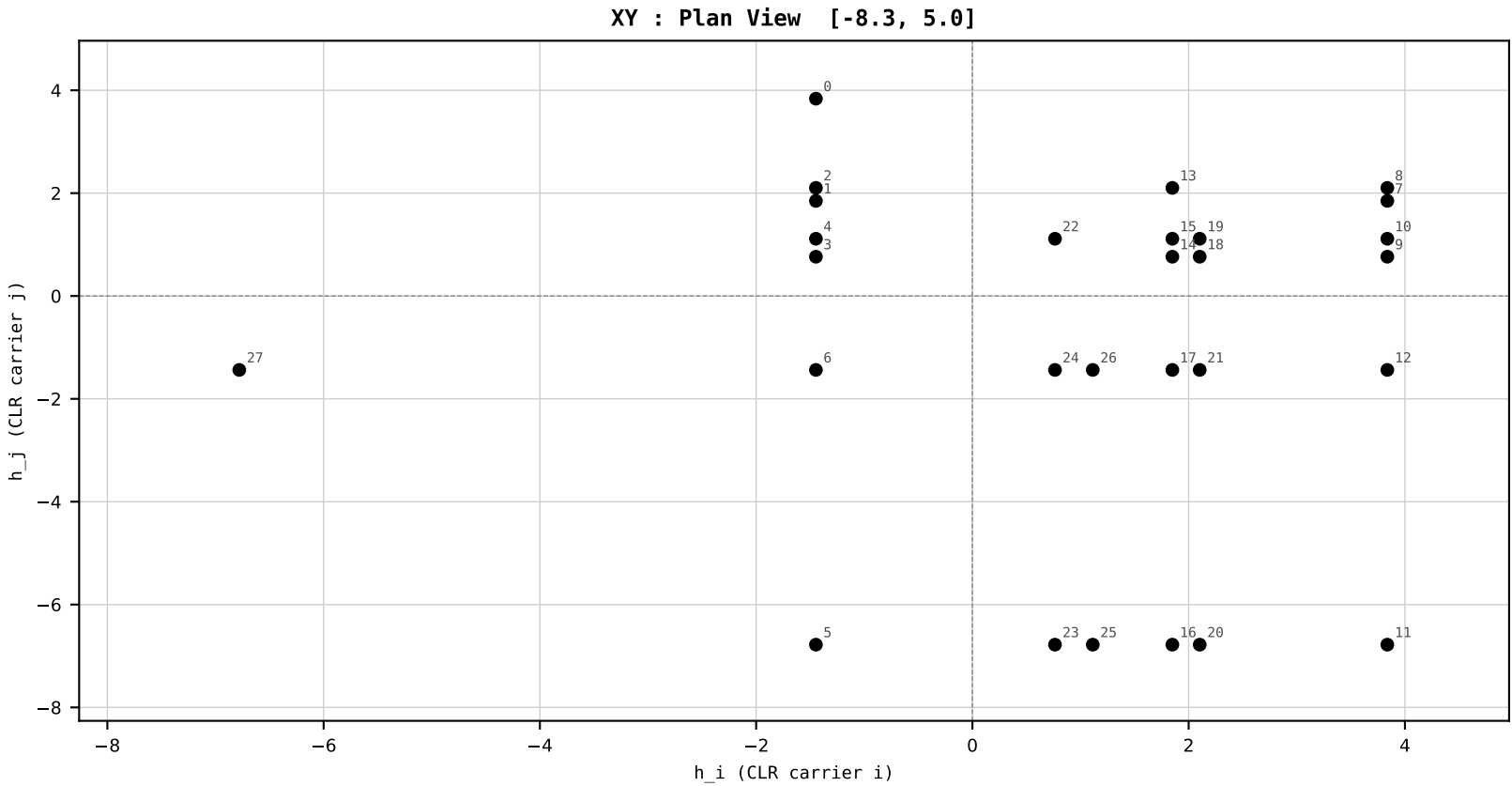
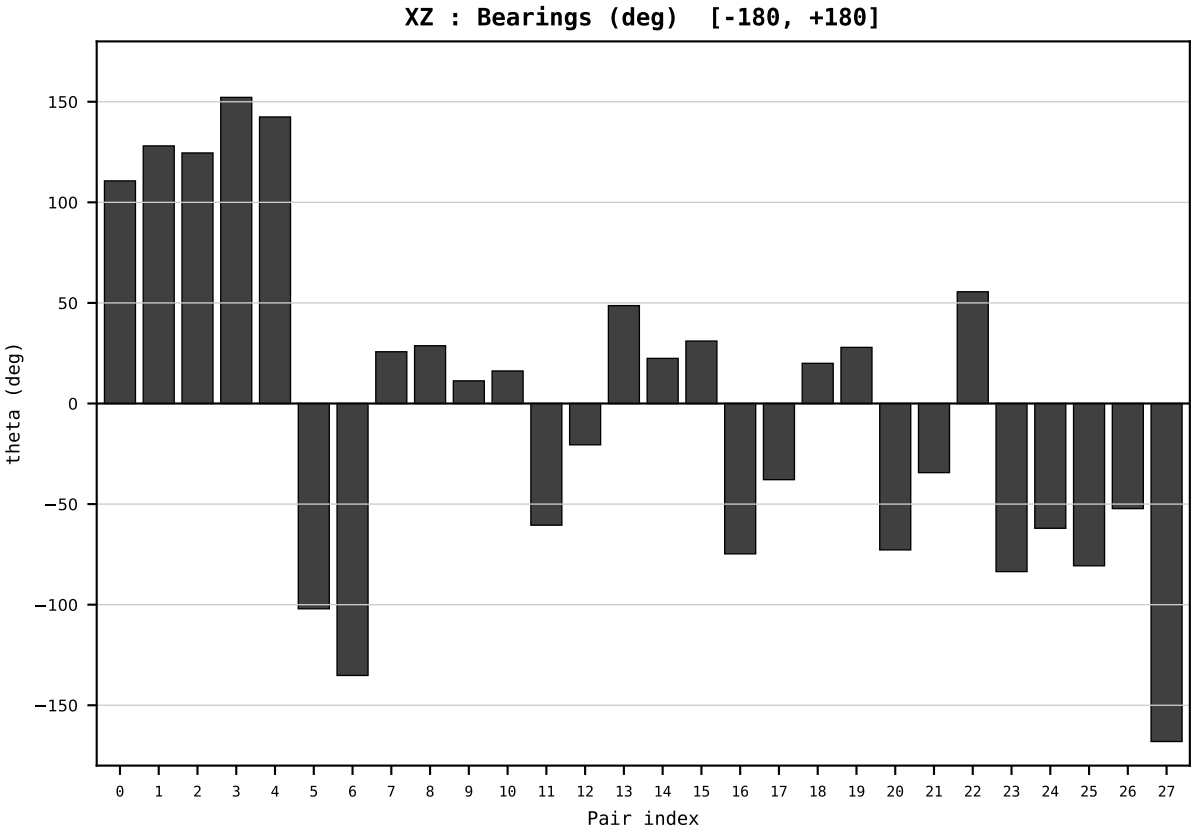
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2001
t : 1 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.506780
Ring = Hs-4
E_metric= 74.5149
kappa_HS= 40829.00
omega = 2.2964 deg
d_A = 0.356346
Helm = Wind
Helm d = +0.210203
DR = 10.617 HLR
DR ratio= 40829.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=2 | Year 2002 | D=8 N=26 pairs=28

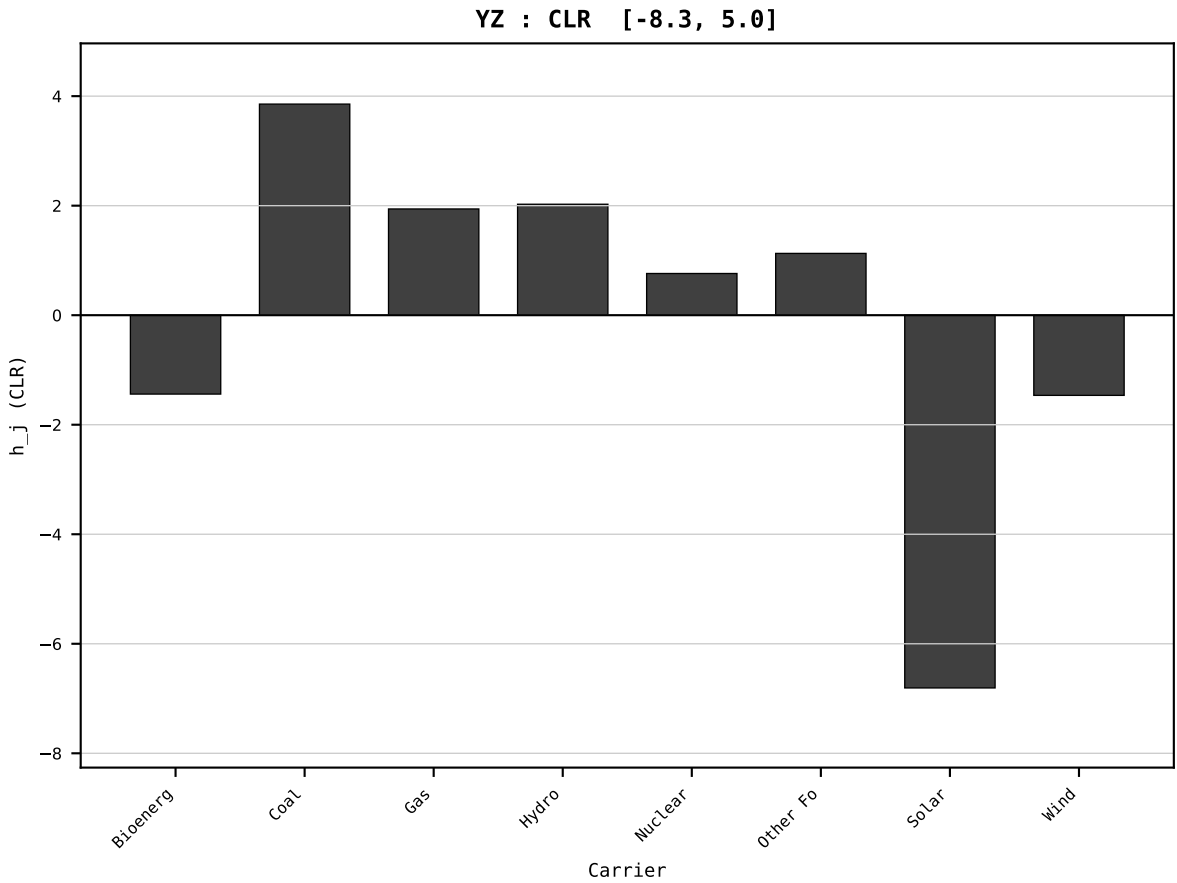
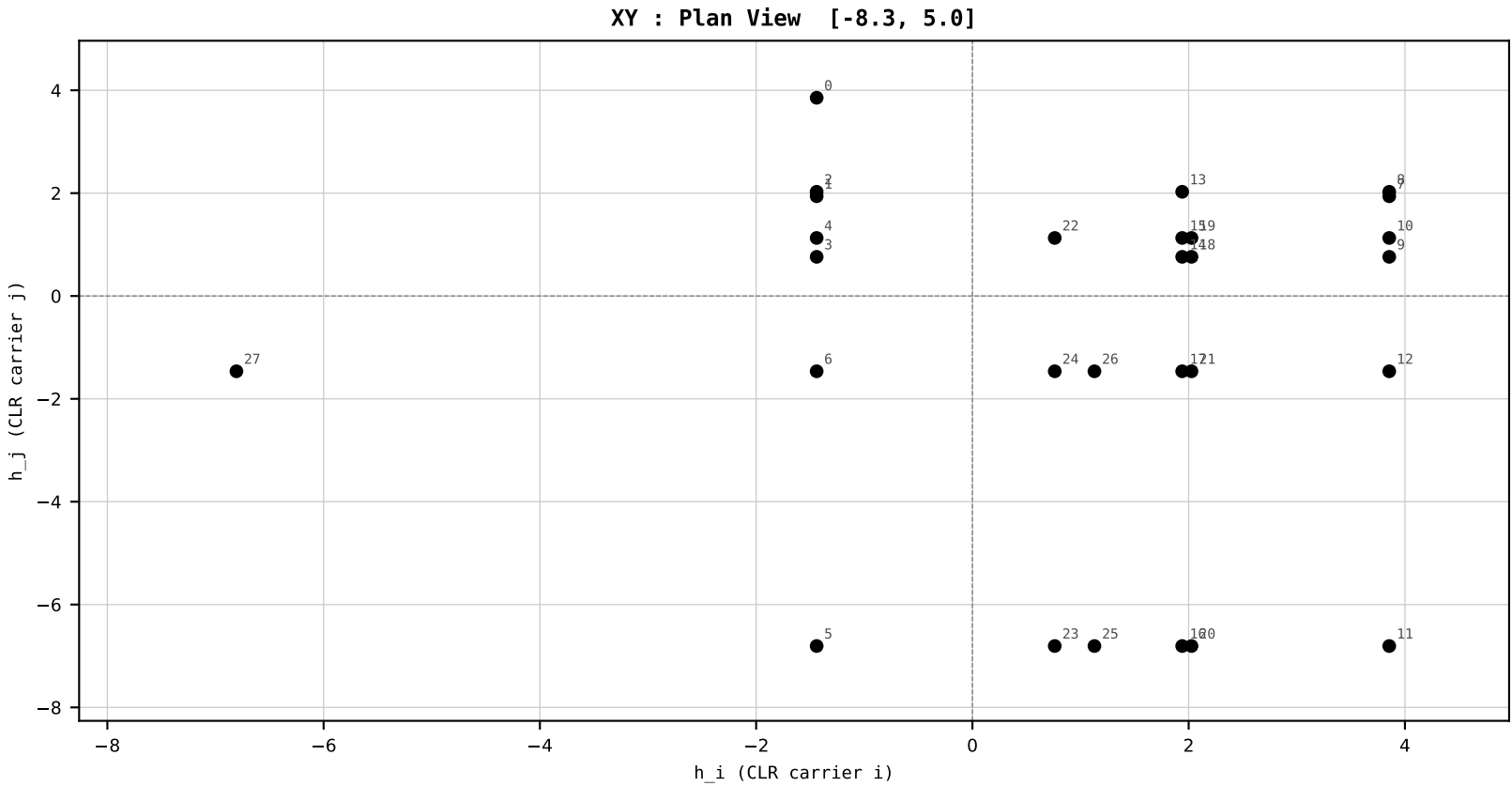
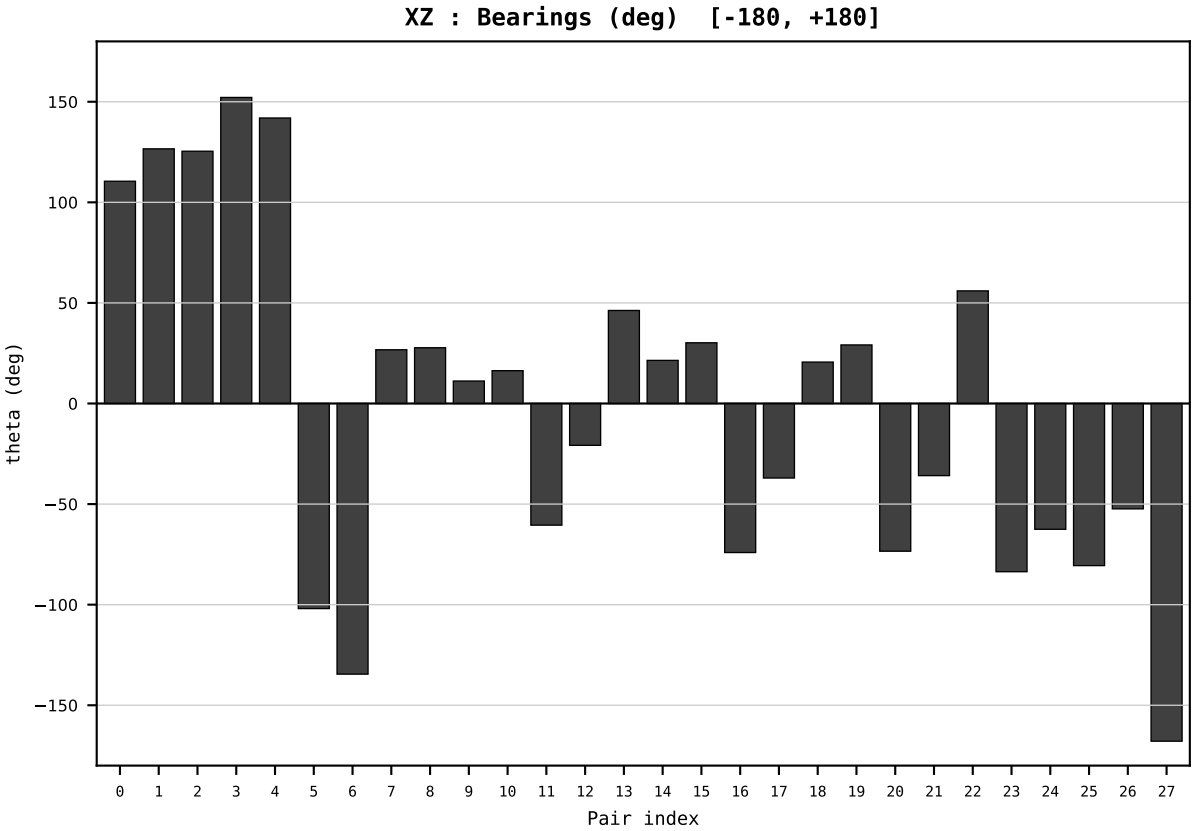
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2002
t : 2 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.509498
Ring = Hs-4
E_metric= 75.1276
kappa_HS= 42670.00
omega = 0.8041 deg
d_A = 0.126449
Helm = Gas
Helm d = +0.090790
DR = 10.661 HLR
DR ratio= 42670.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=3 | Year 2003 | D=8 N=26 pairs=28

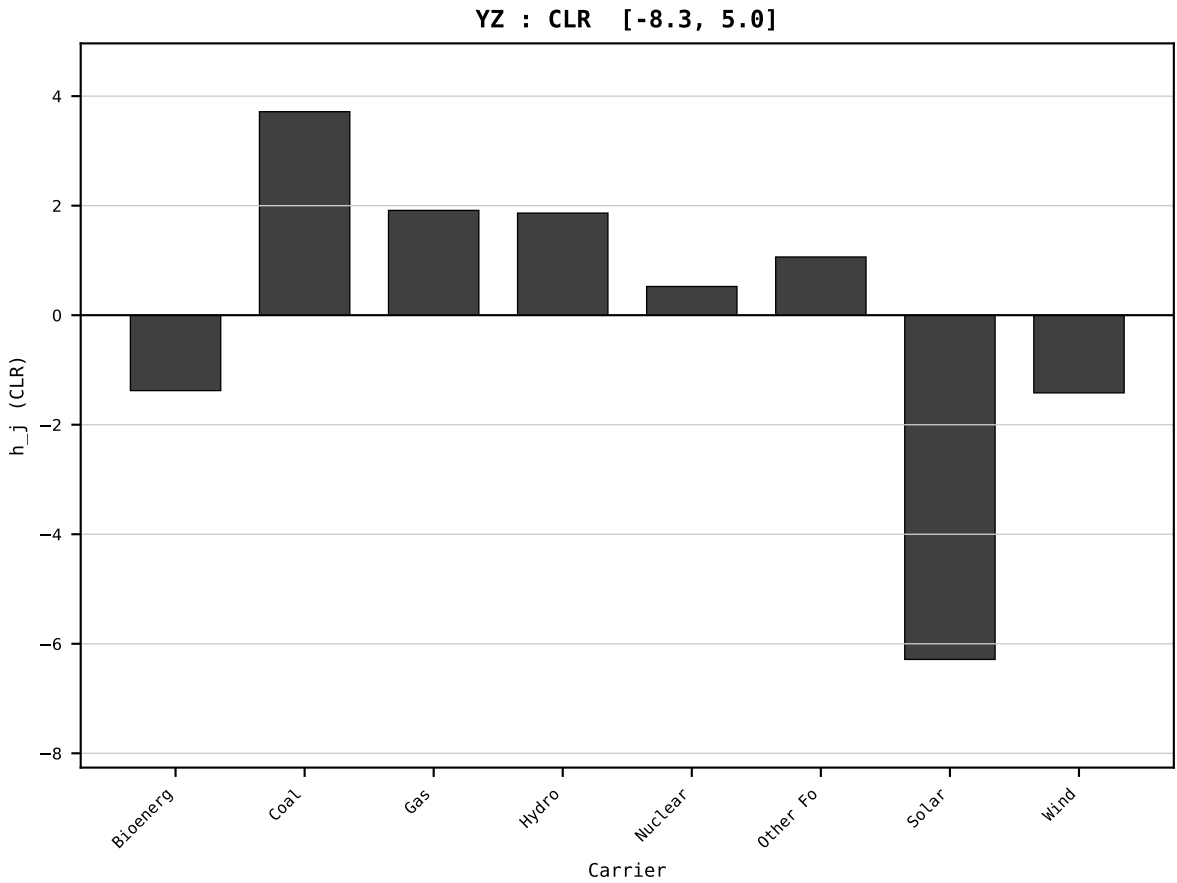
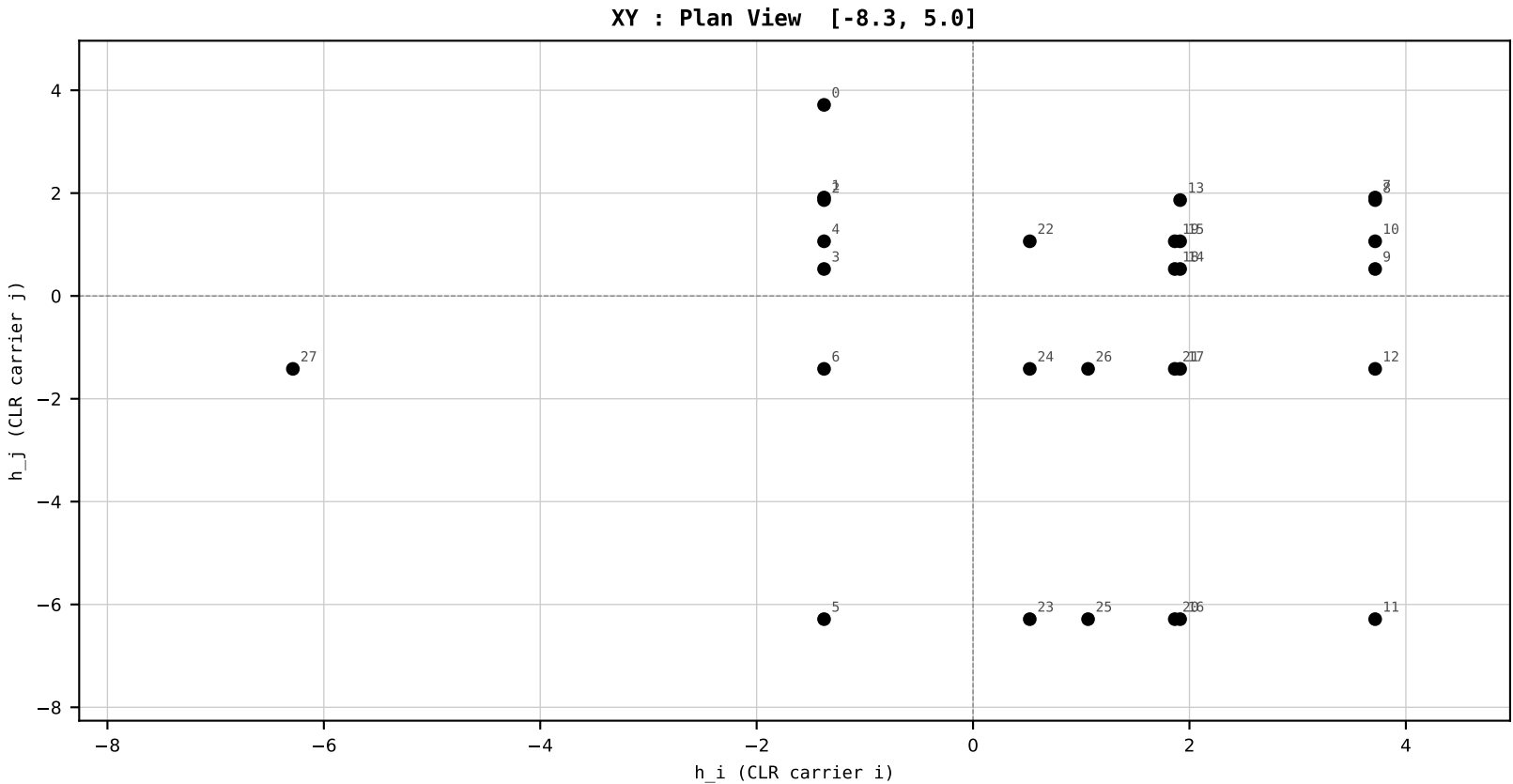
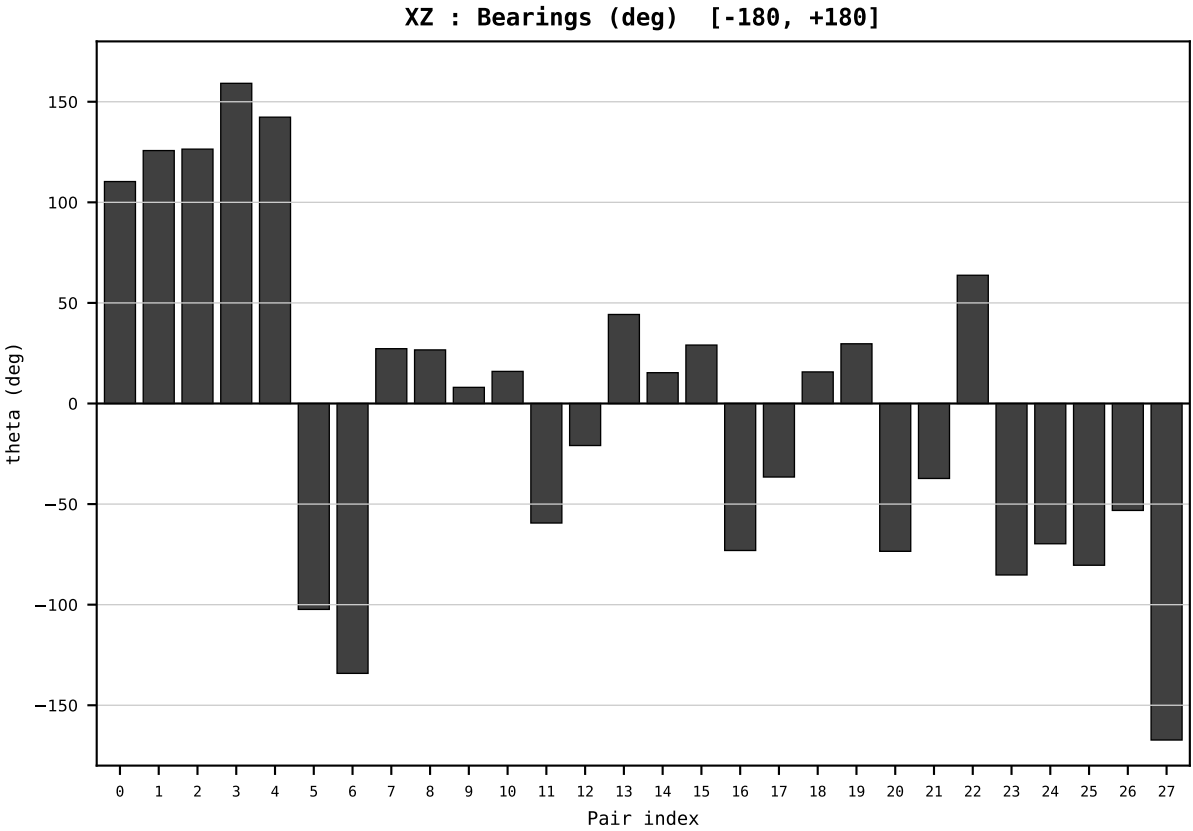
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2003
t : 3 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.500822
Ring = Hs-4
E_metric= 65.7748
kappa_HS= 22073.50
omega = 1.8448 deg
d_A = 0.619370
Helm = Solar
Helm d = +0.520423
DR = 10.002 HLR
DR ratio= 22073.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=4 | Year 2004 | D=8 N=26 pairs=28

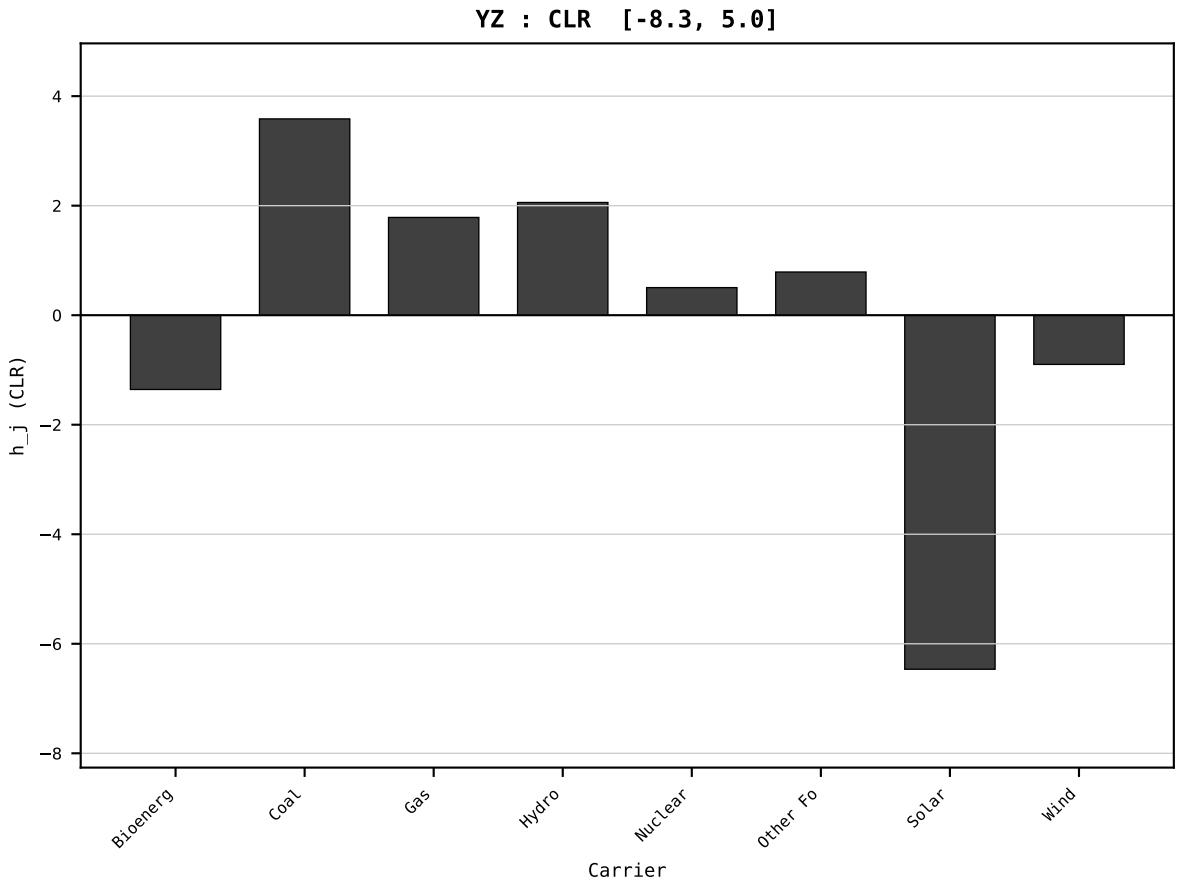
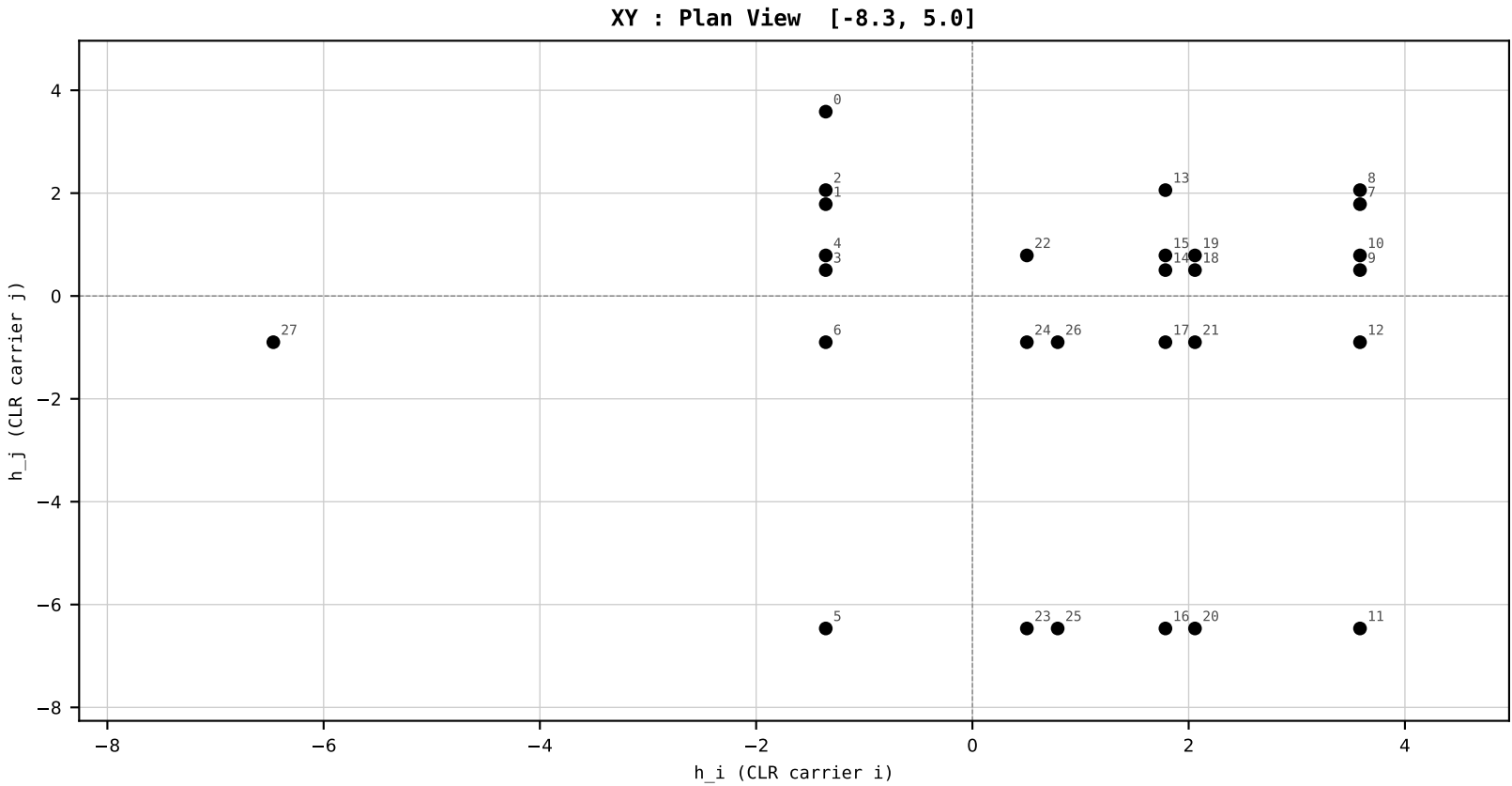
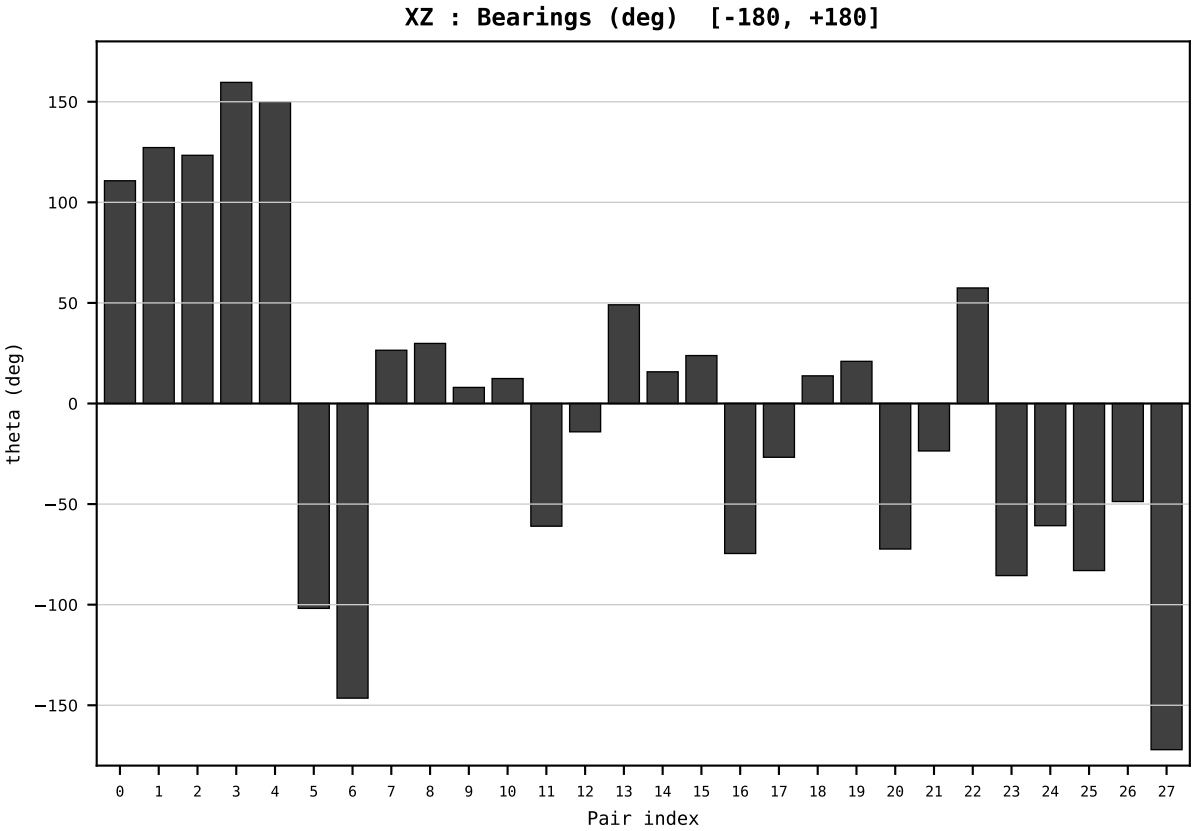
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2004
t : 4 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.474628
Ring = Hs-3
E_metric= 65.5944
kappa_HS= 23150.50
omega = 4.7416 deg
d_A = 0.670616
Helm = Wind
Helm d = +0.519664
DR = 10.050 HLR
DR ratio= 23150.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=5 | Year 2005 | D=8 N=26 pairs=28

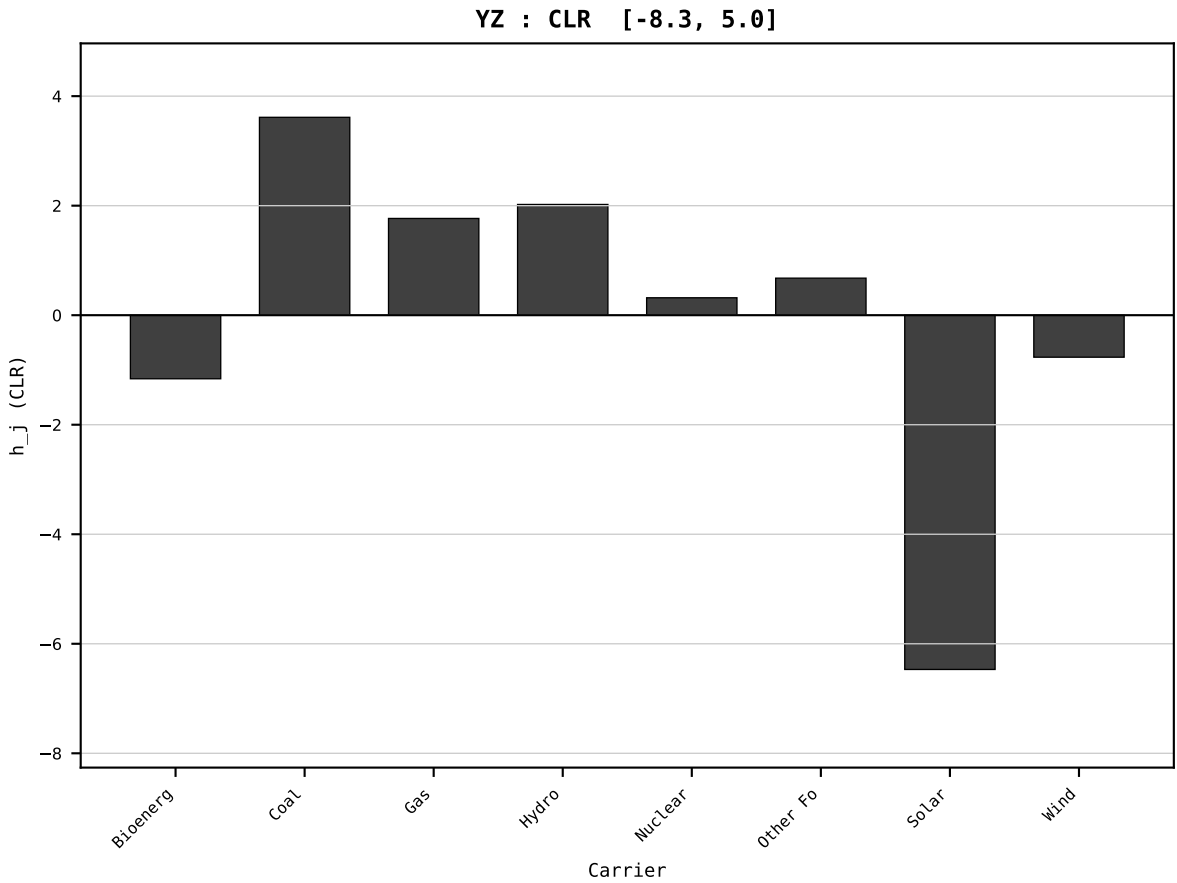
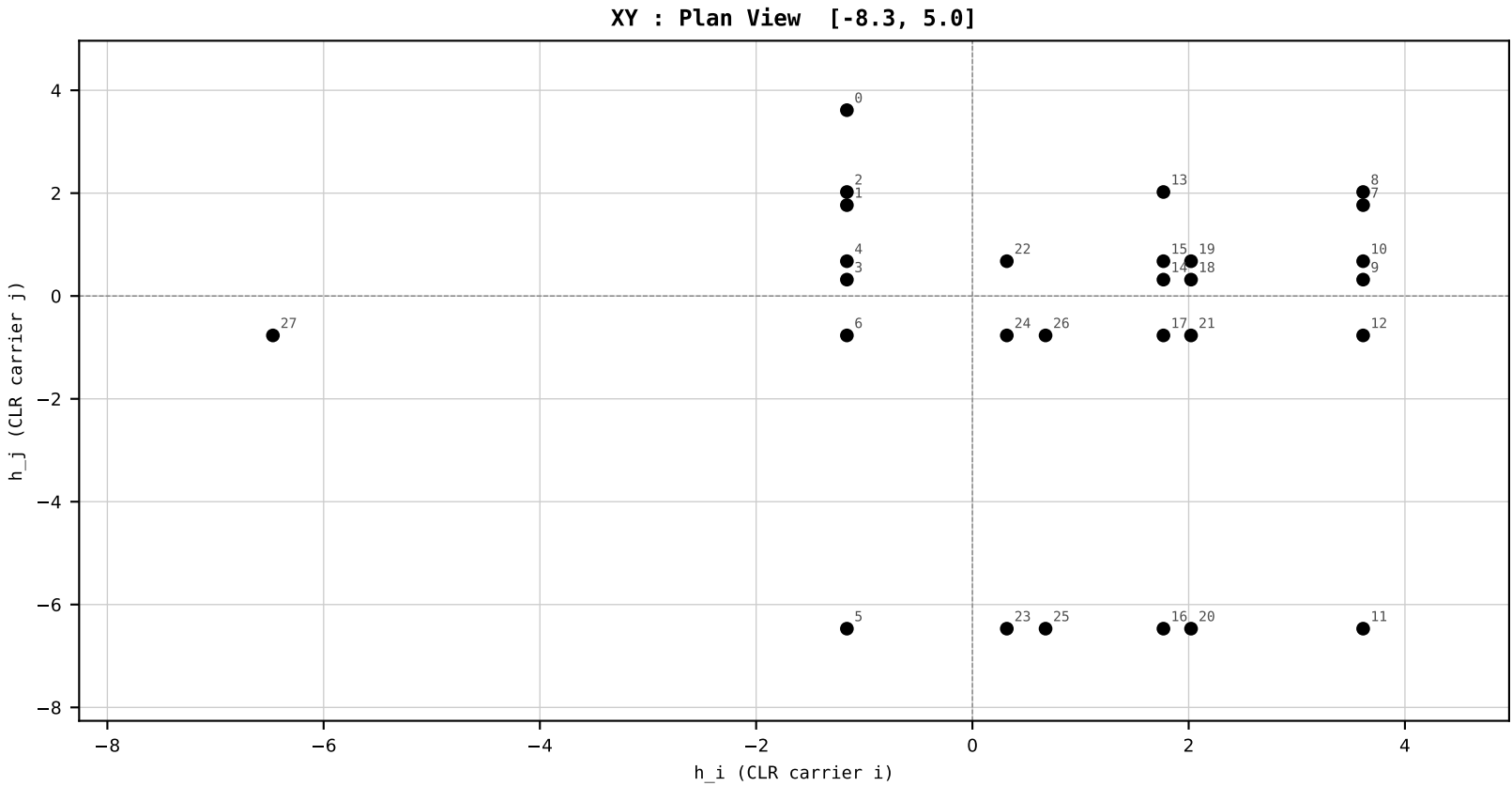
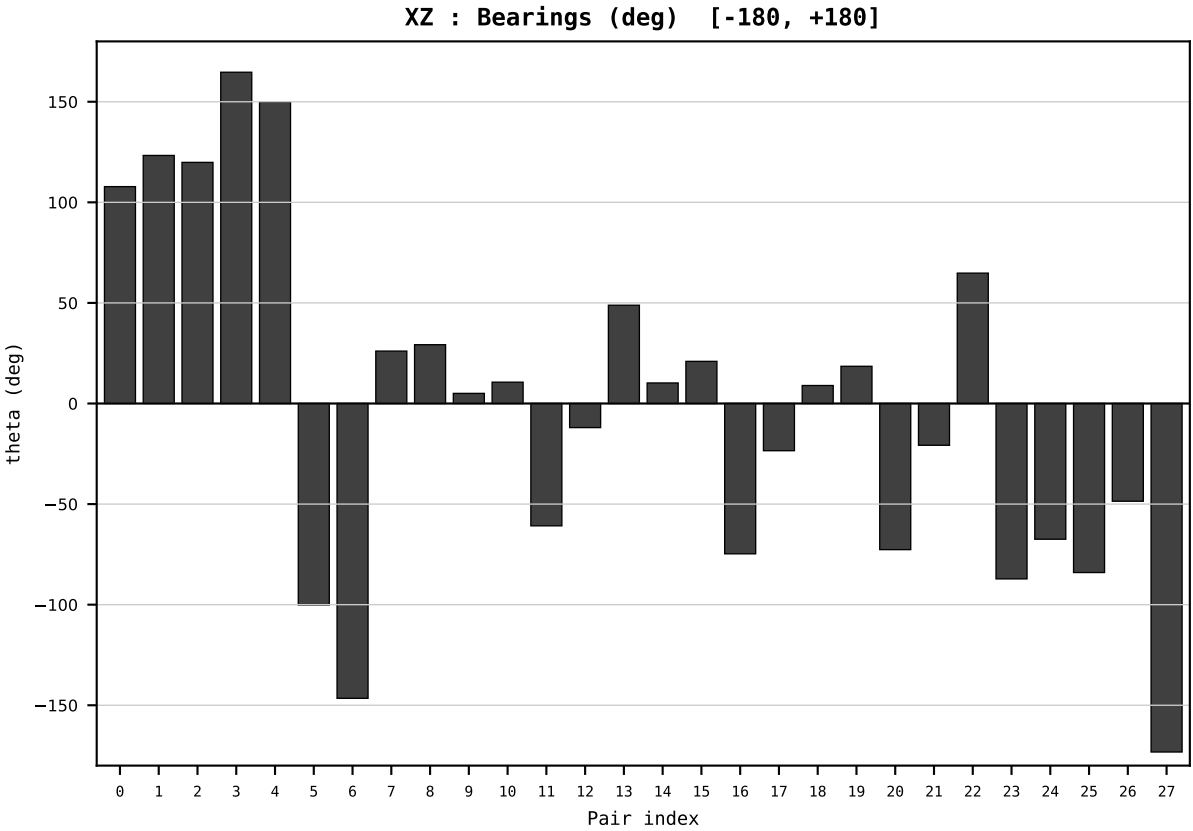
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2005
t : 5 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.491003
Ring = Hs-3
E_metric= 64.6110
kappa_HS= 23924.00
omega = 2.2606 deg
d_A = 0.324101
Helm = Bioenergy
Helm d = +0.195431
DR = 10.083 HLR
DR ratio= 23924.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=6 | Year 2006 | D=8 N=26 pairs=28

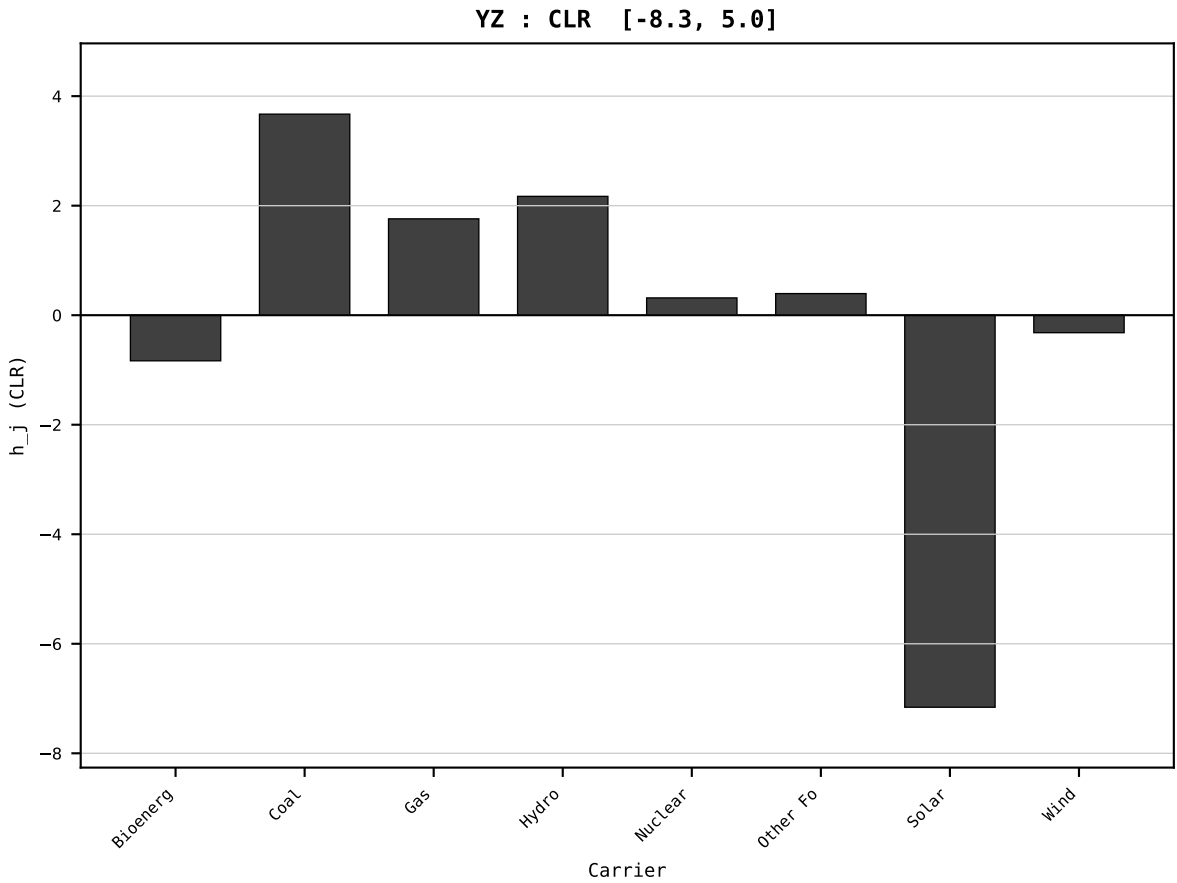
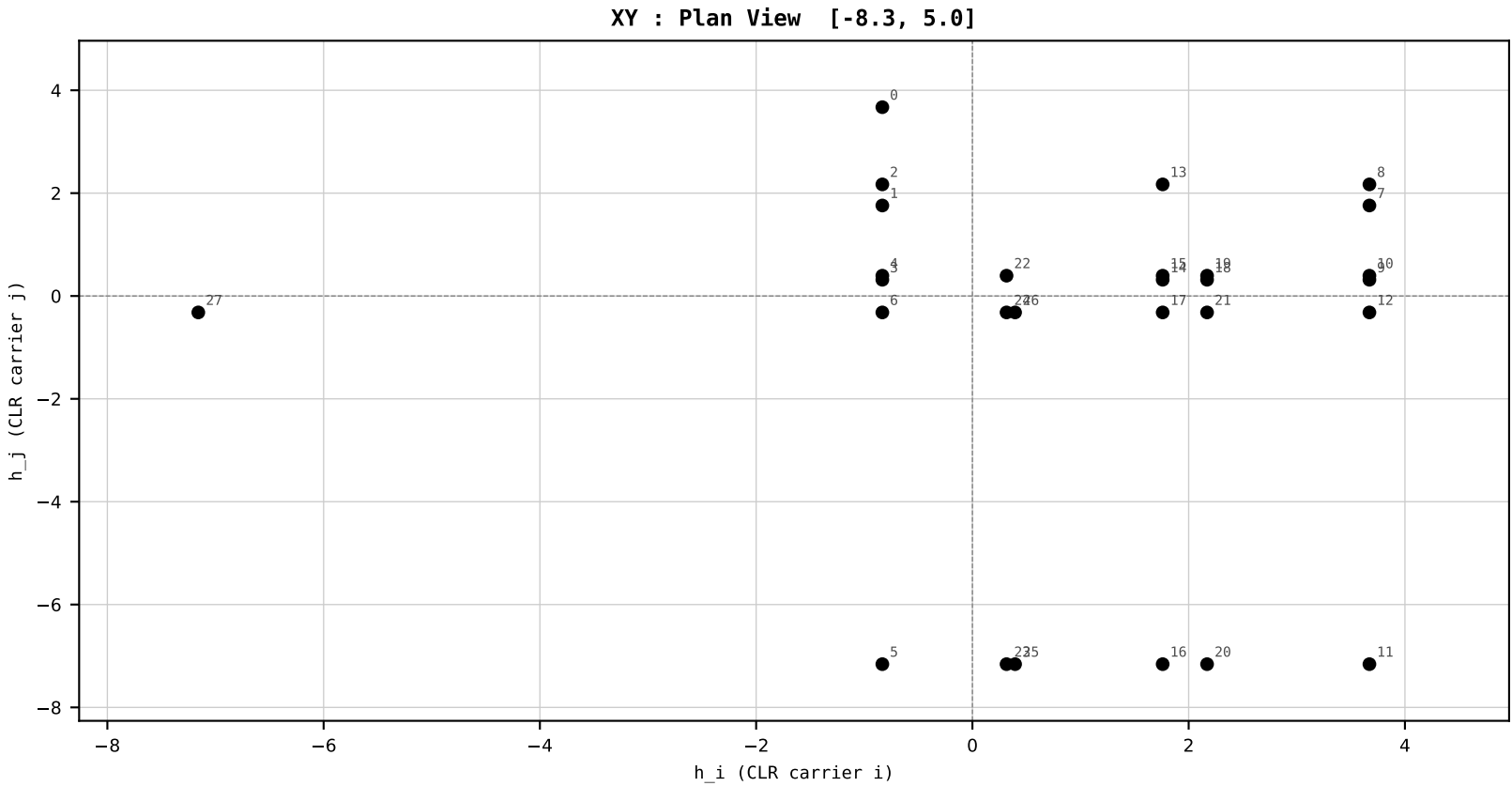
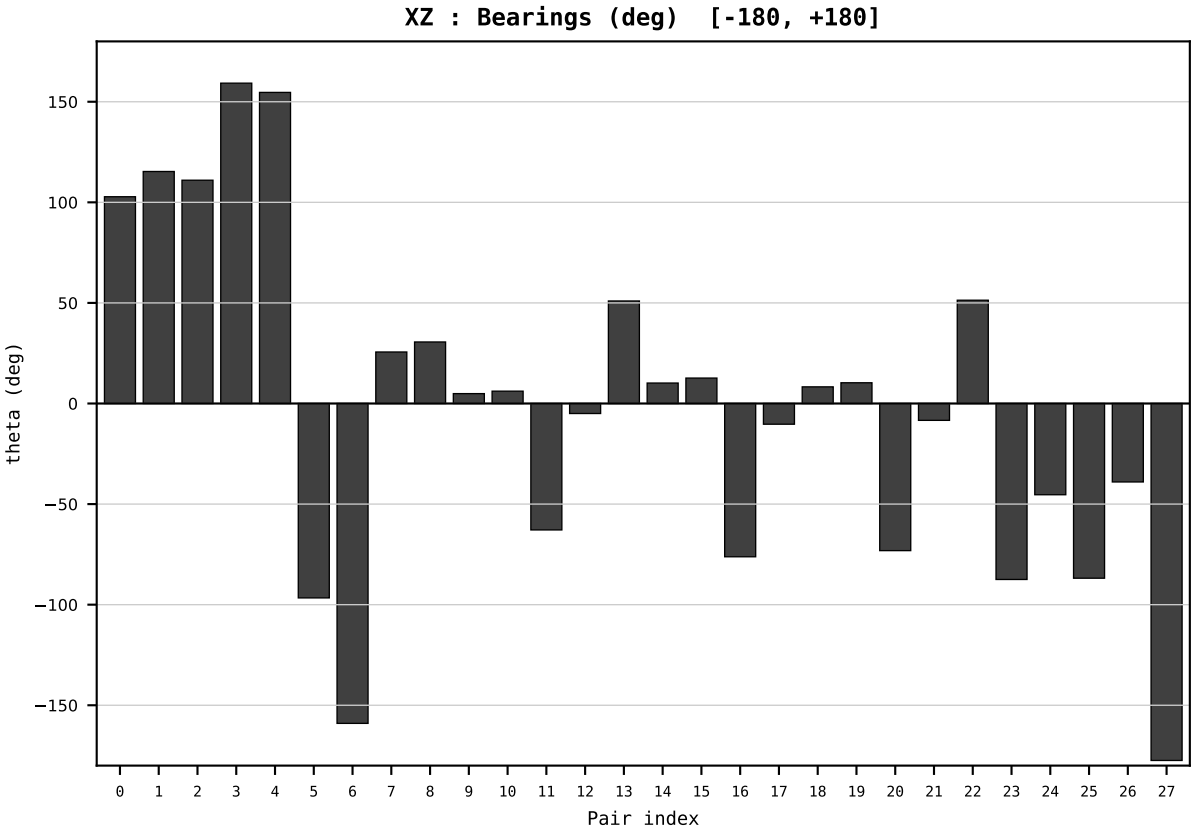
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2006
t : 6 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.493296
Ring = Hs-3
E_metric= 73.5896
kappa_HS= 50546.00
omega = 5.3311 deg
d_A = 0.942605
Helm = Solar
Helm d = -0.689998
DR = 10.831 HLR
DR ratio= 50546.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=7 | Year 2007 | D=8 N=26 pairs=28

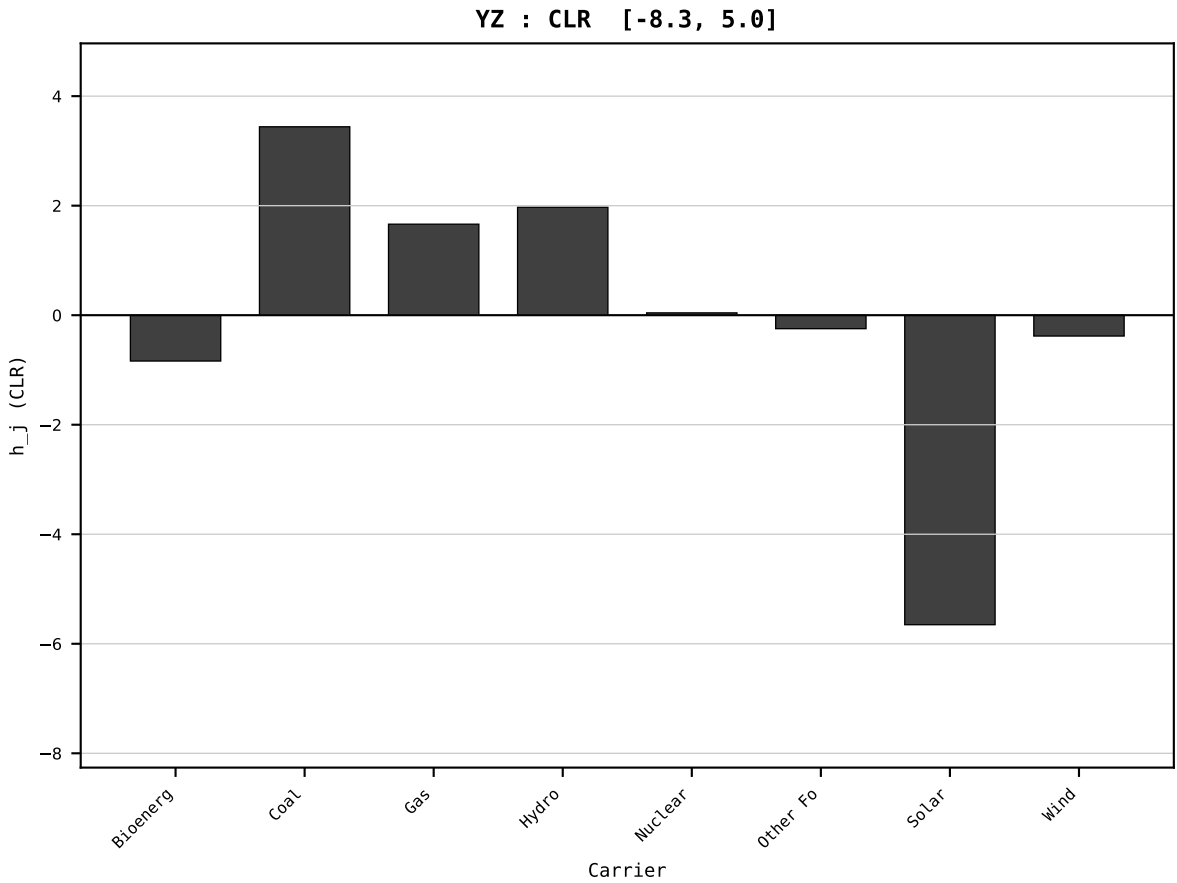
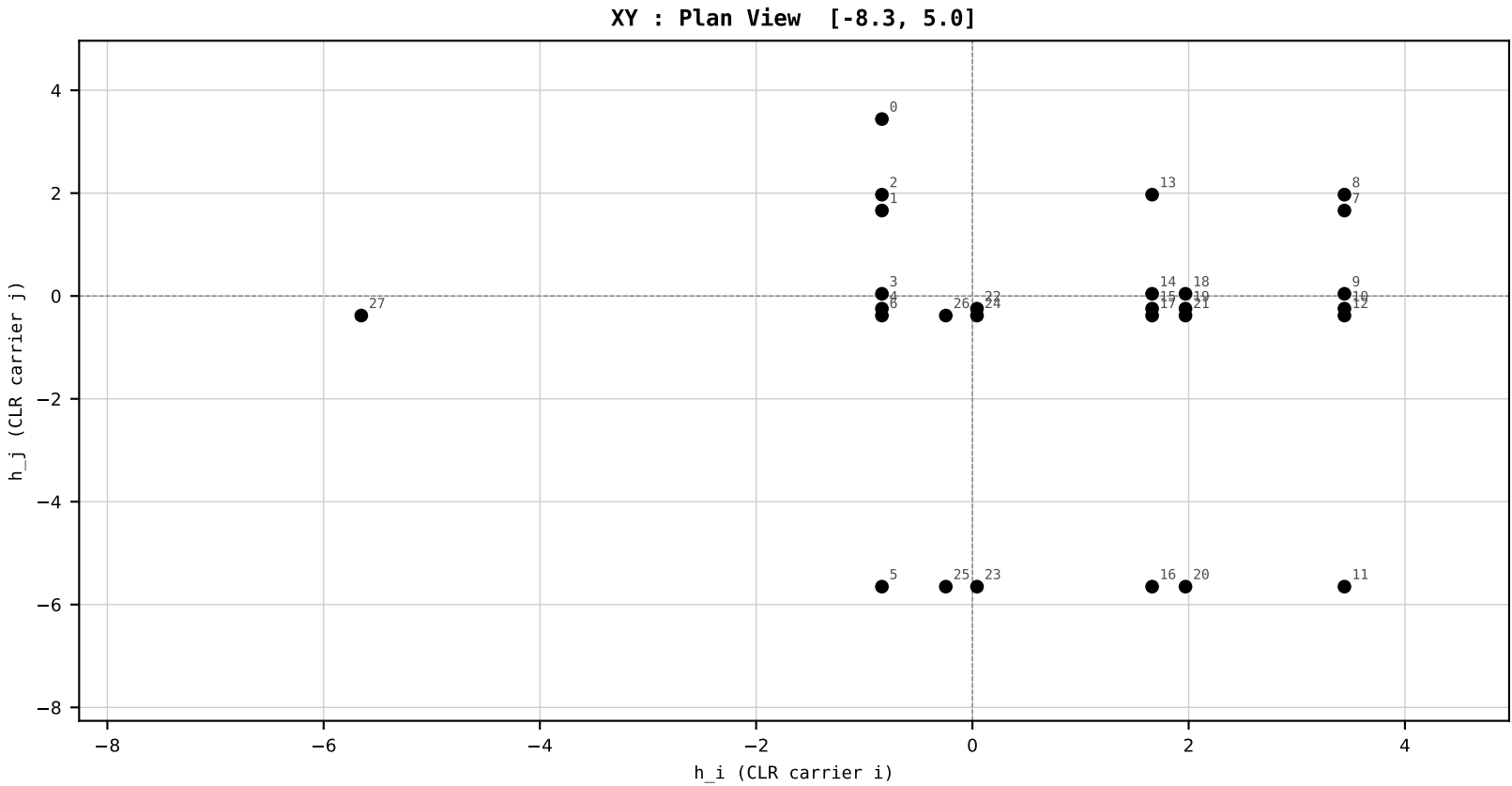
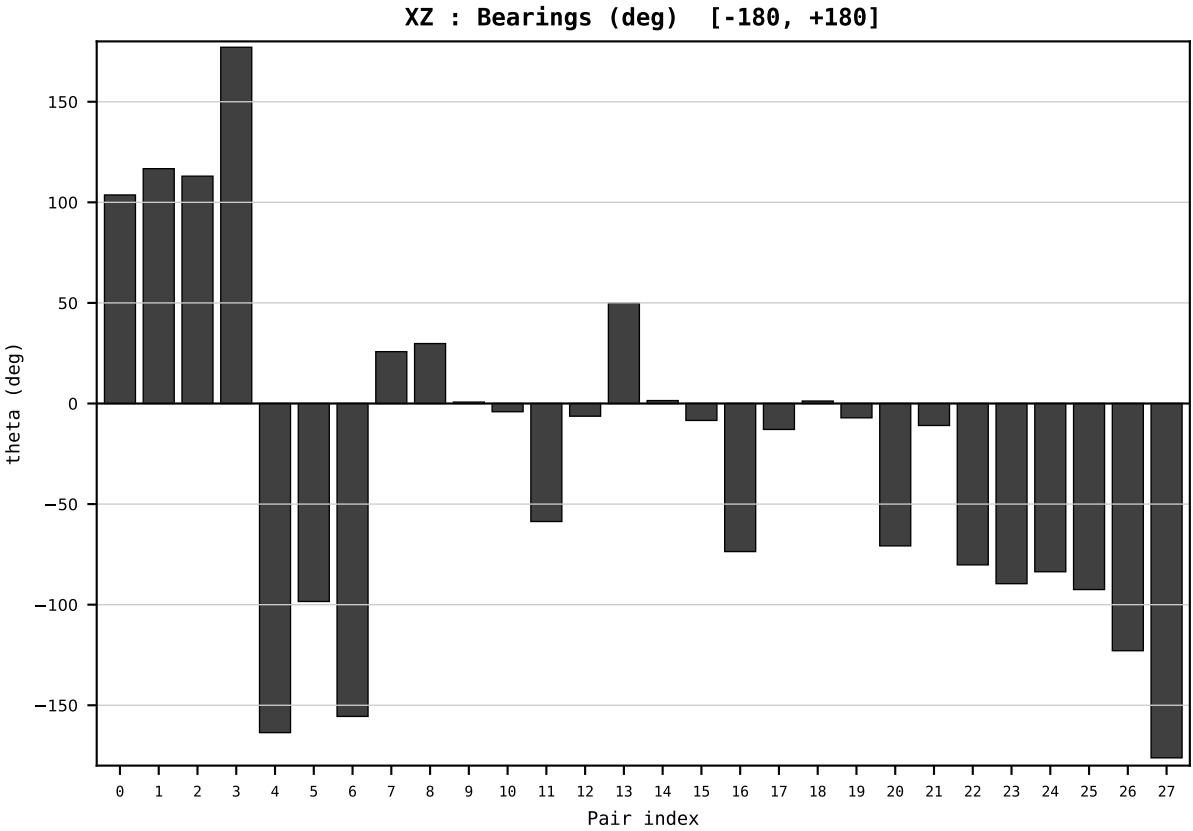
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2007
t : 7 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.488879
Ring = Hs-3
E_metric= 51.3339
kappa_HS= 8887.83
omega = 6.8016 deg
d_A = 1.692216
Helm = Solar
Helm d = +1.507238
DR = 9.092 HLR
DR ratio= 8887.8x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=8 | Year 2008 | D=8 N=26 pairs=28

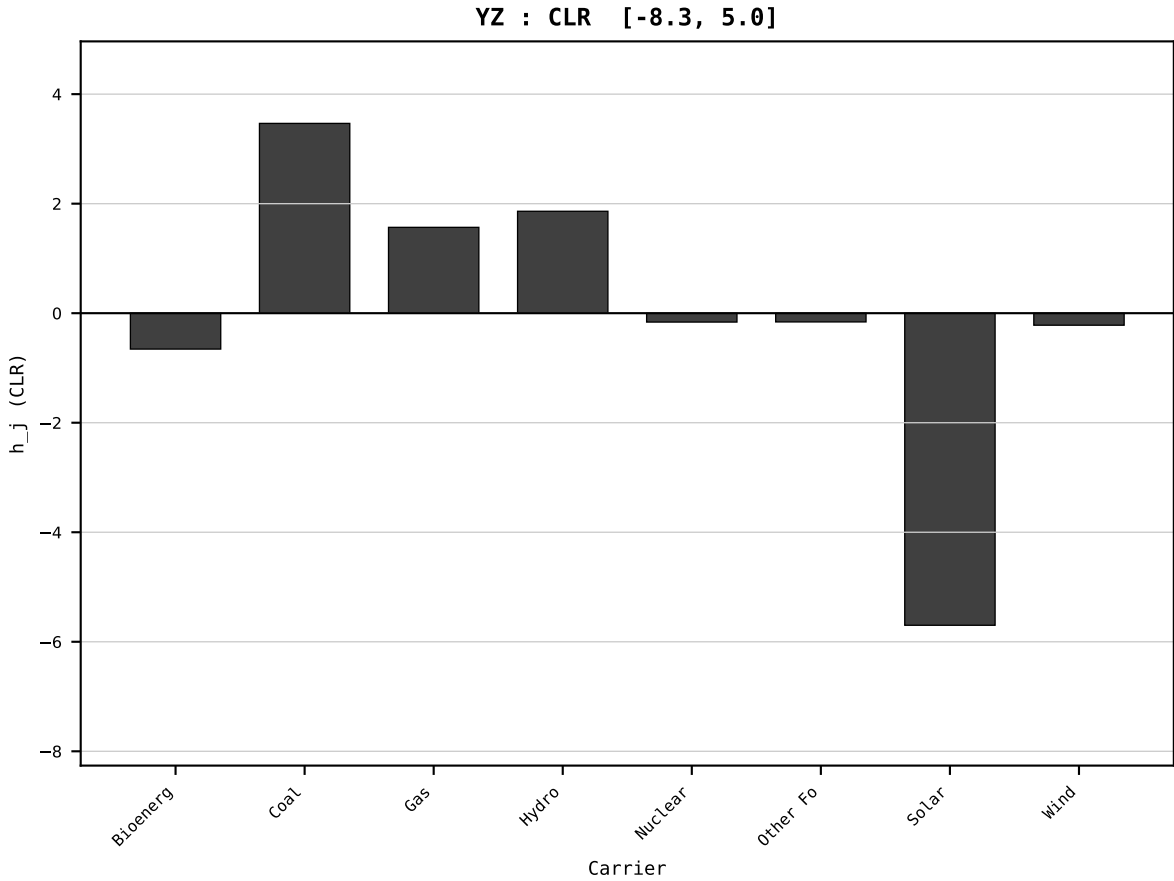
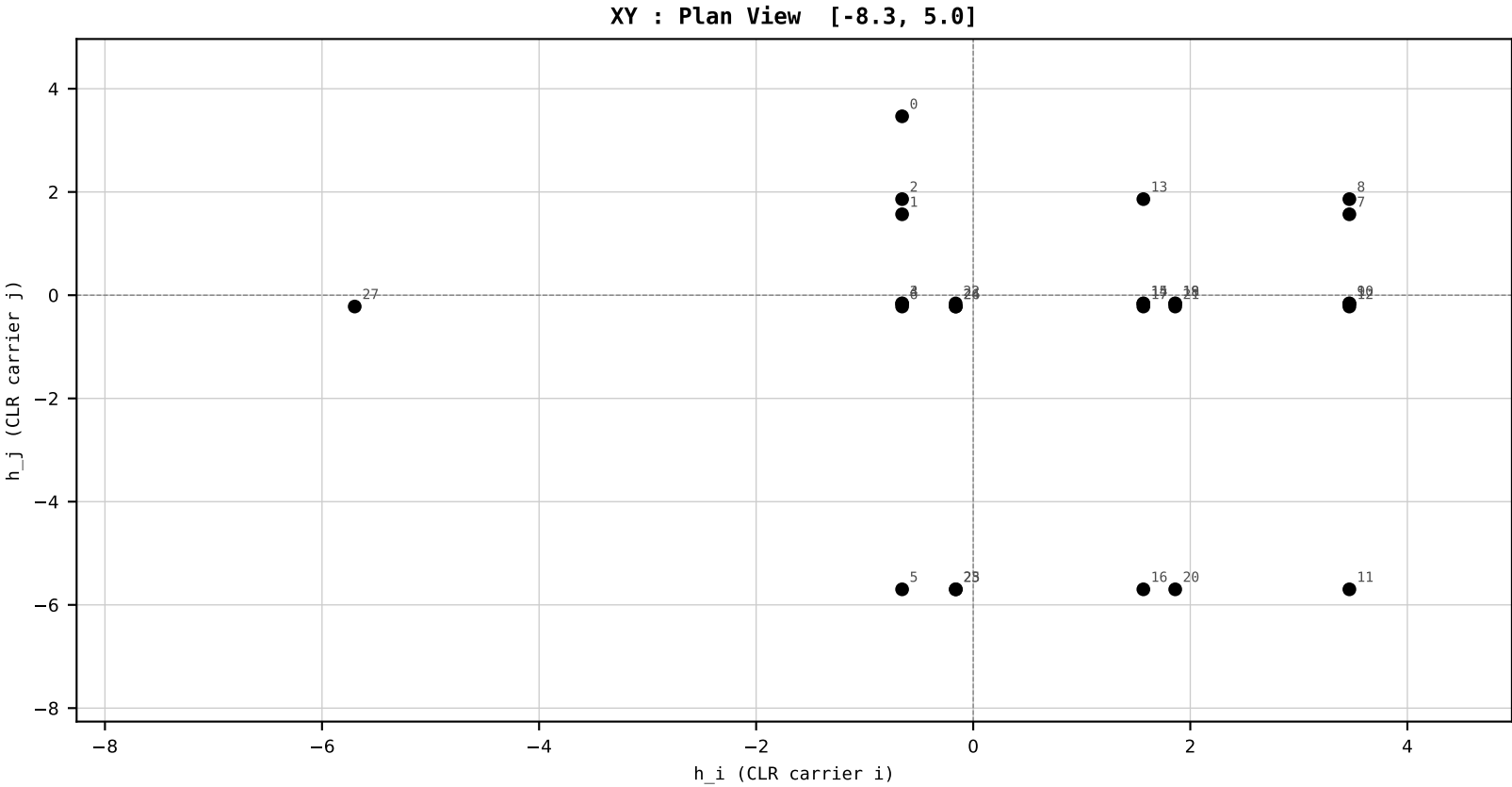
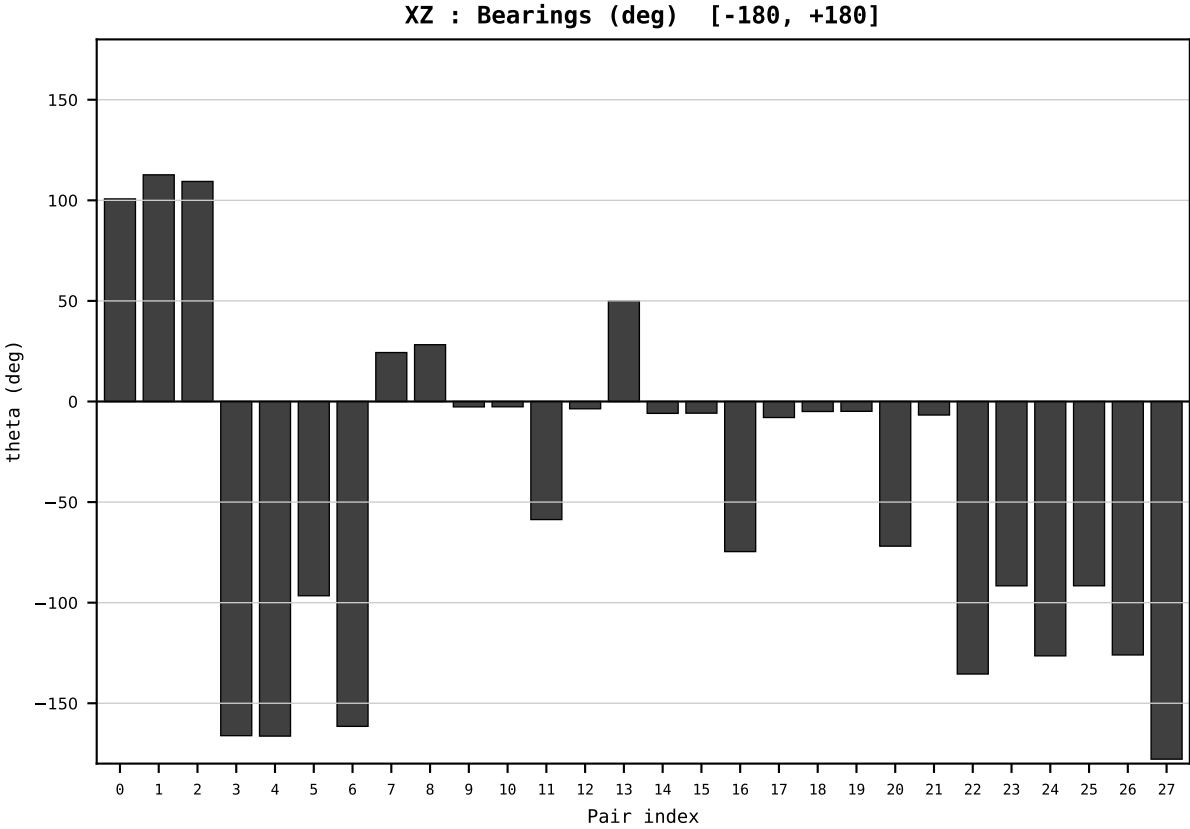
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2008
t : 8 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.503376
Ring = Hs-4
E_metric= 50.9375
kappa_HS= 9552.00
omega = 2.9028 deg
d_A = 0.363314
Helm = Nuclear
Helm d = -0.204435
DR = 9.165 HLR
DR ratio= 9552.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=9 | Year 2009 | D=8 N=26 pairs=28

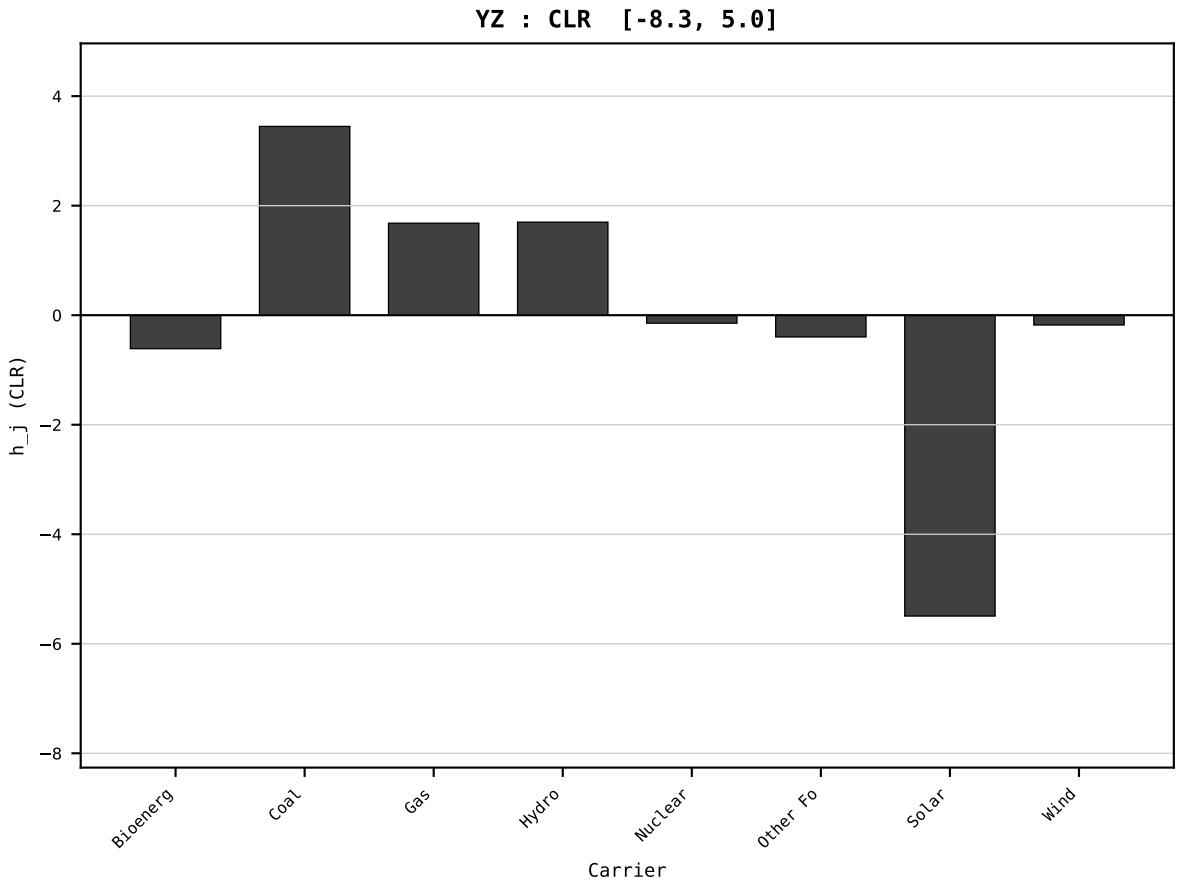
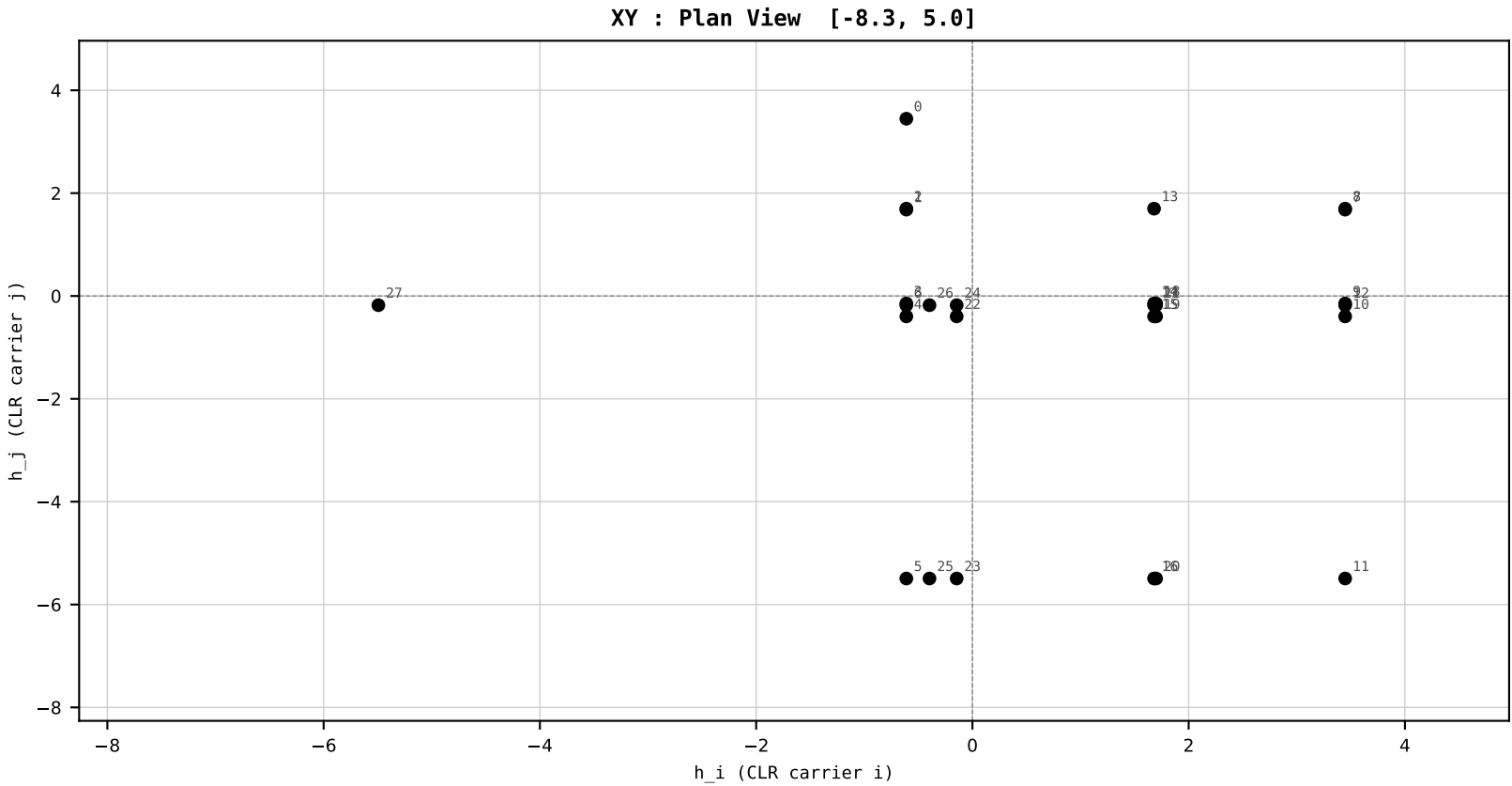
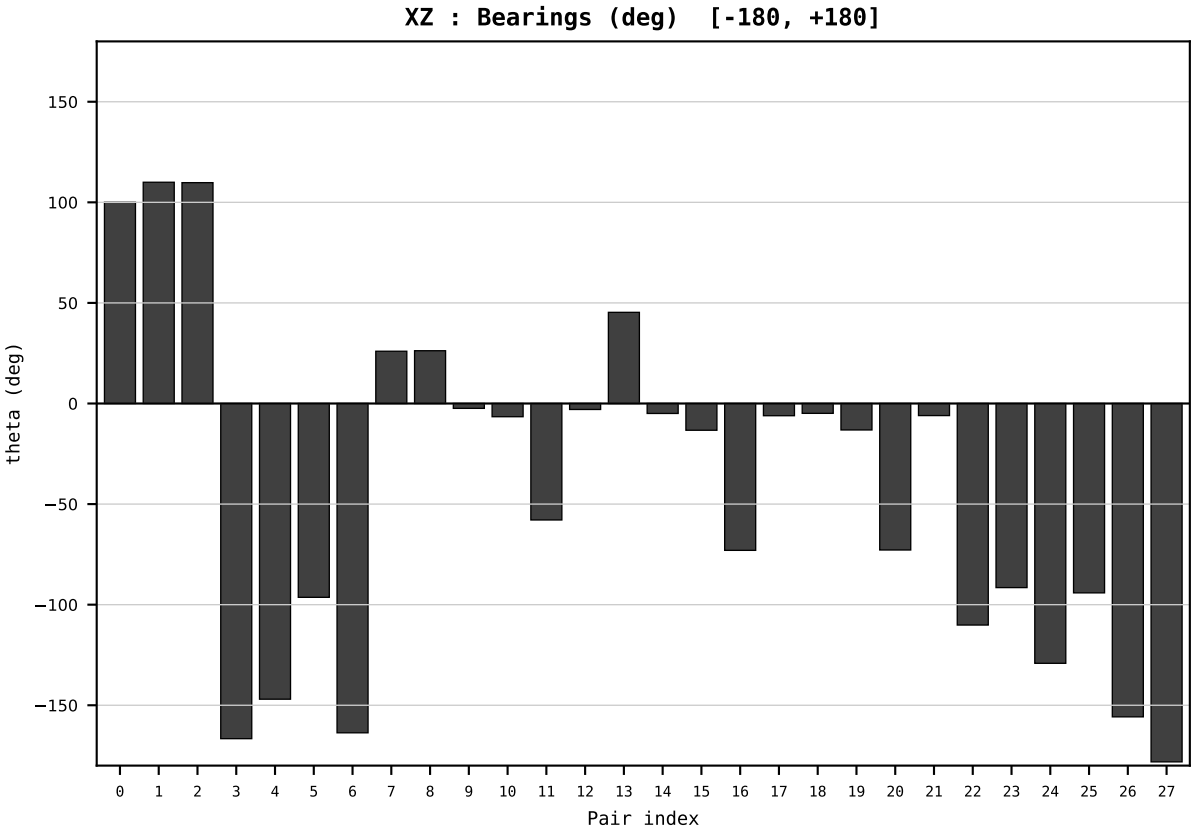
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2009
t : 9 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.505927
Ring = Hs-4
E_metric= 48.3562
kappa_HS= 7638.88
omega = 2.6745 deg
d_A = 0.376406
Helm = Other Fossil
Helm d = -0.237572
DR = 8.941 HLR
DR ratio= 7638.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=10 | Year 2010 | D=8 N=26 pairs=28

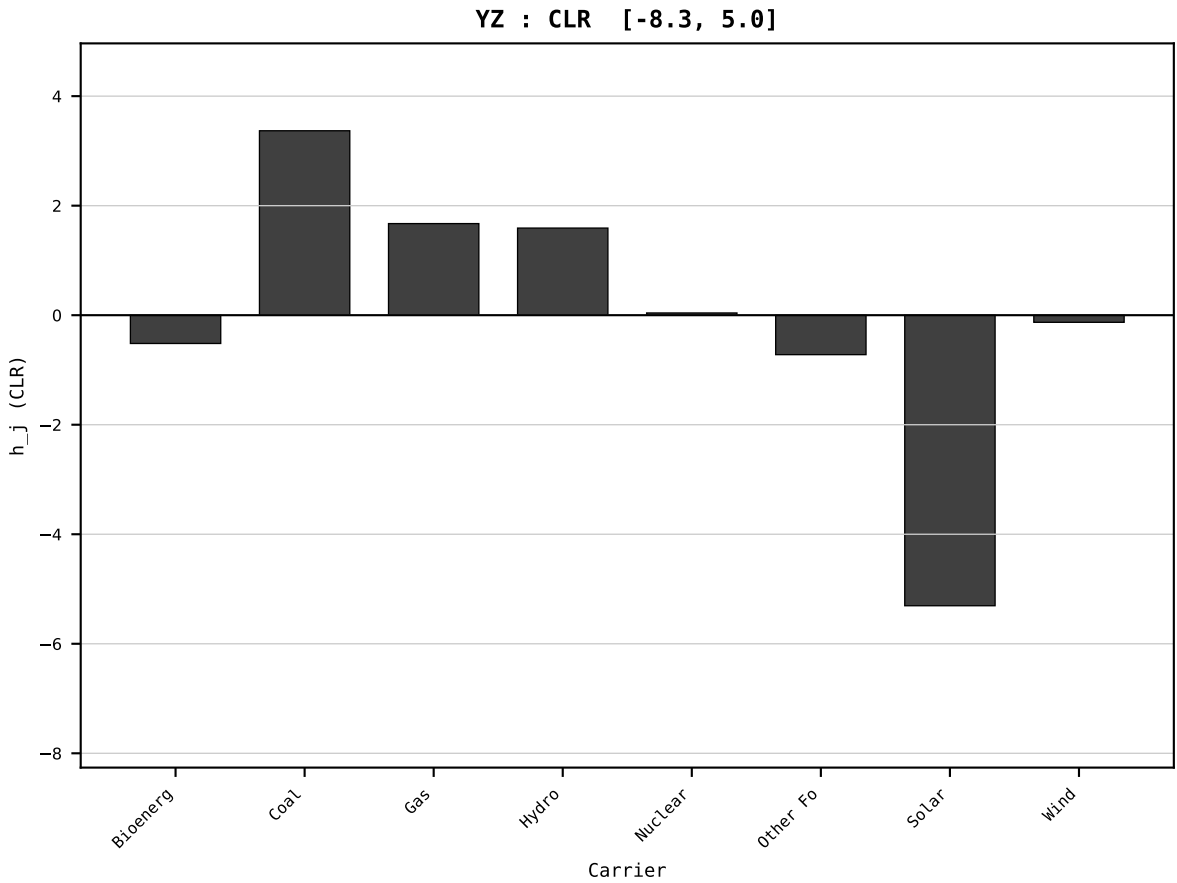
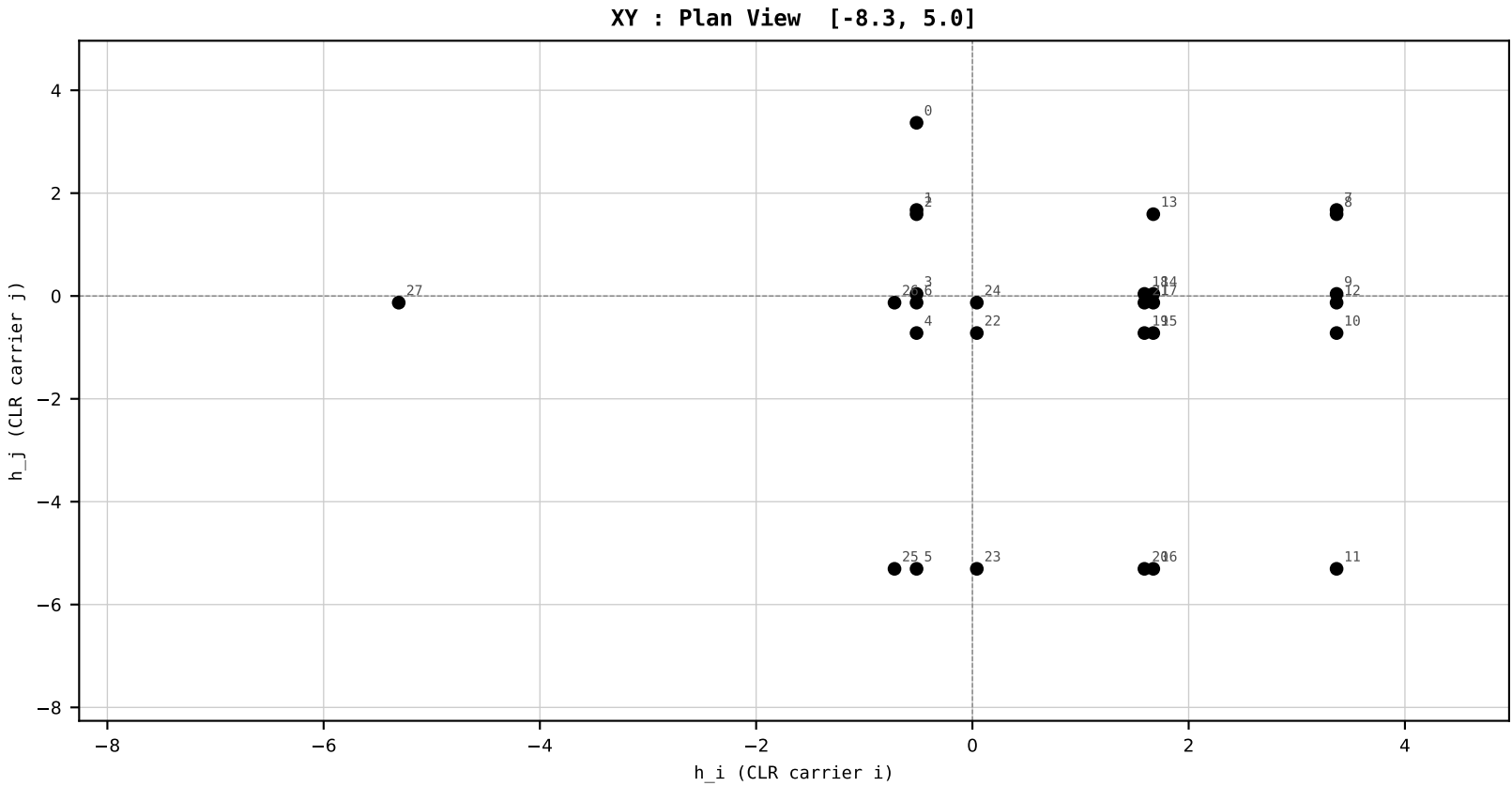
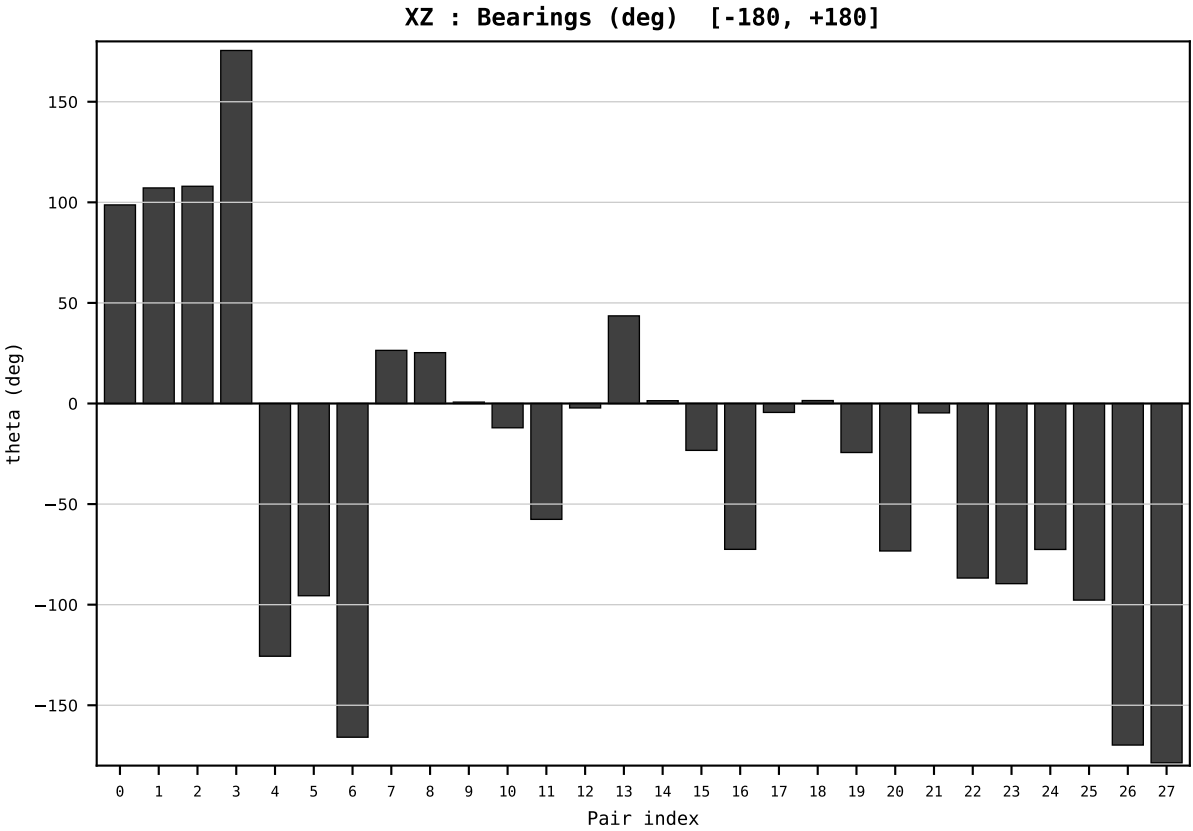
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2010
t : 10 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.493325
Ring = Hs-3
E_metric= 45.6238
kappa_HS= 5845.18
omega = 3.3901 deg
d_A = 0.451796
Helm = Other Fossil
Helm d = -0.323553
DR = 8.673 HLR
DR ratio= 5845.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=11 | Year 2011 | D=8 N=26 pairs=28

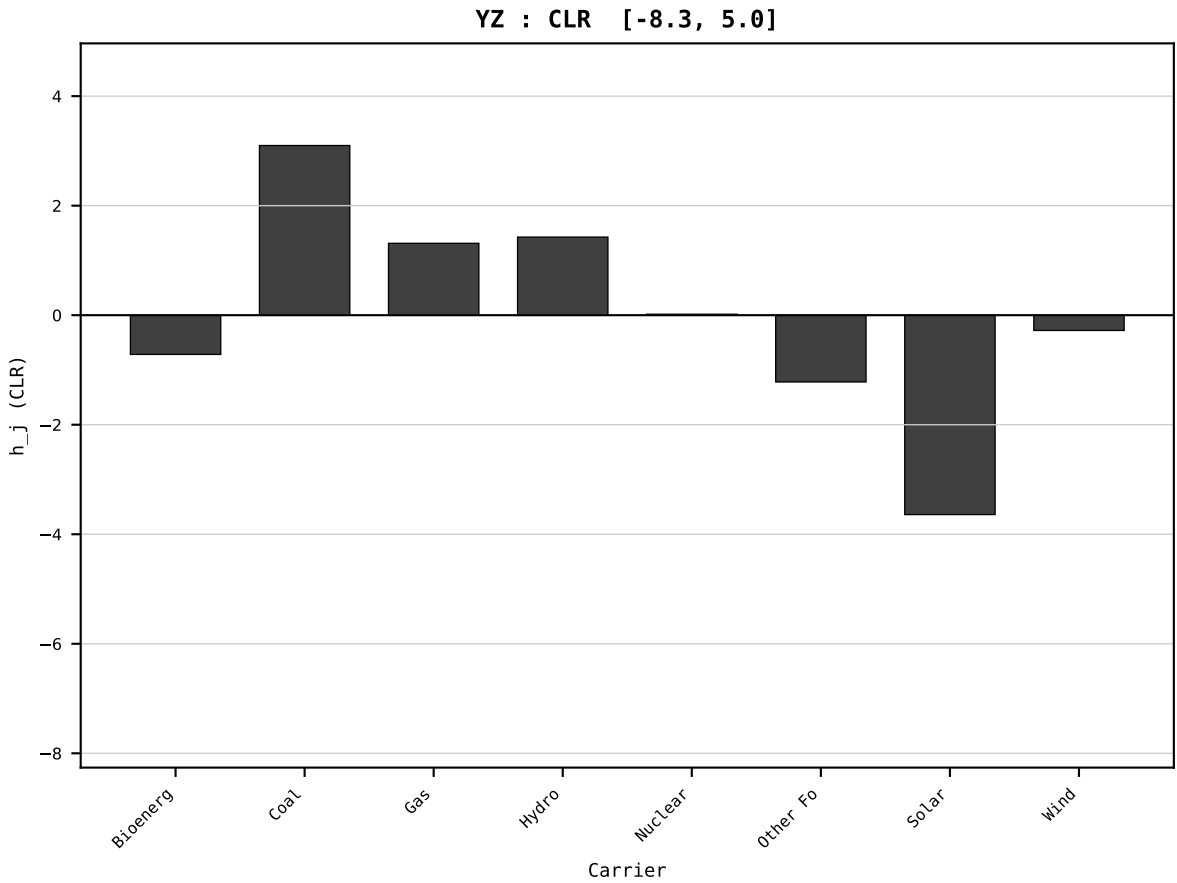
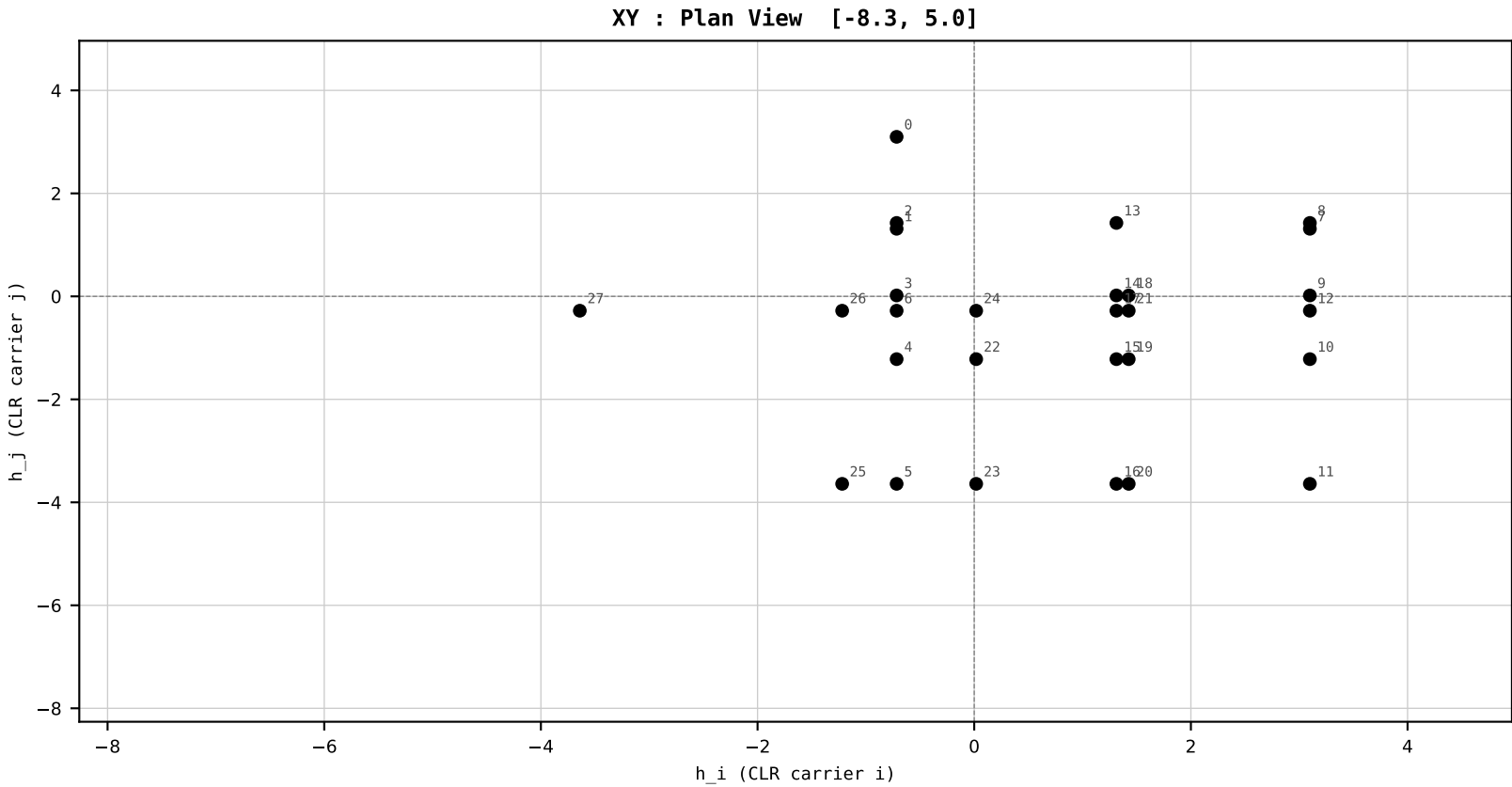
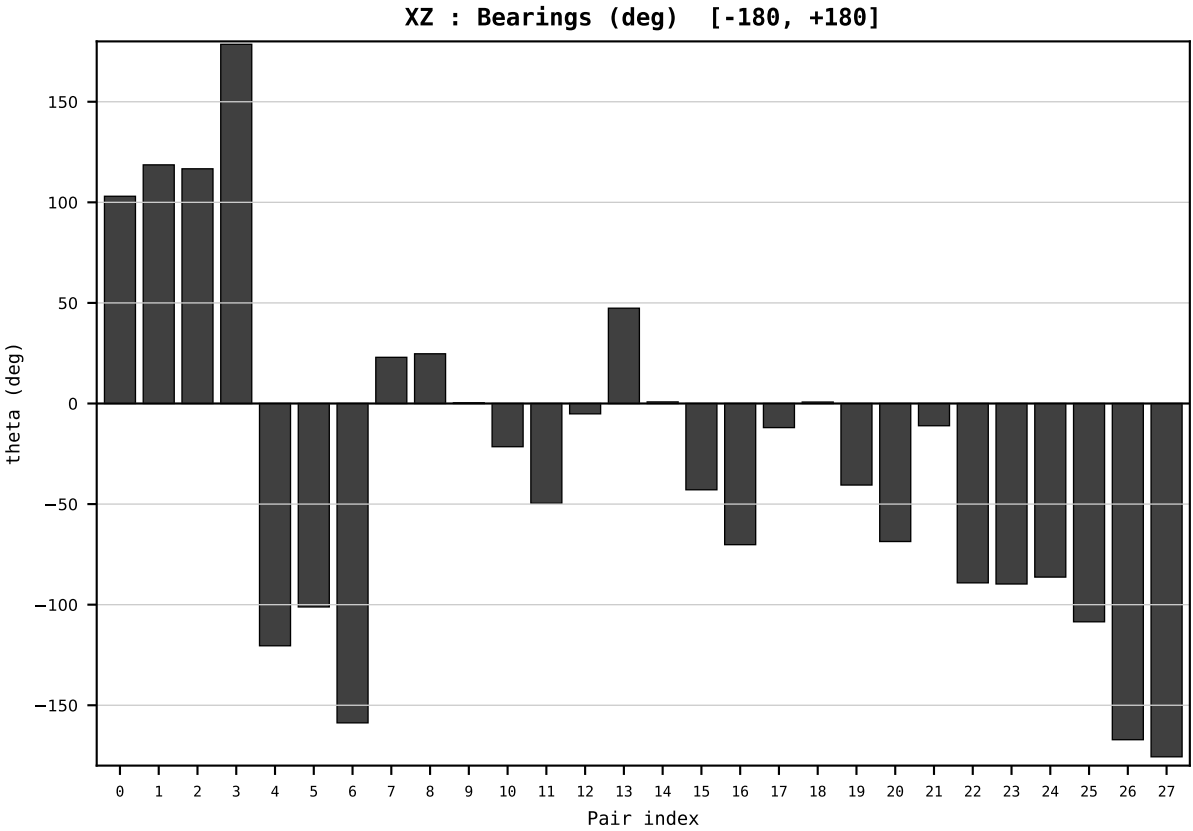
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2011
t : 11 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.480839
Ring = Hs-3
E_metric= 28.6888
kappa_HS= 844.88
omega = 11.1115 deg
d_A = 1.819832
Helm = Solar
Helm d = +1.664605
DR = 6.739 HLR
DR ratio= 844.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=12 | Year 2012 | D=8 N=26 pairs=28

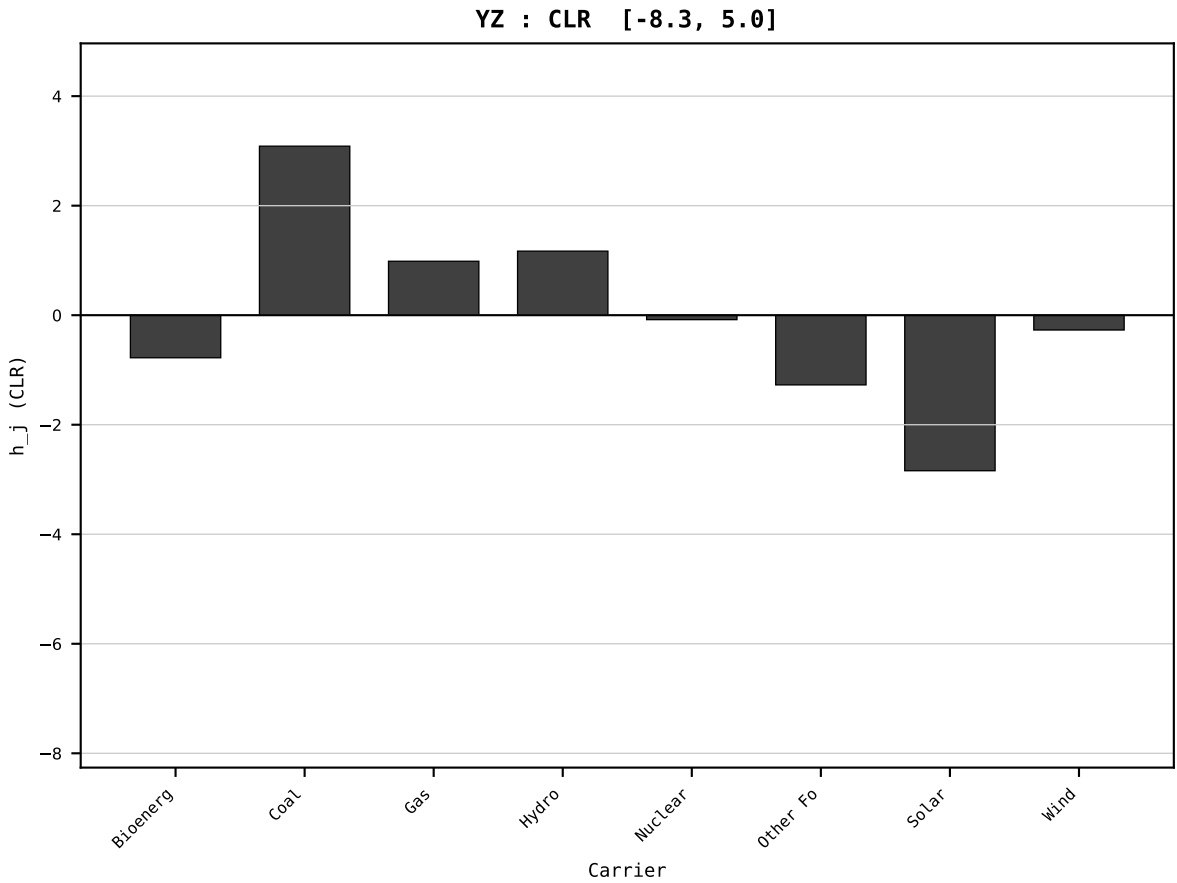
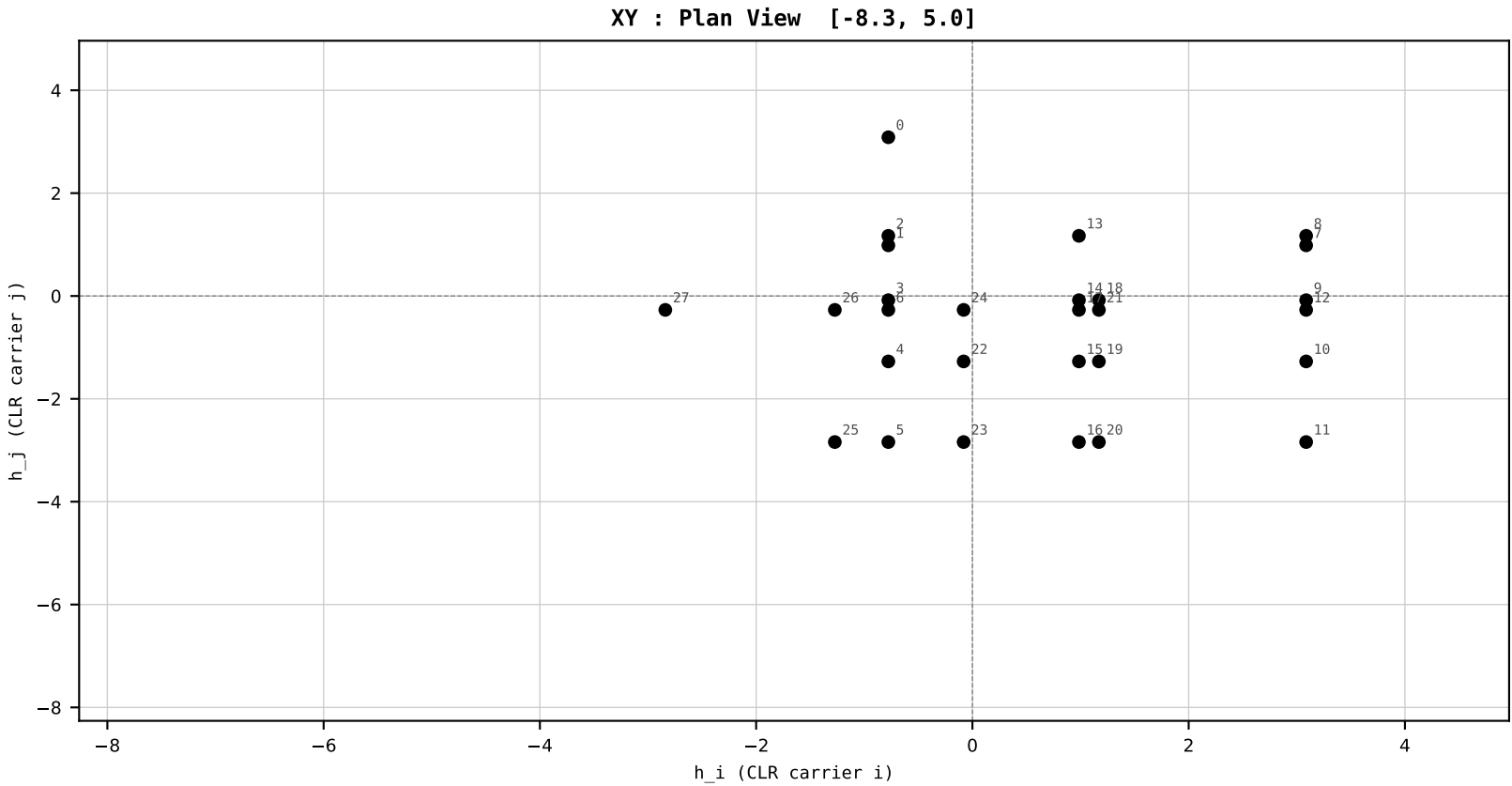
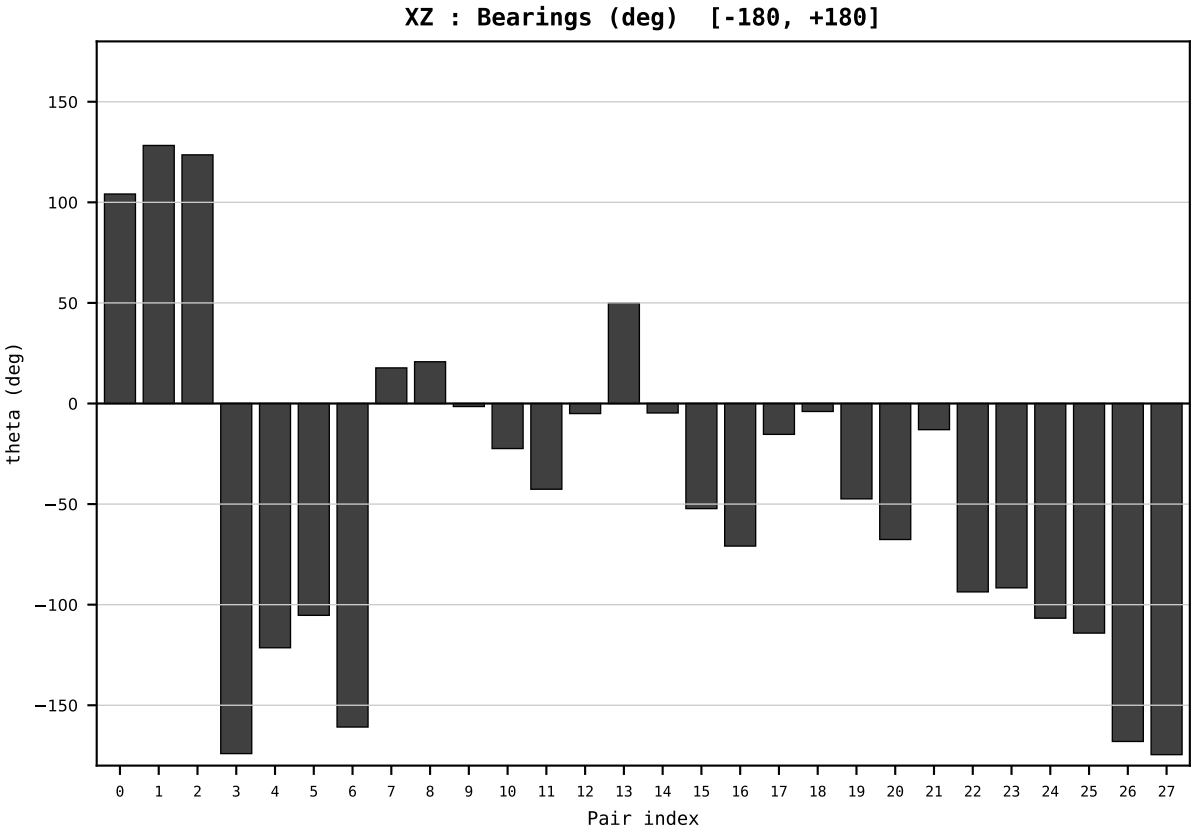
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2012
t : 12 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.516572
Ring = Hs-4
E_metric= 22.2333
kappa_HS= 374.82
omega = 7.3940 deg
d_A = 0.911514
Helm = Solar
Helm d = +0.800856
DR = 5.926 HLR
DR ratio= 374.8x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=13 | Year 2013 | D=8 N=26 pairs=28

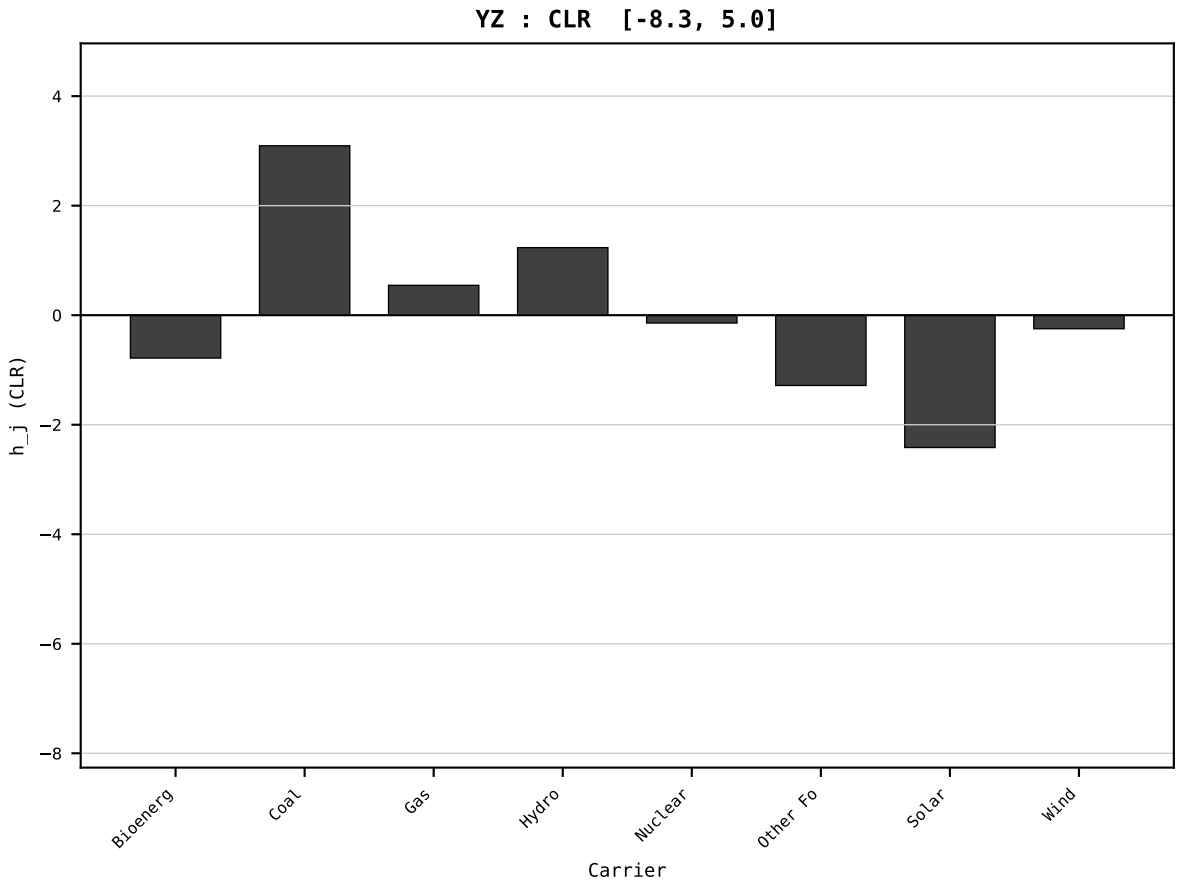
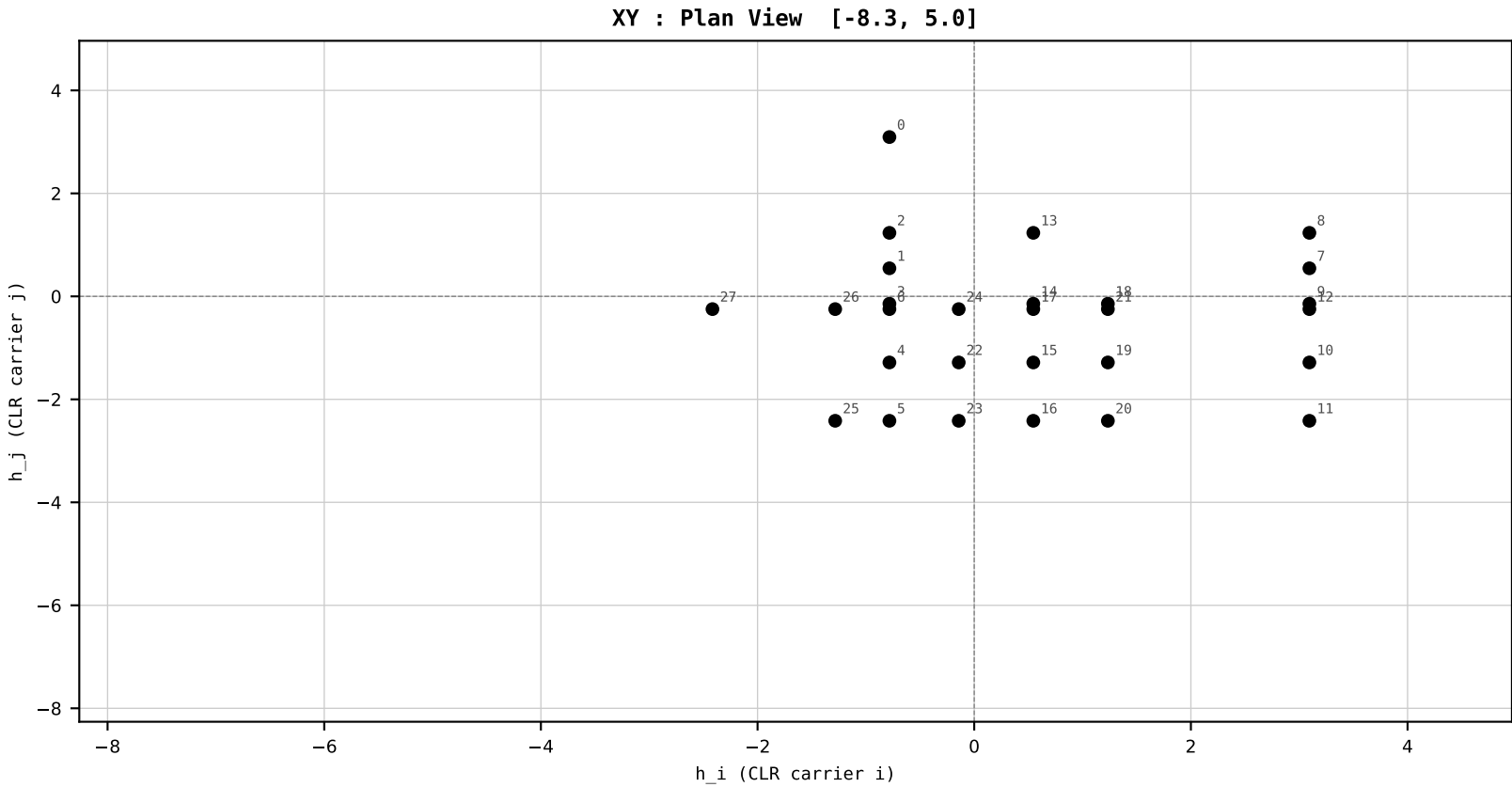
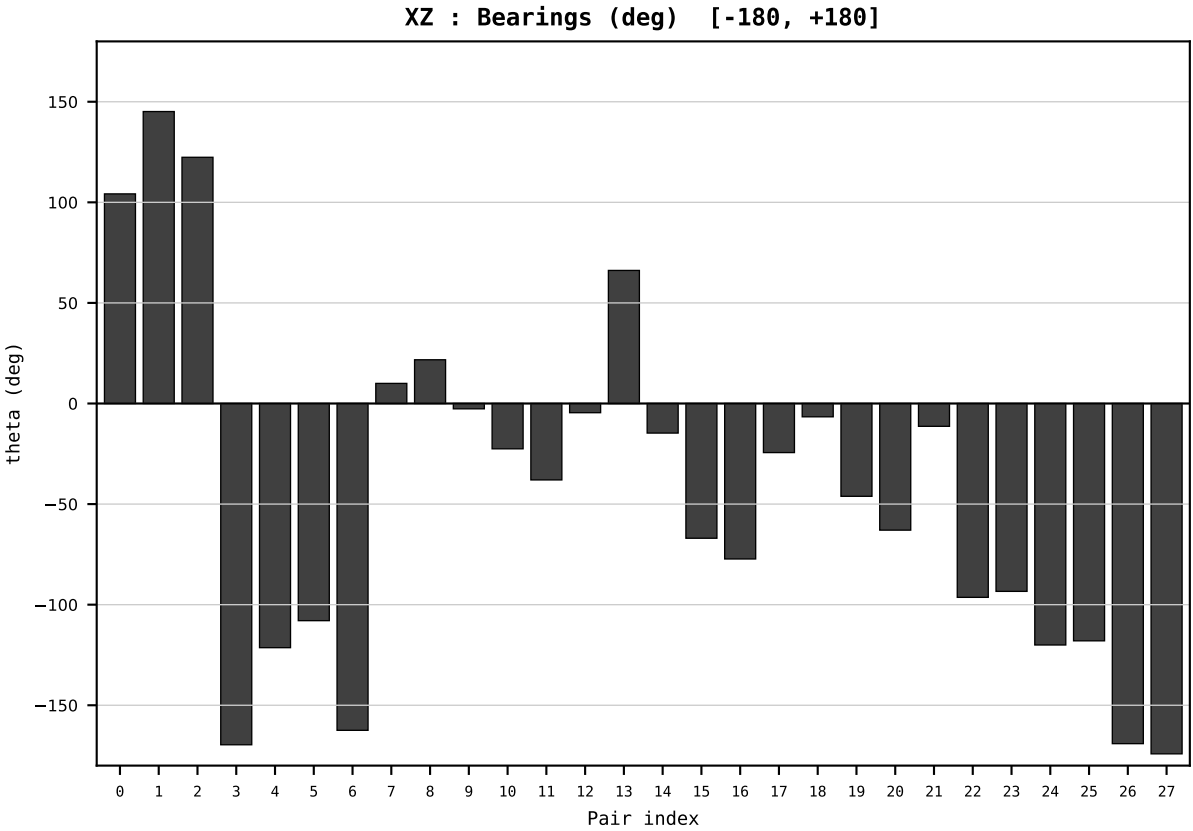
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.538501
Ring = Hs-4
E_metric= 19.5721
kappa_HS= 247.22
omega = 6.8358 deg
d_A = 0.617545
Helm = Gas
Helm d = -0.439615
DR = 5.510 HLR
DR ratio= 247.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=14 | Year 2014 | D=8 N=26 pairs=28

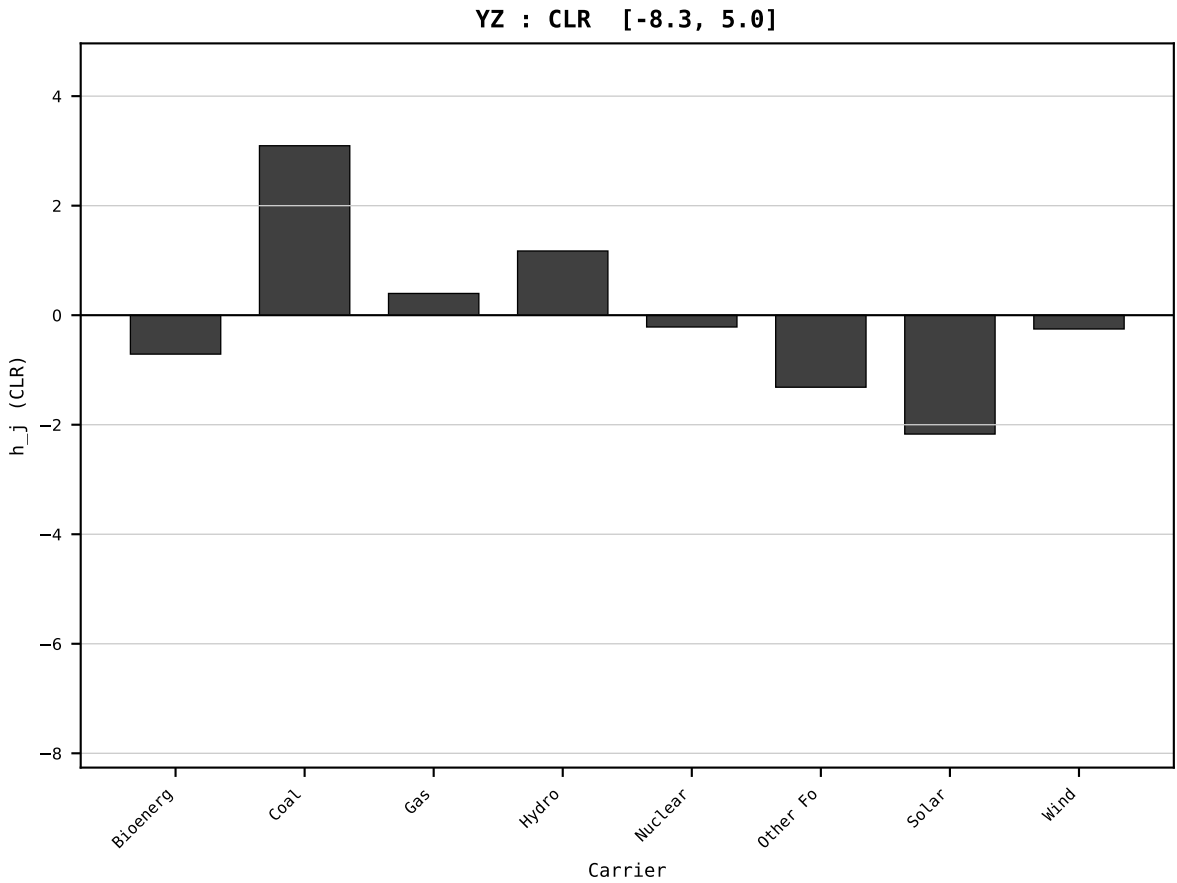
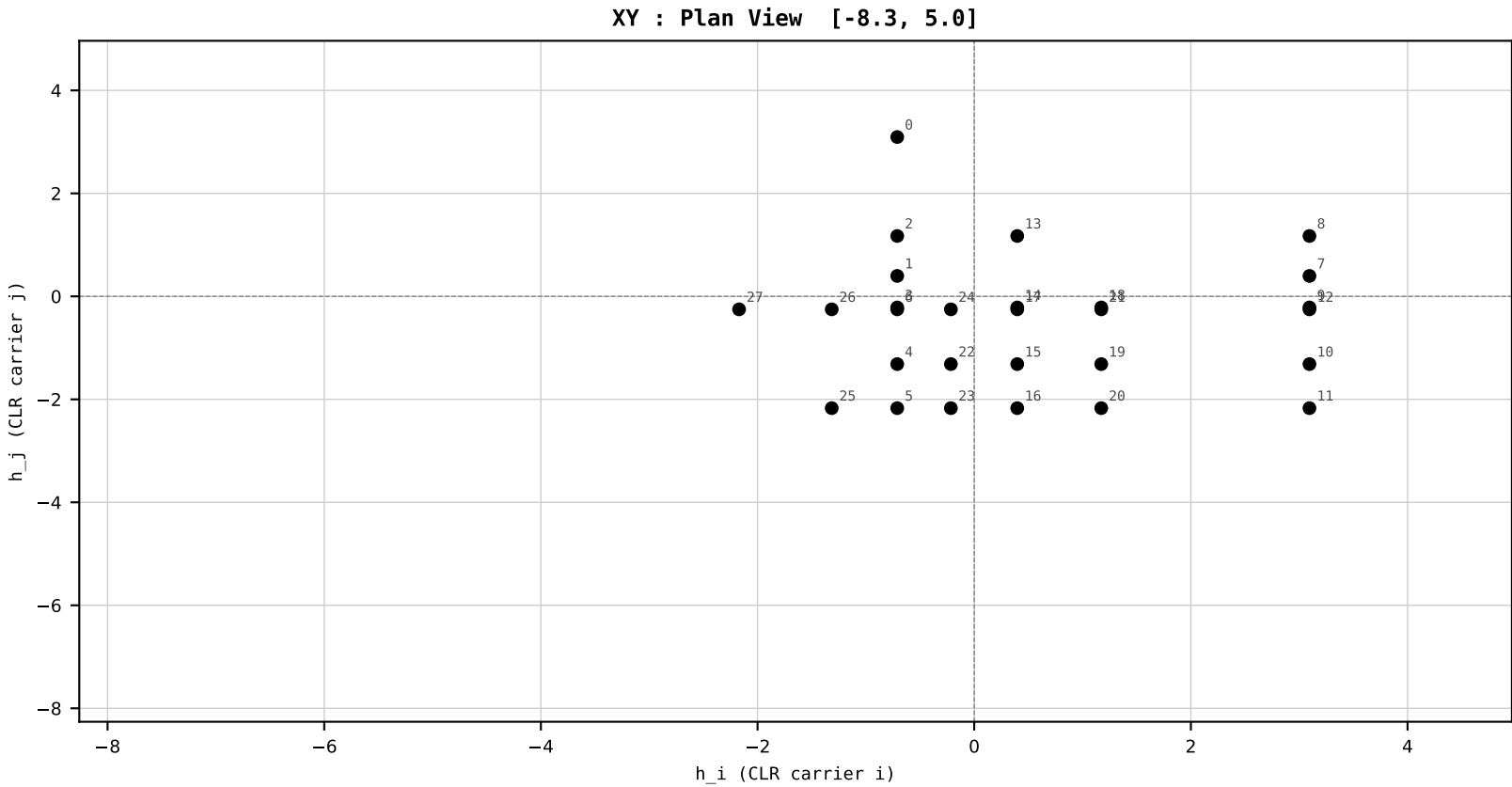
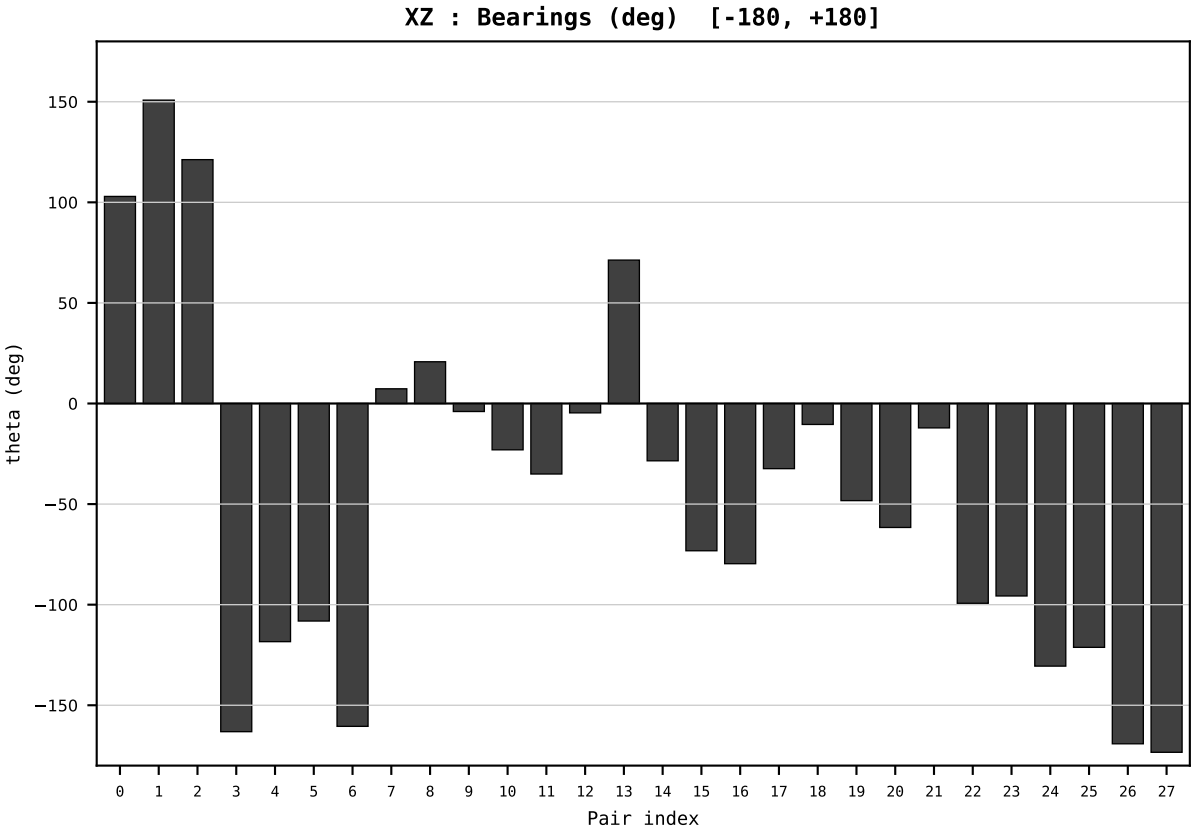
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2014
t : 14 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.549920
Ring = Hs-4
E_metric= 18.1649
kappa_HS= 193.47
omega = 3.5245 deg
d_A = 0.312363
Helm = Solar
Helm d = +0.245715
DR = 5.265 HLR
DR ratio= 193.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=15 | Year 2015 | D=8 N=26 pairs=28

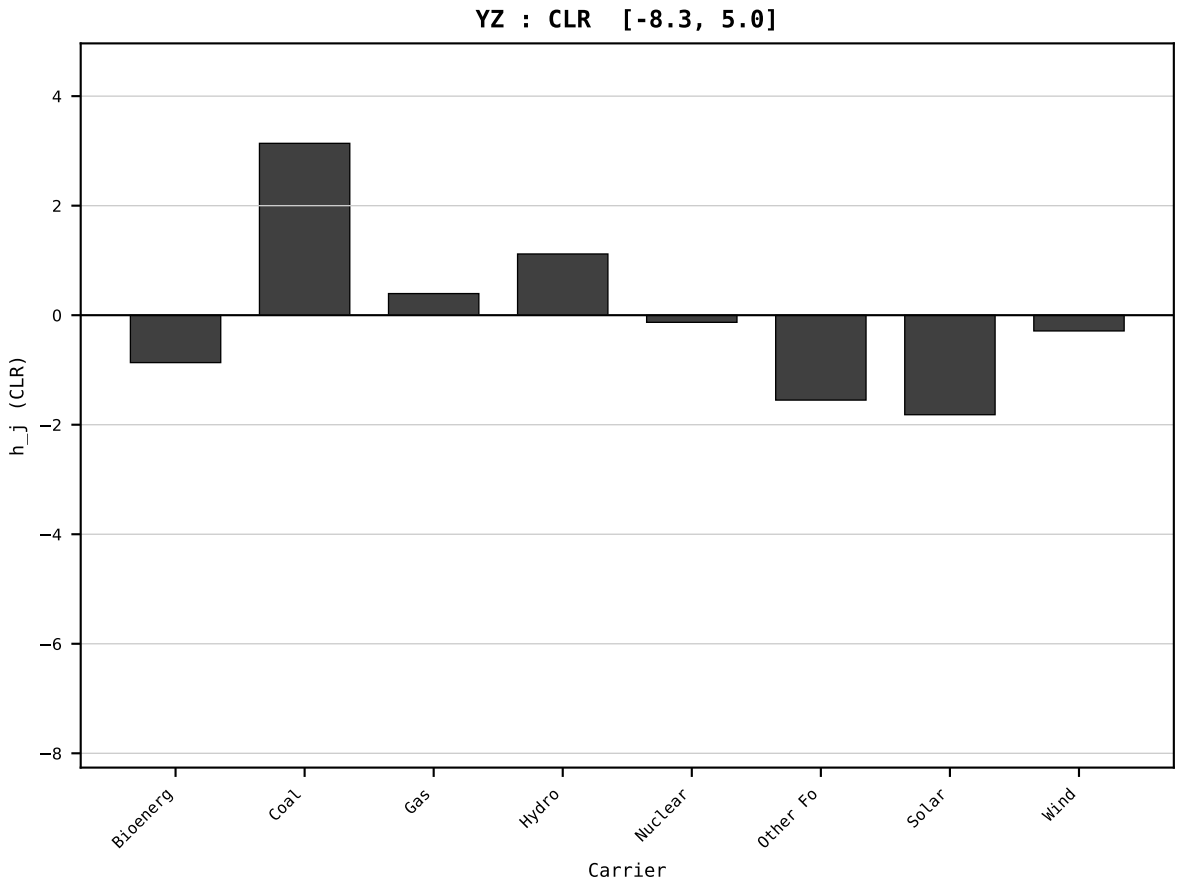
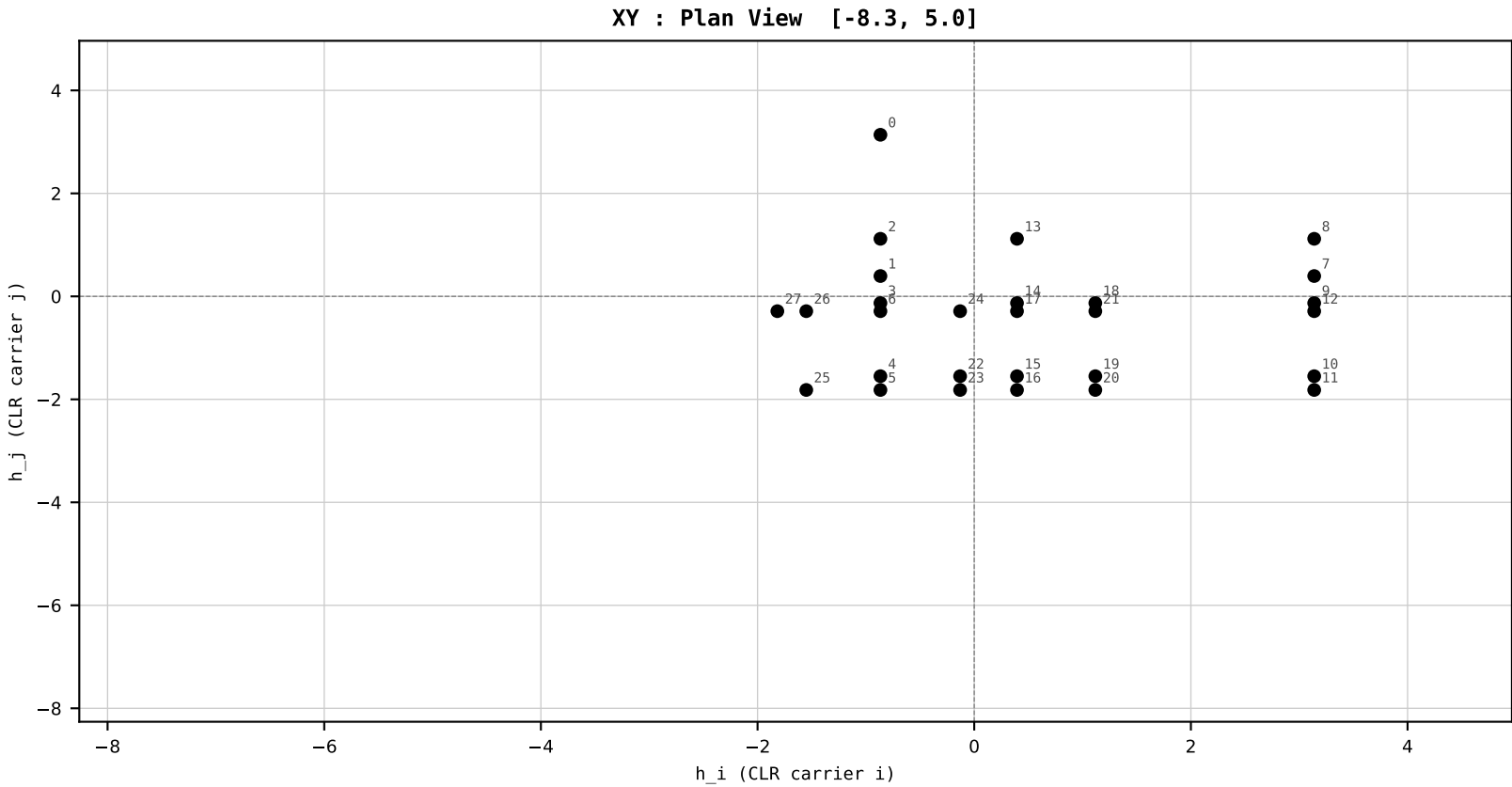
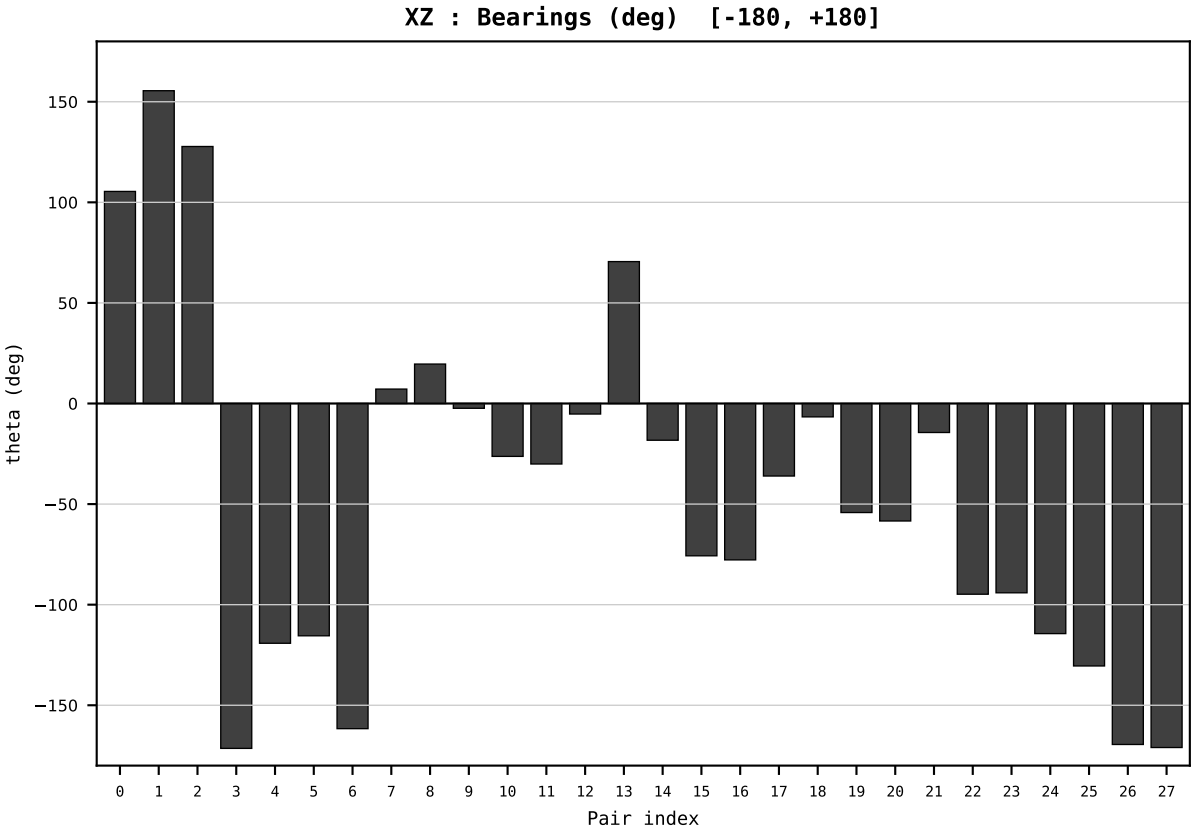
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2015
t : 15 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.566019
Ring = Hs-4
E_metric= 17.8098
kappa_HS= 141.97
omega = 6.2793 deg
d_A = 0.466443
Helm = Solar
Helm d = +0.353359
DR = 4.956 HLR
DR ratio= 142.0x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=16 | Year 2016 | D=8 N=26 pairs=28

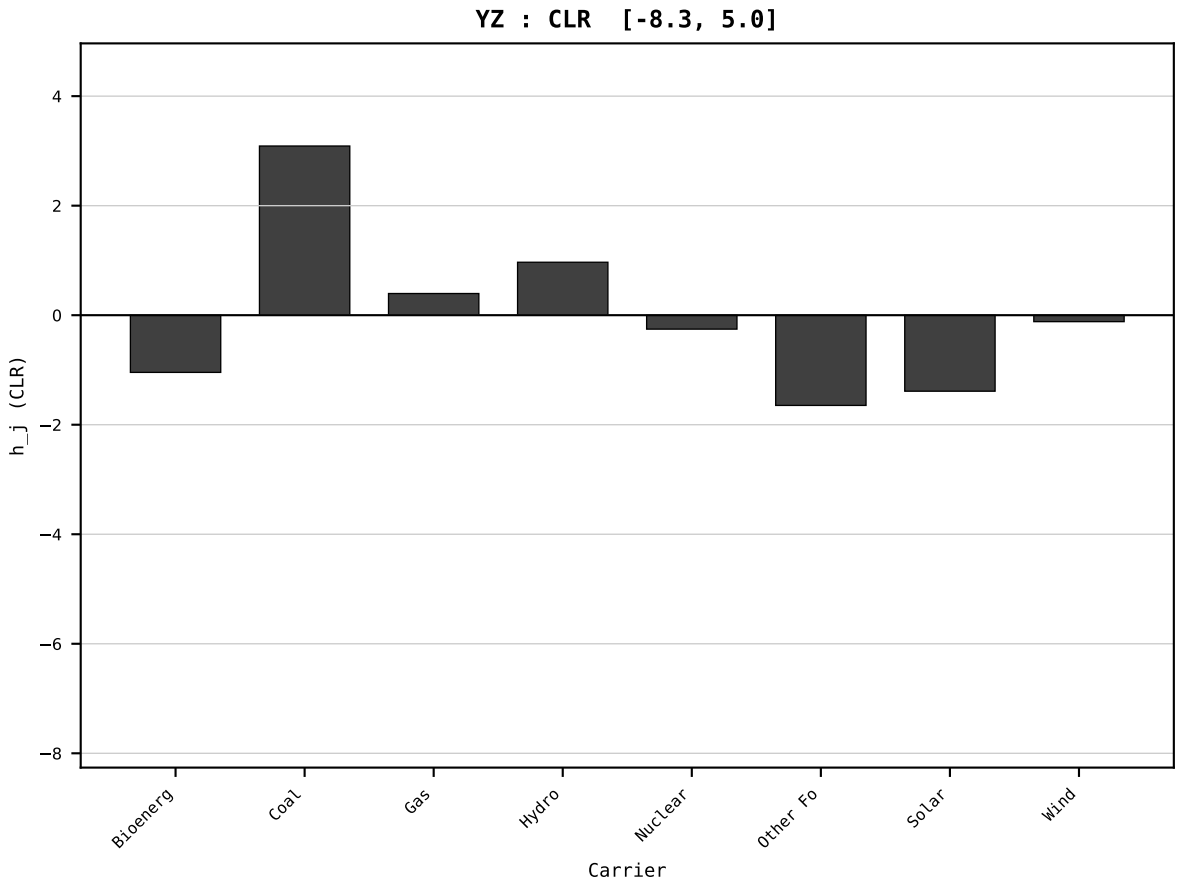
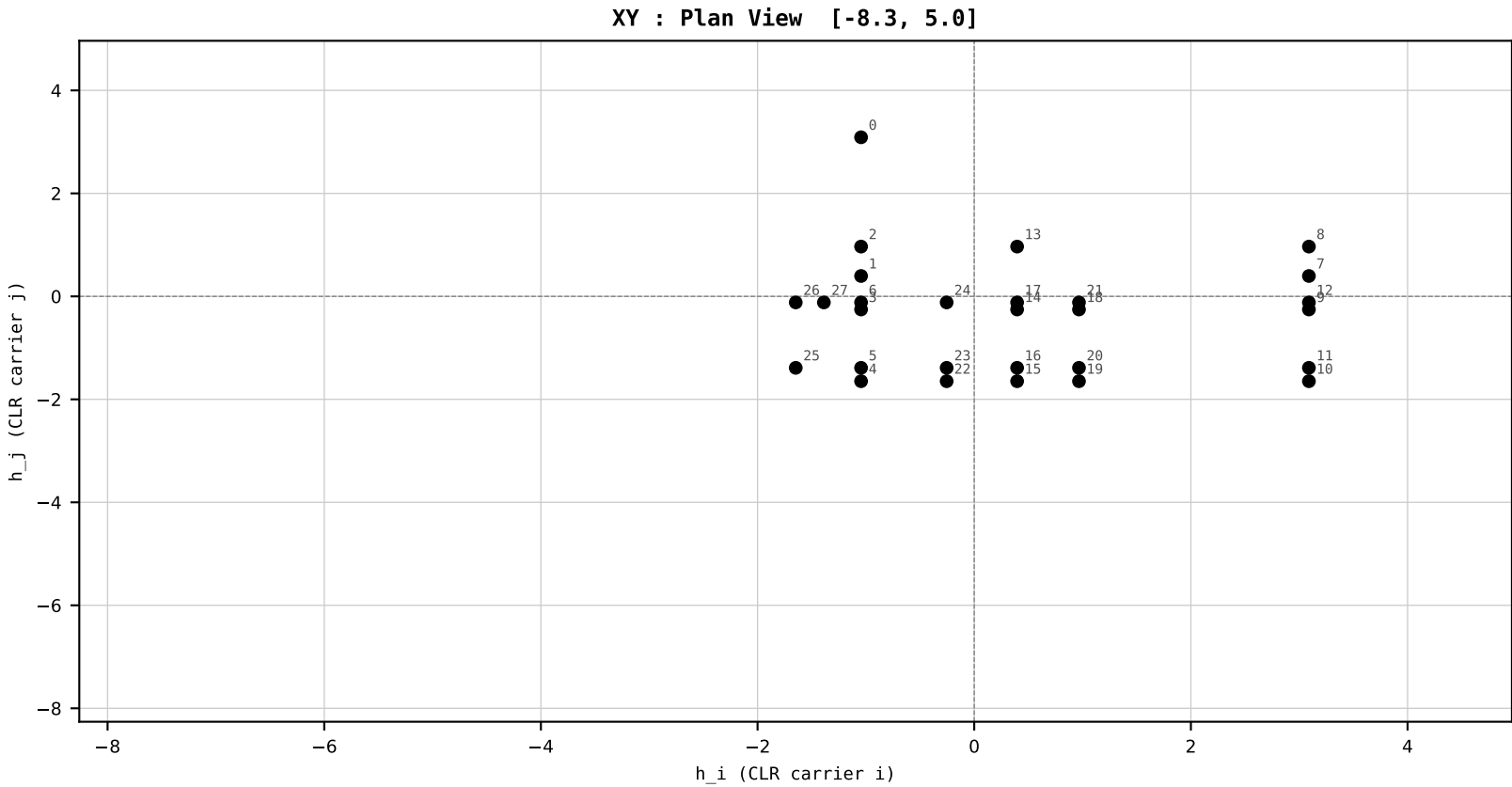
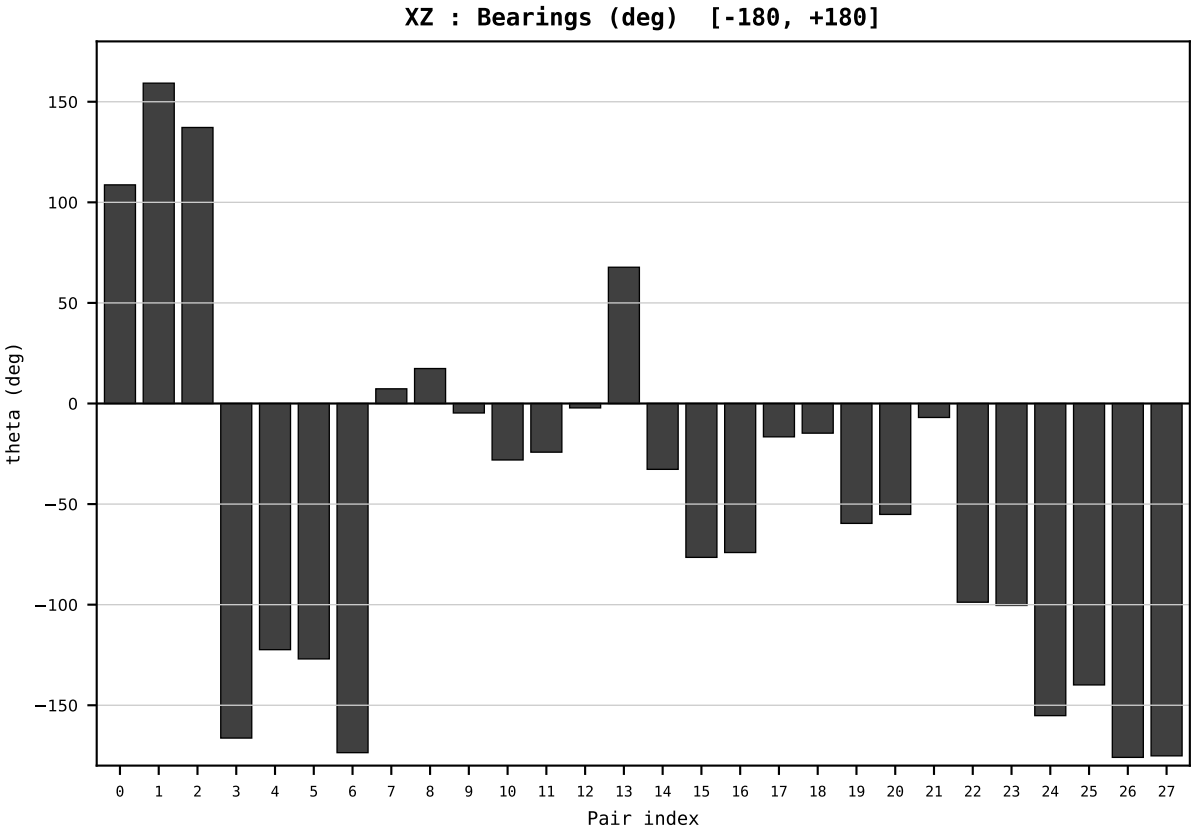
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2016
t : 16 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.562151
Ring = Hs-4
E_metric= 16.4465
kappa_HS= 114.09
omega = 7.1743 deg
d_A = 0.543254
Helm = Solar
Helm d = +0.429368
DR = 4.737 HLR
DR ratio= 114.1x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=17 | Year 2017 | D=8 N=26 pairs=28

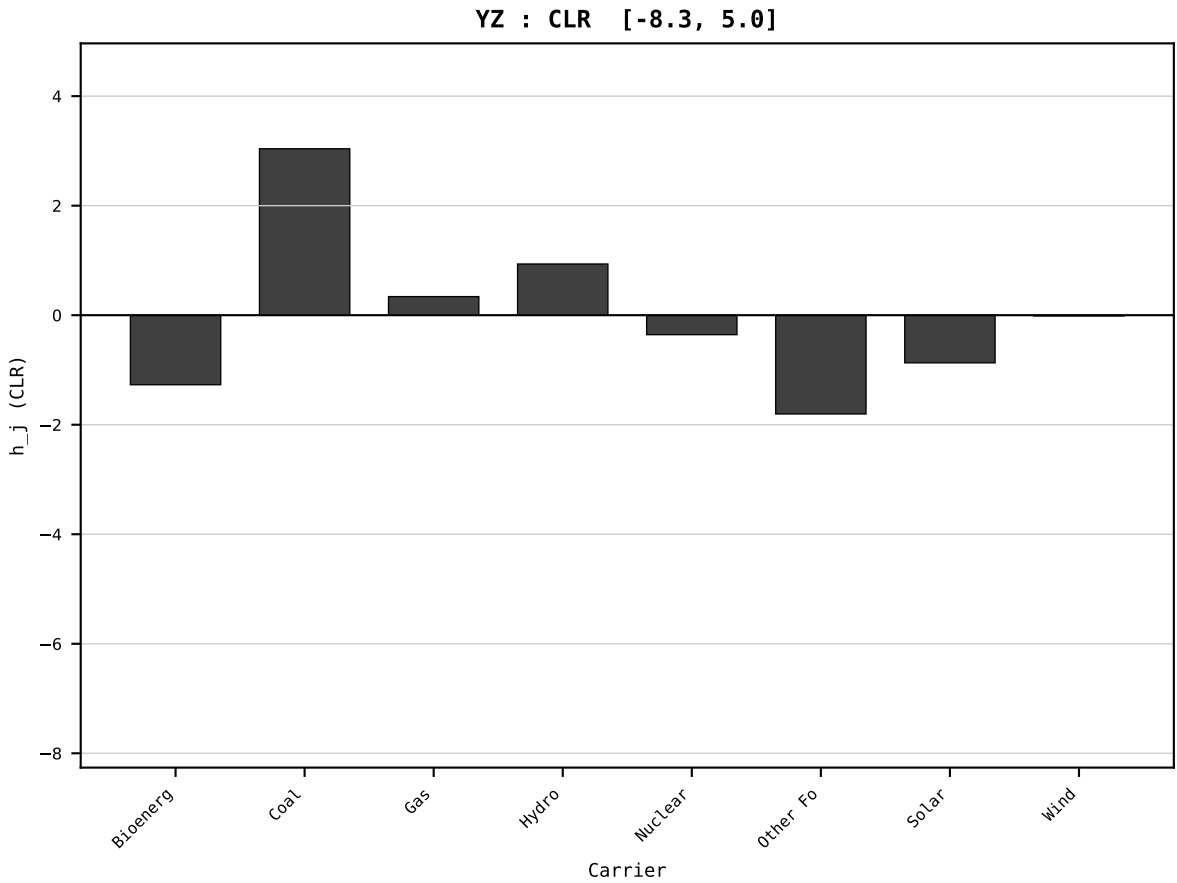
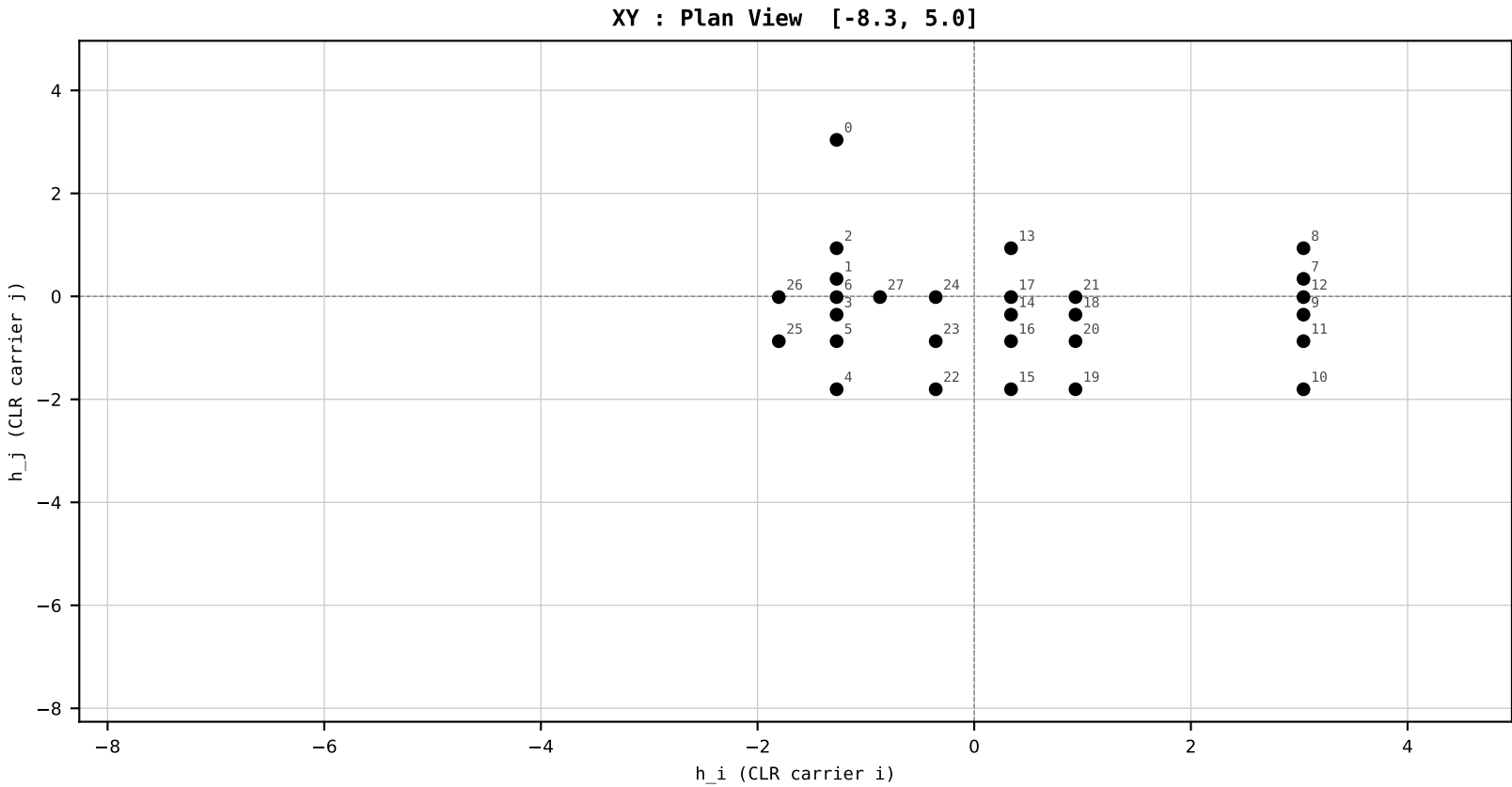
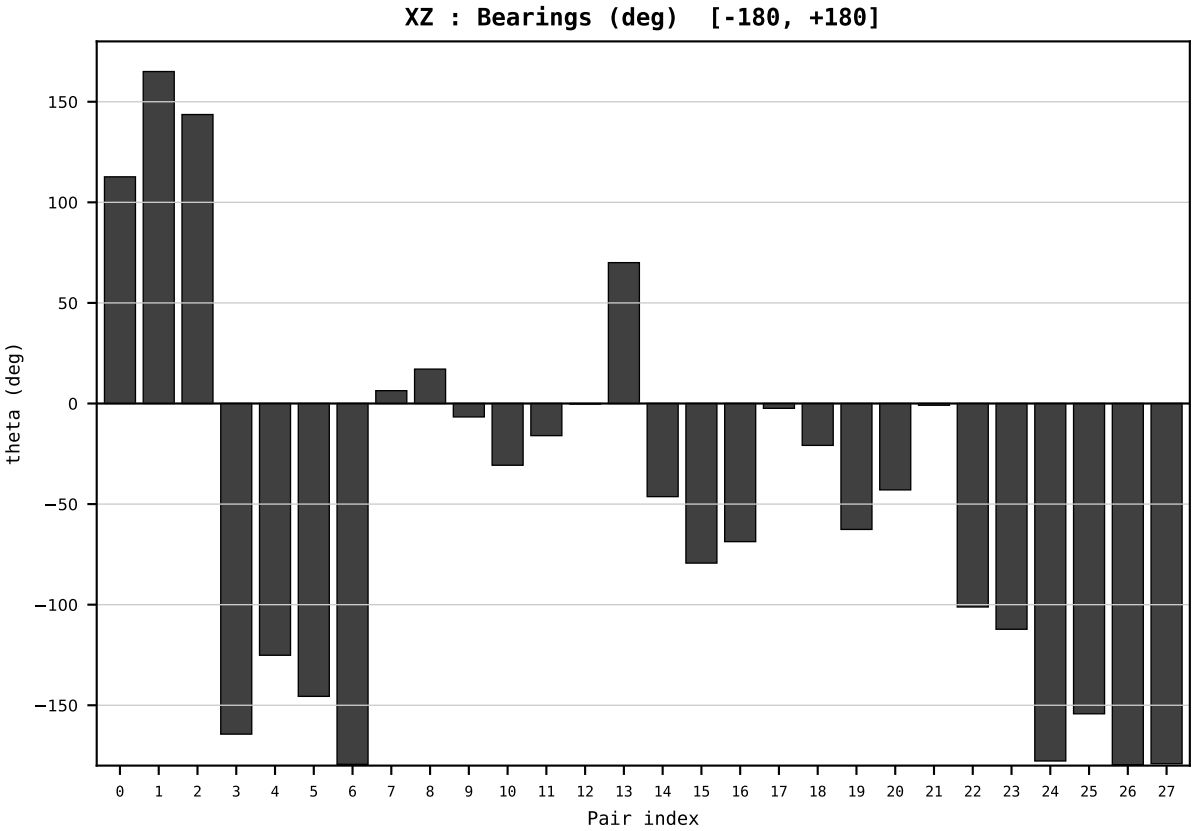
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2017
t : 17 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.550408
Ring = Hs-4
E_metric= 15.9784
kappa_HS= 126.88
omega = 8.6362 deg
d_A = 0.609086
Helm = Solar
Helm d = +0.517691
DR = 4.843 HLR
DR ratio= 126.9x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=18 | Year 2018 | D=8 N=26 pairs=28

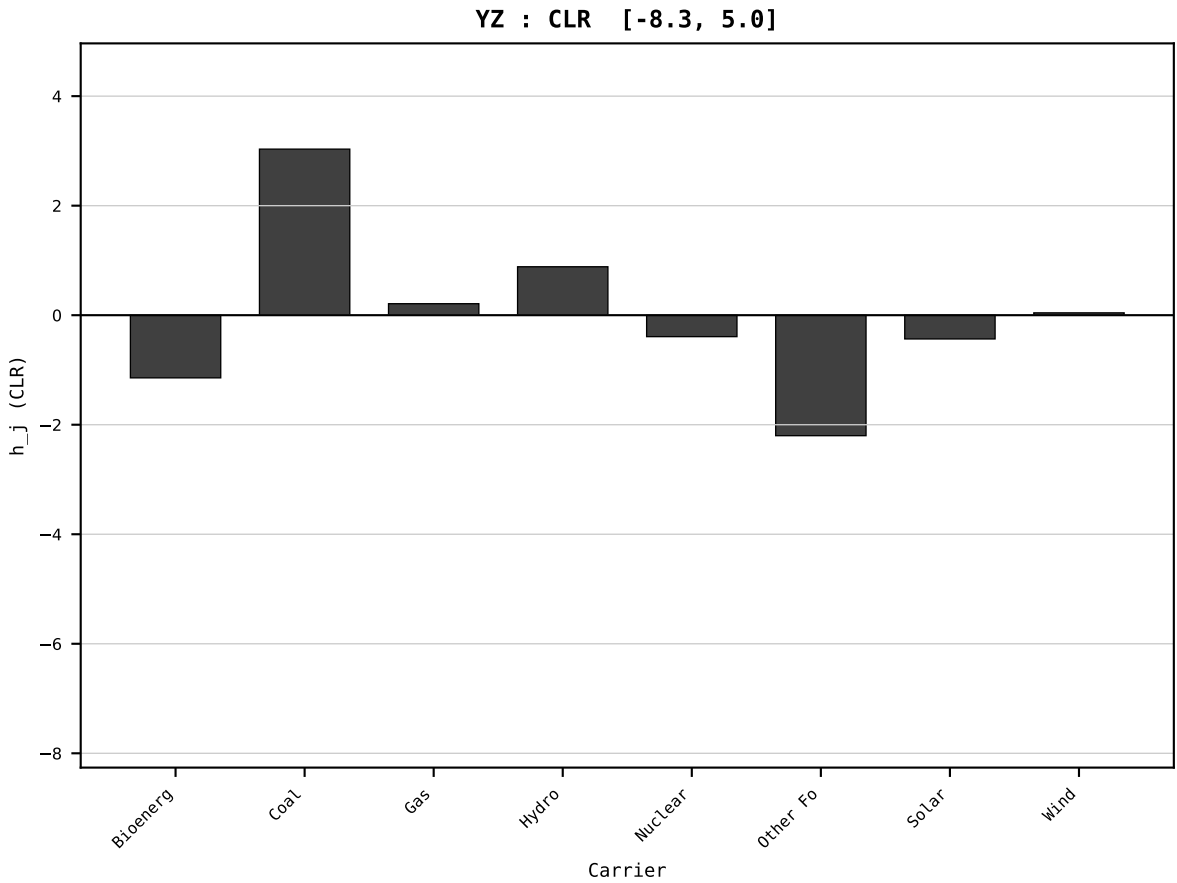
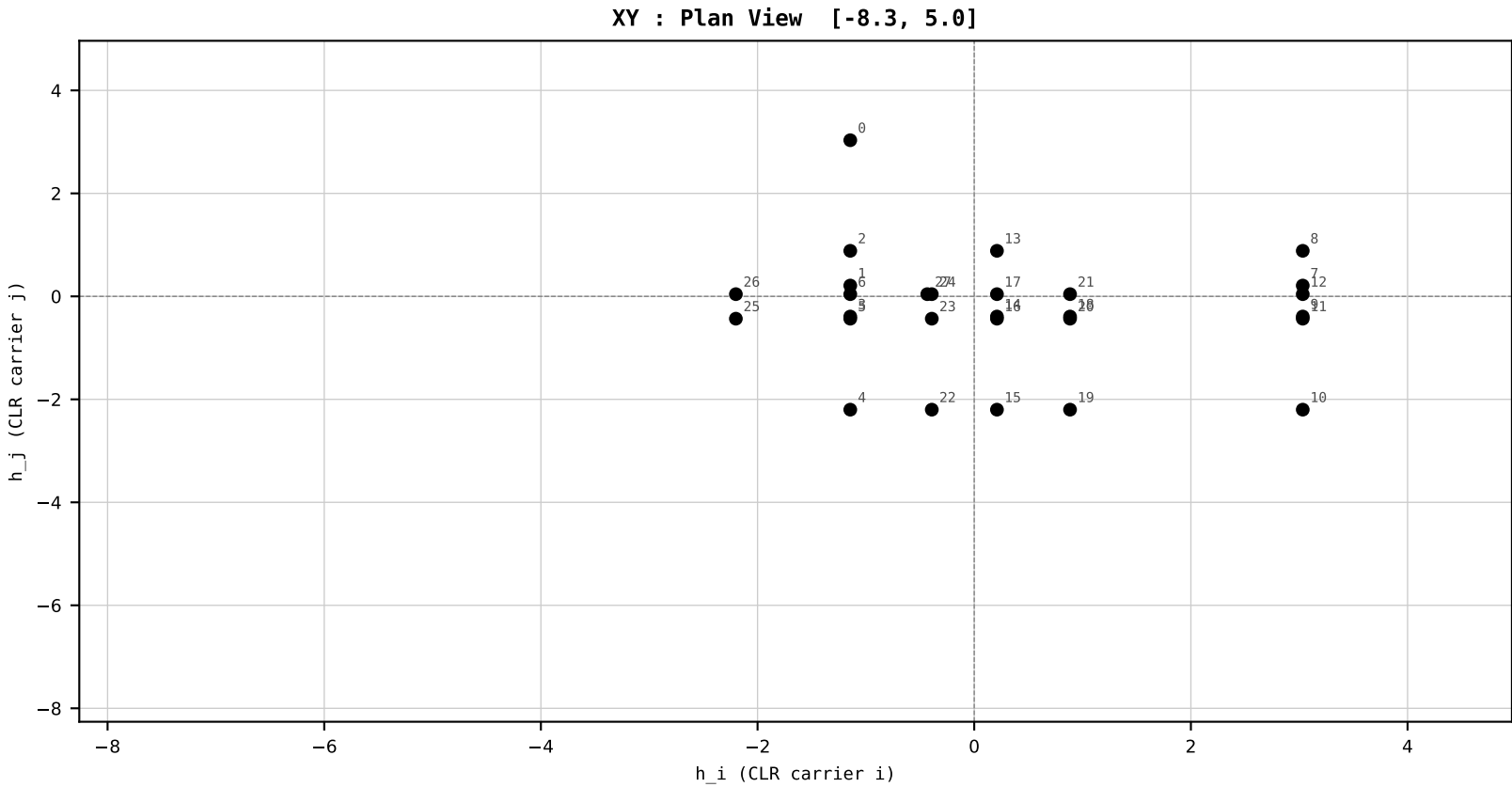
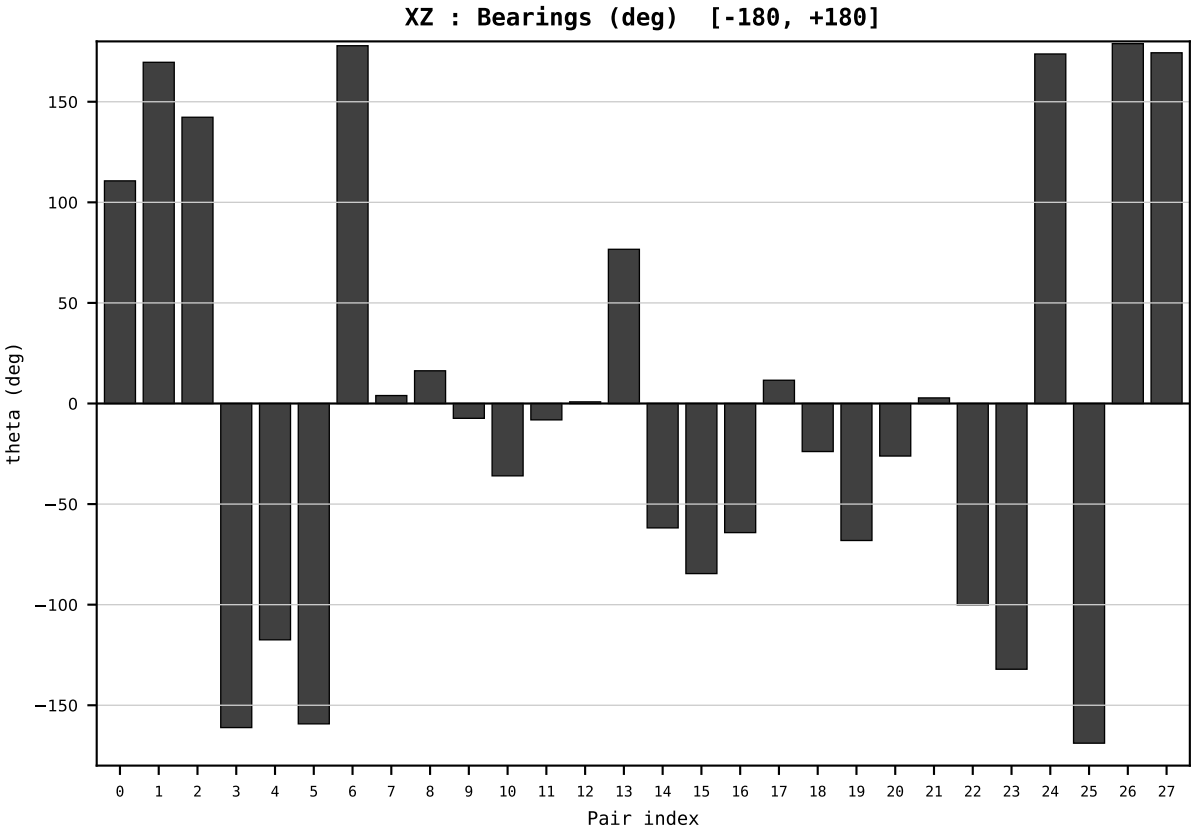
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2018
t : 18 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.546713
Ring = Hs-4
E_metric= 16.5164
kappa_HS= 187.32
omega = 8.8095 deg
d_A = 0.622688
Helm = Solar
Helm d = +0.436690
DR = 5.233 HLR
DR ratio= 187.3x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=19 | Year 2019 | D=8 N=26 pairs=28

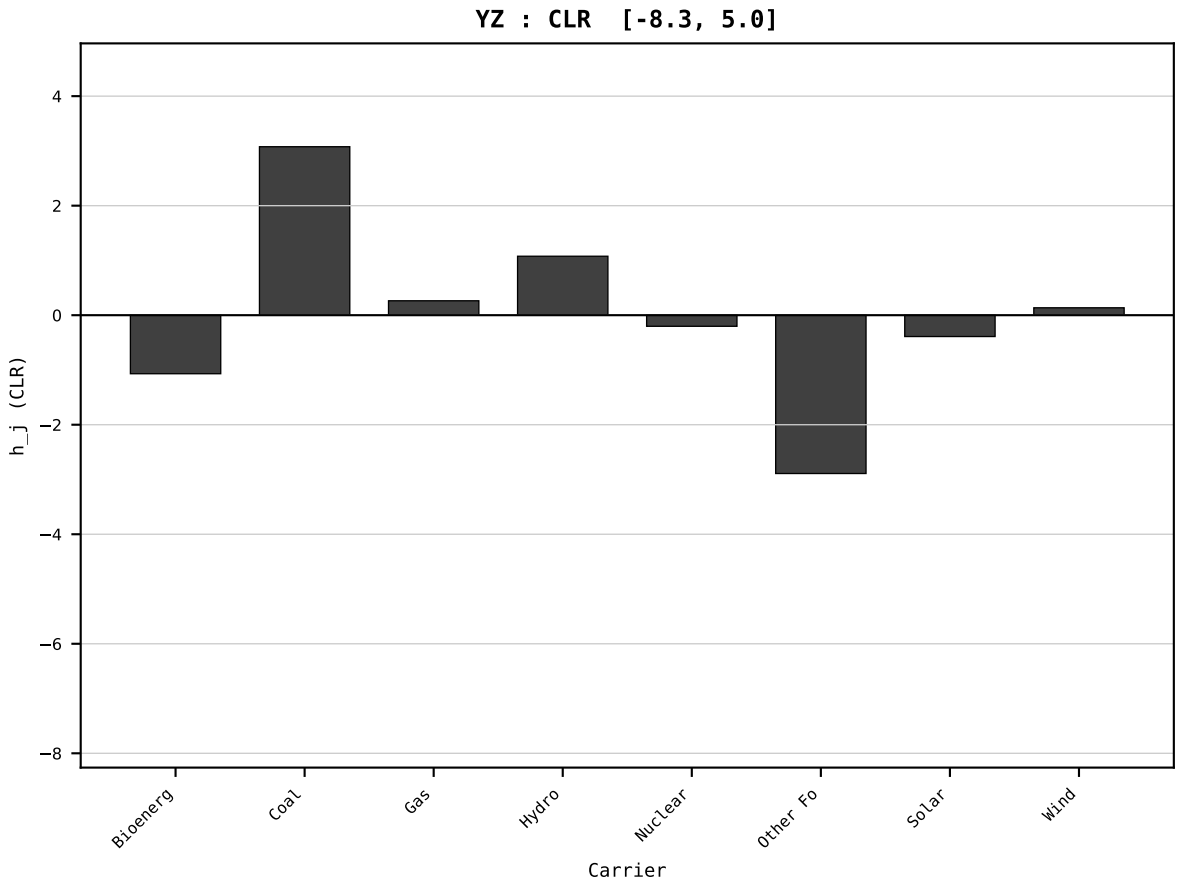
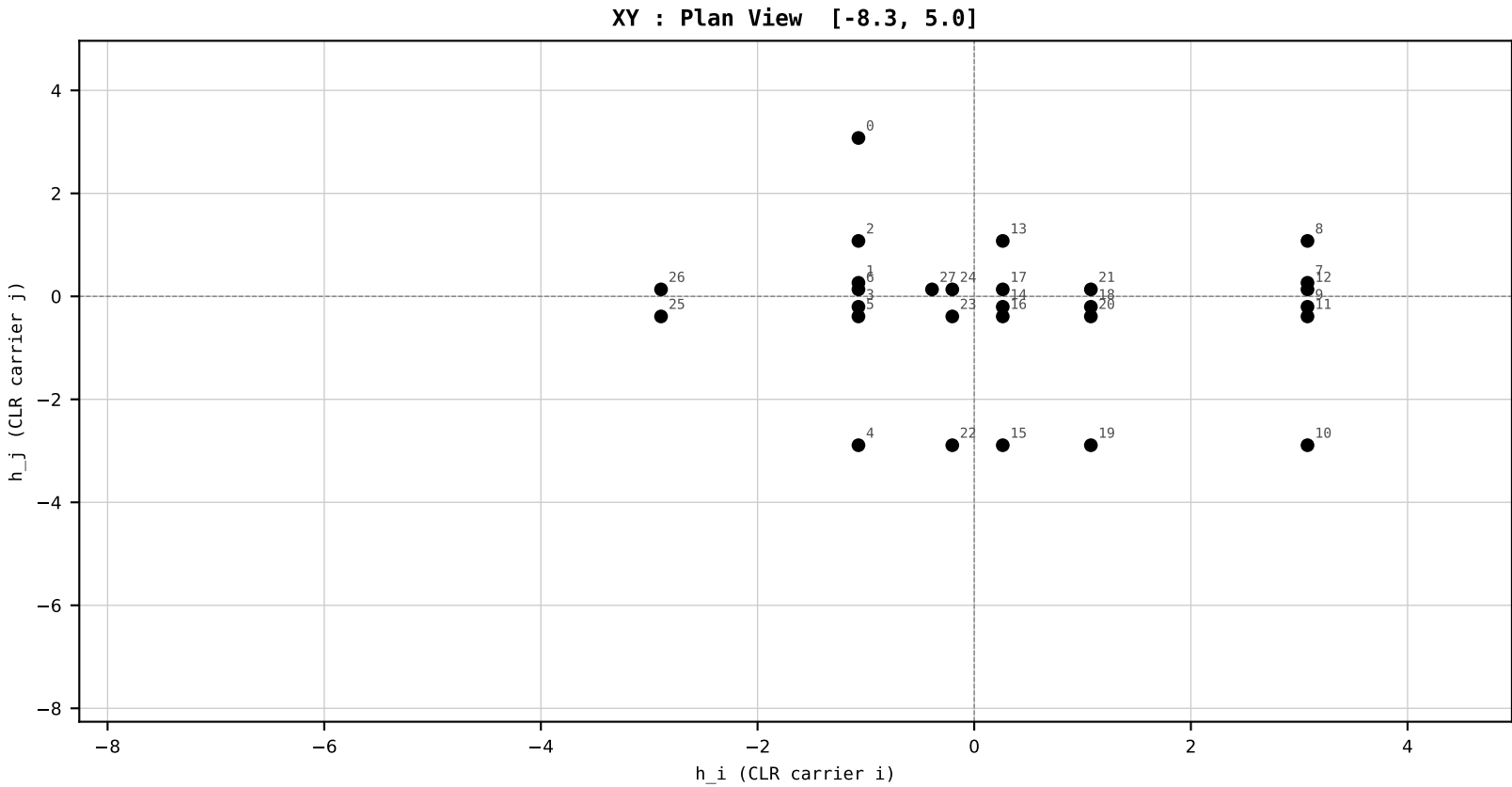
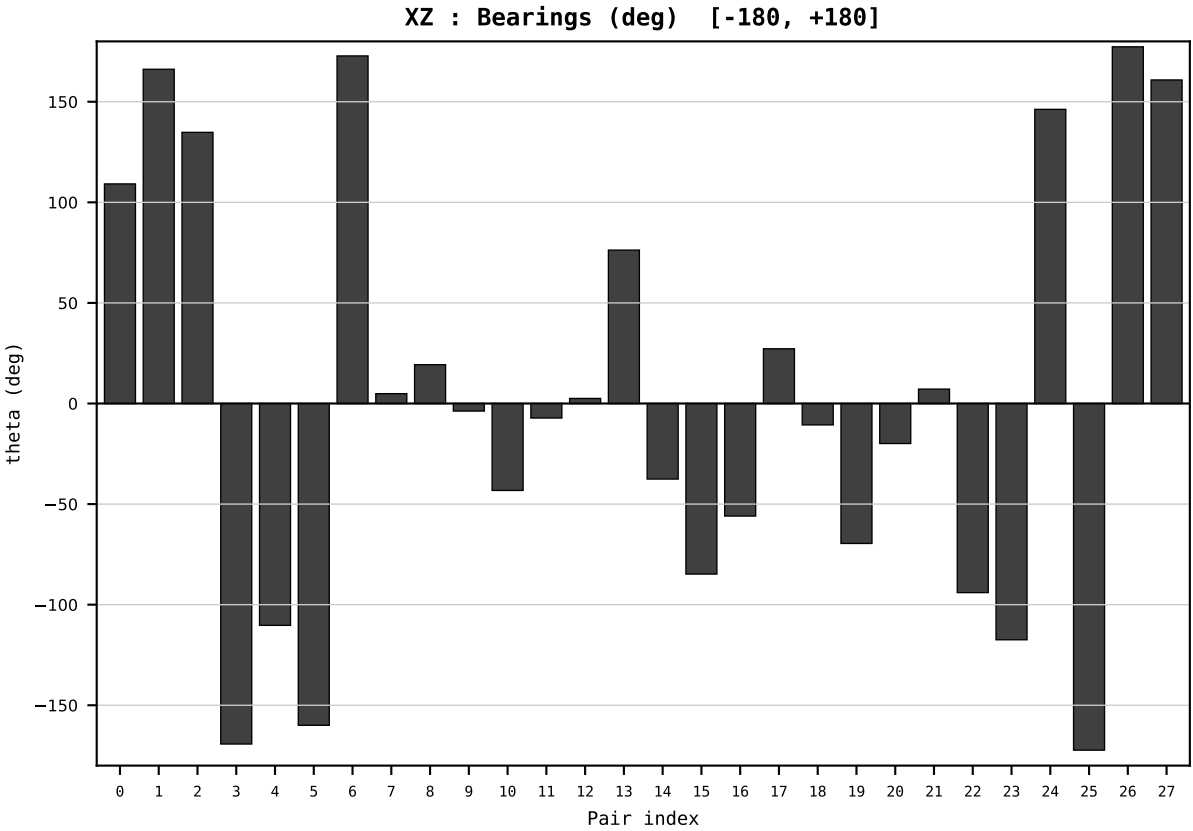
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2019
t : 19 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.534043
Ring = Hs-4
E_metric= 20.4039
kappa_HS= 390.48
omega = 8.0922 deg
d_A = 0.755525
Helm = Other Fossil
Helm d = -0.690711
DR = 5.967 HLR
DR ratio= 390.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=20 | Year 2020 | D=8 N=26 pairs=28

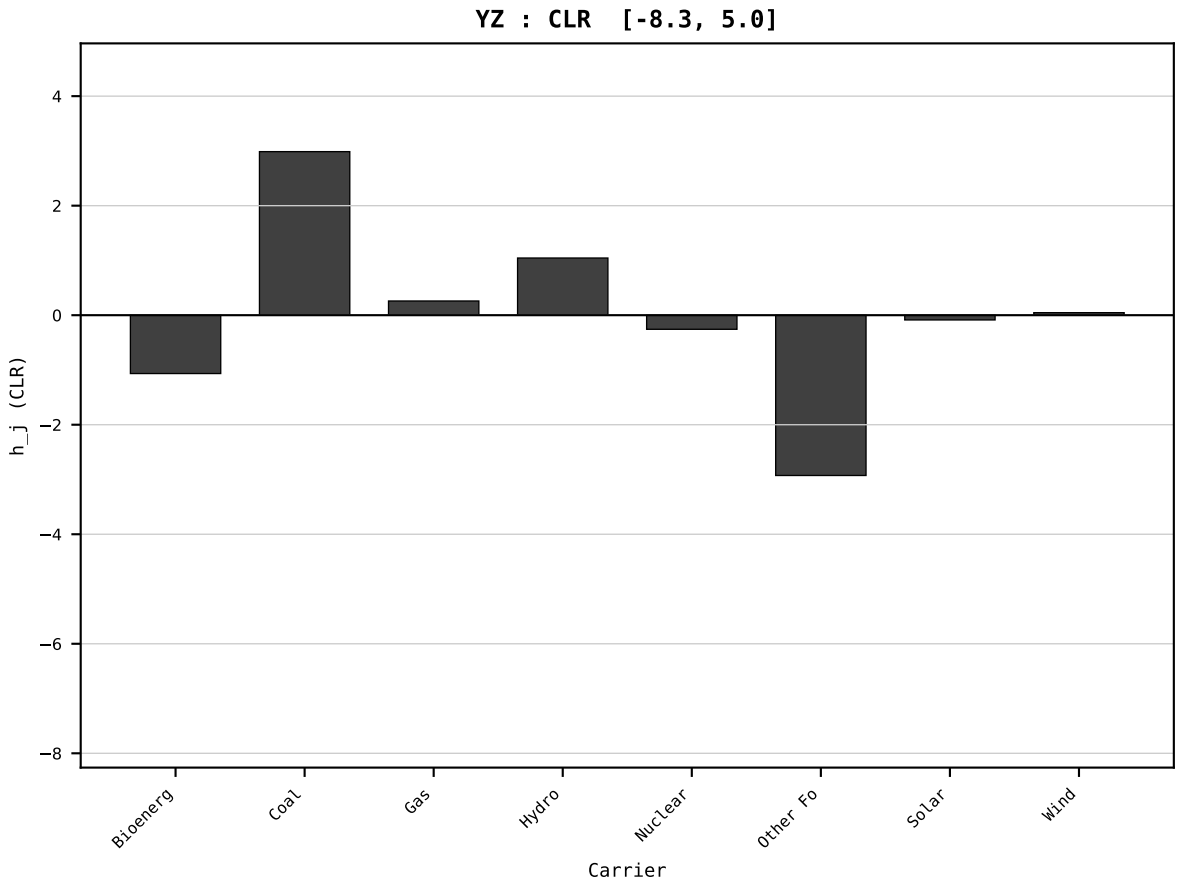
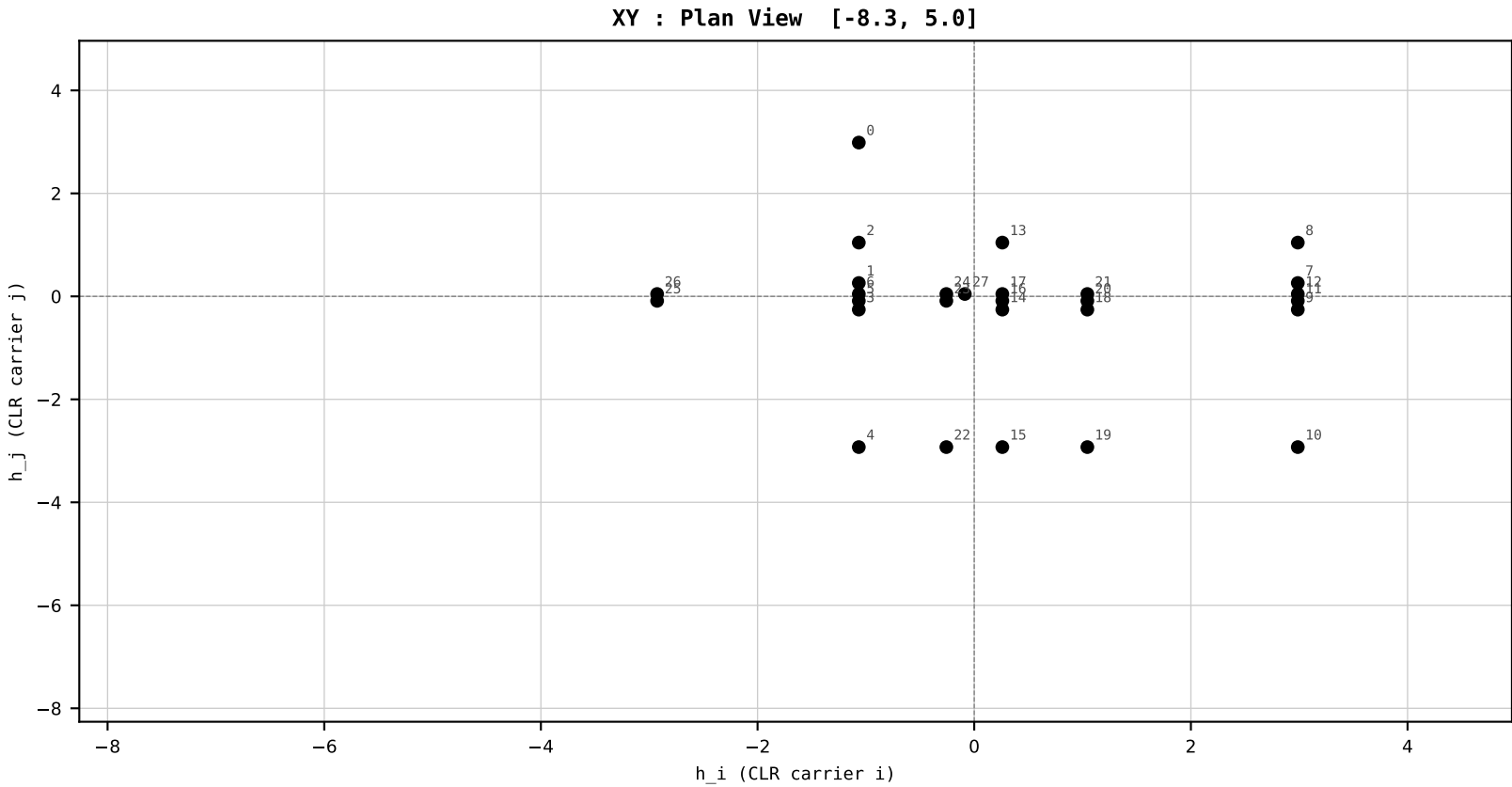
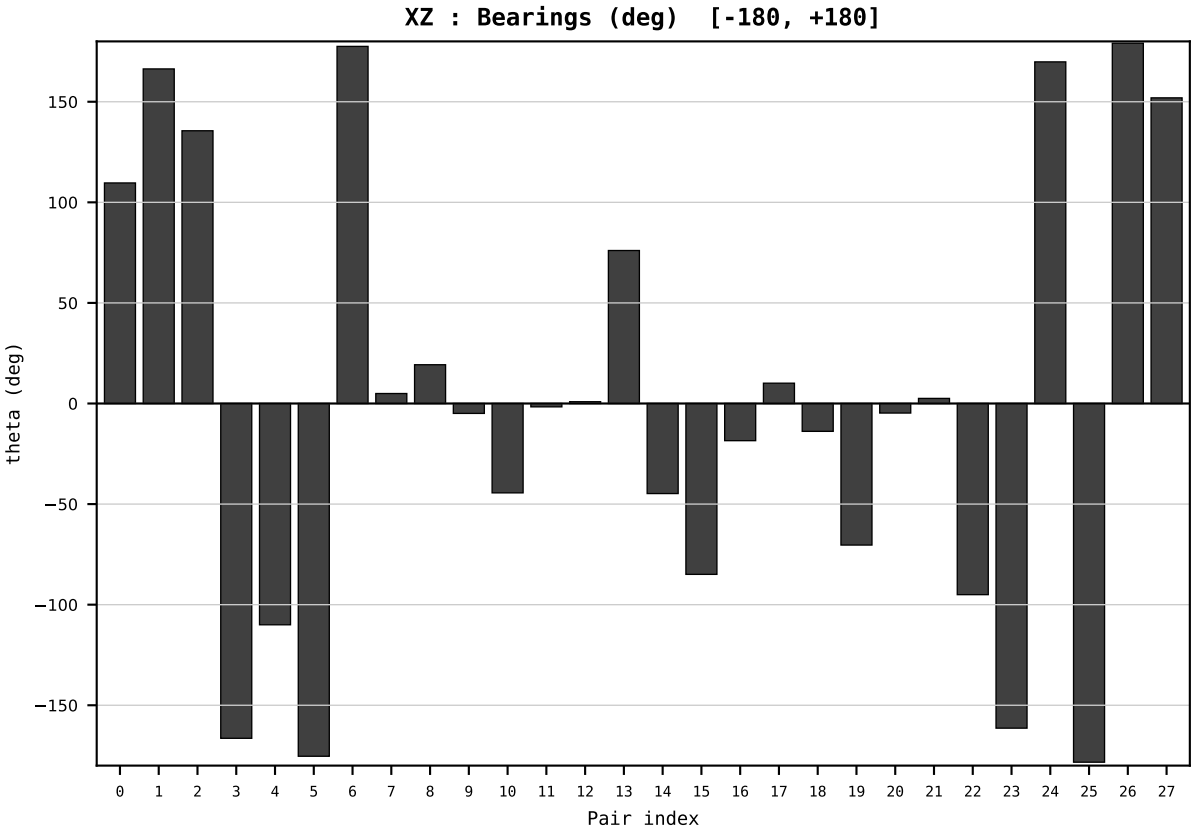
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2020
t : 20 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.509401
Ring = Hs-4
E_metric= 19.8580
kappa_HS= 370.16
omega = 4.2261 deg
d_A = 0.336395
Helm = Solar
Helm d = +0.302967
DR = 5.914 HLR
DR ratio= 370.2x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=21 | Year 2021 | D=8 N=26 pairs=28

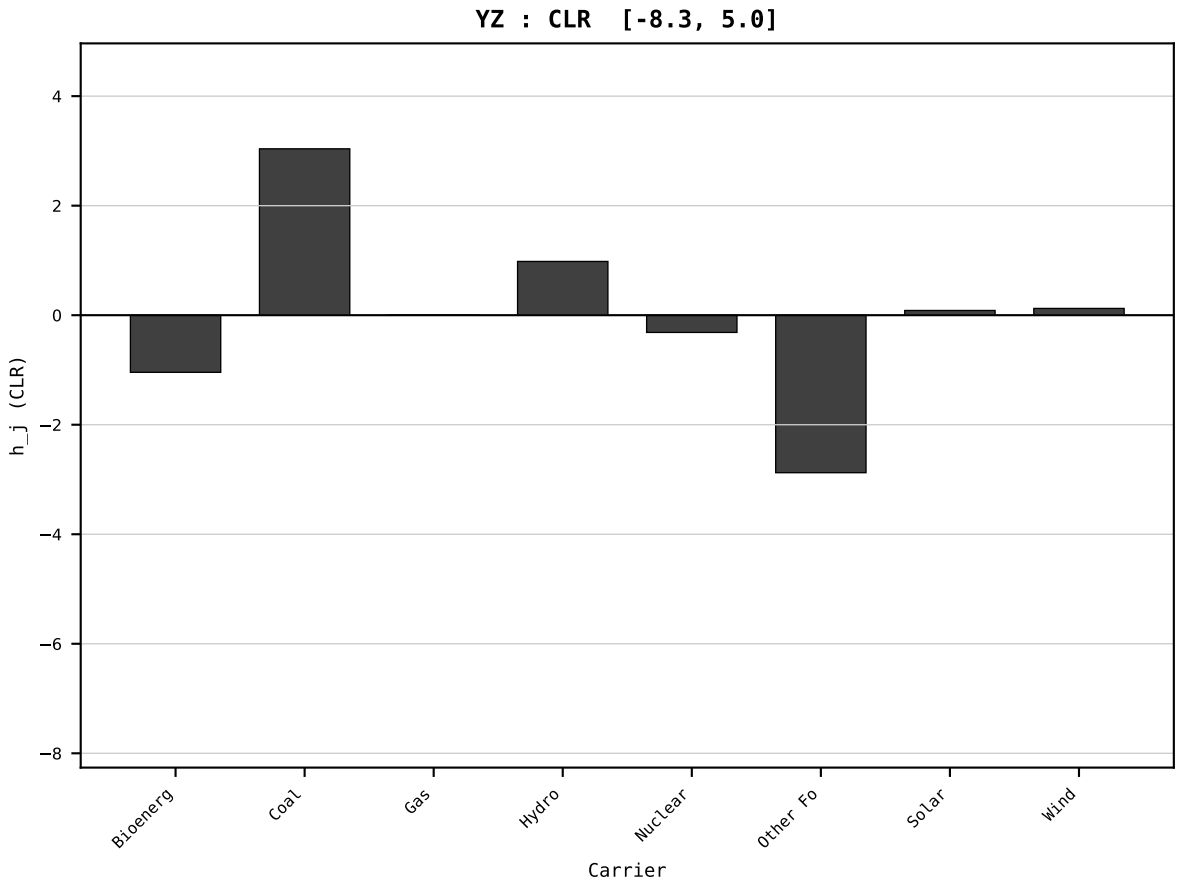
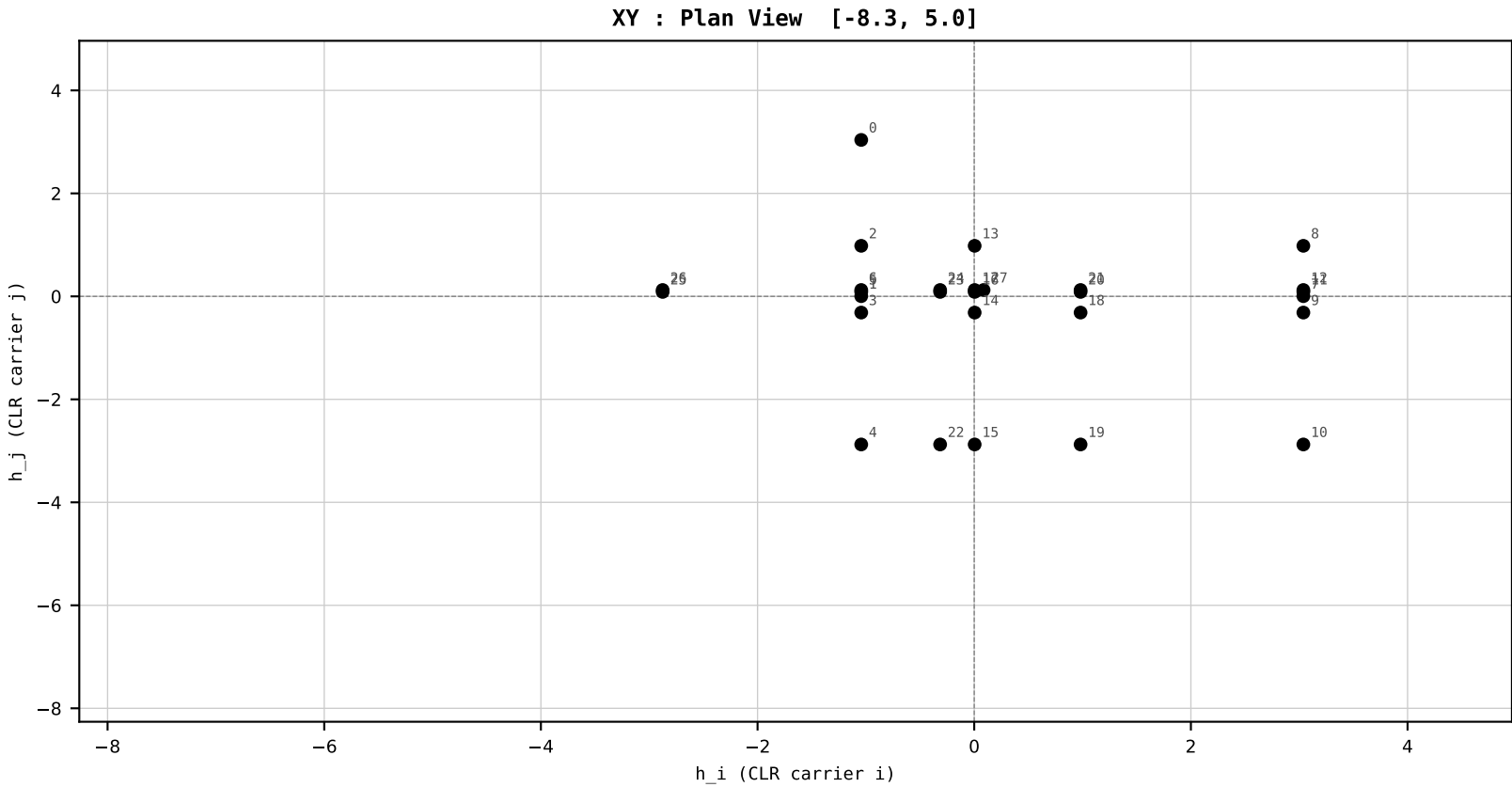
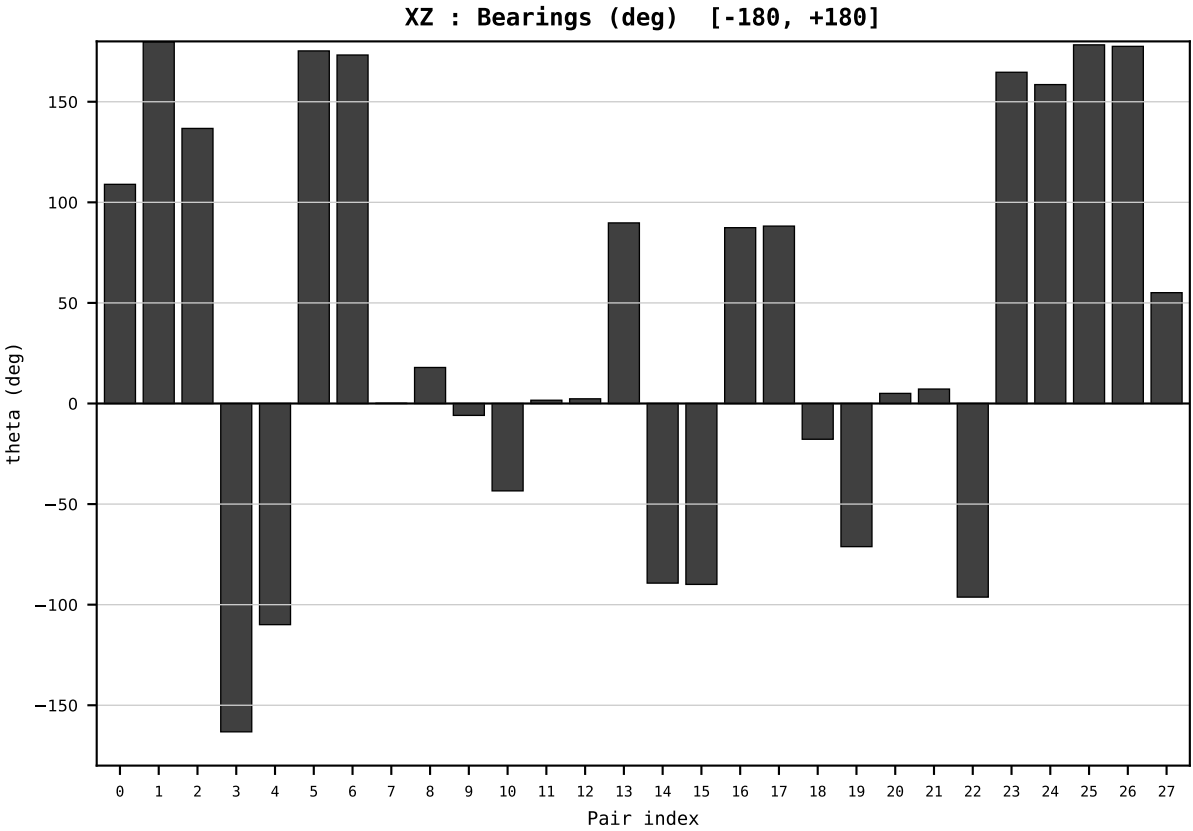
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2021
t : 21 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.528191
Ring = Hs-4
E_metric= 19.6773
kappa_HS= 370.37
omega = 4.3477 deg
d_A = 0.337902
Helm = Gas
Helm d = -0.255484
DR = 5.915 HLR
DR ratio= 370.4x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=22 | Year 2022 | D=8 N=26 pairs=28

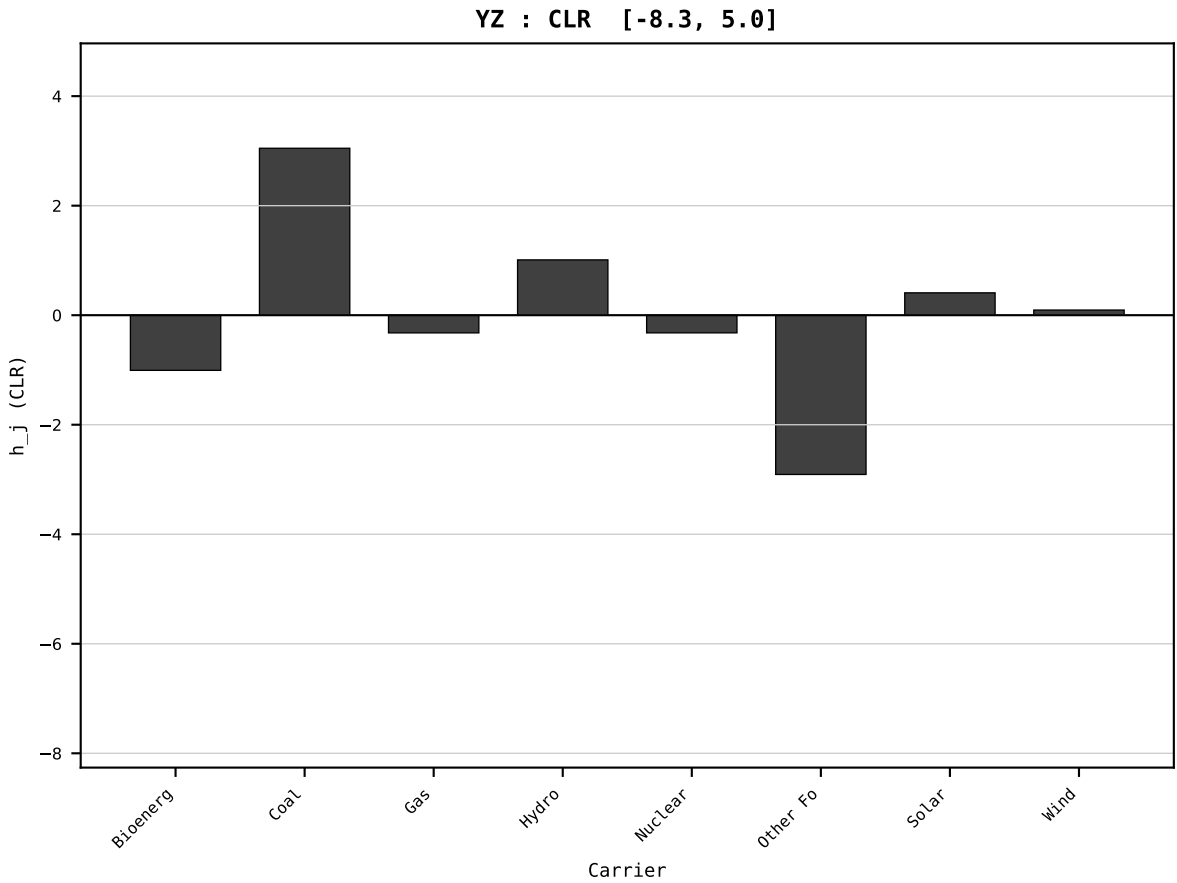
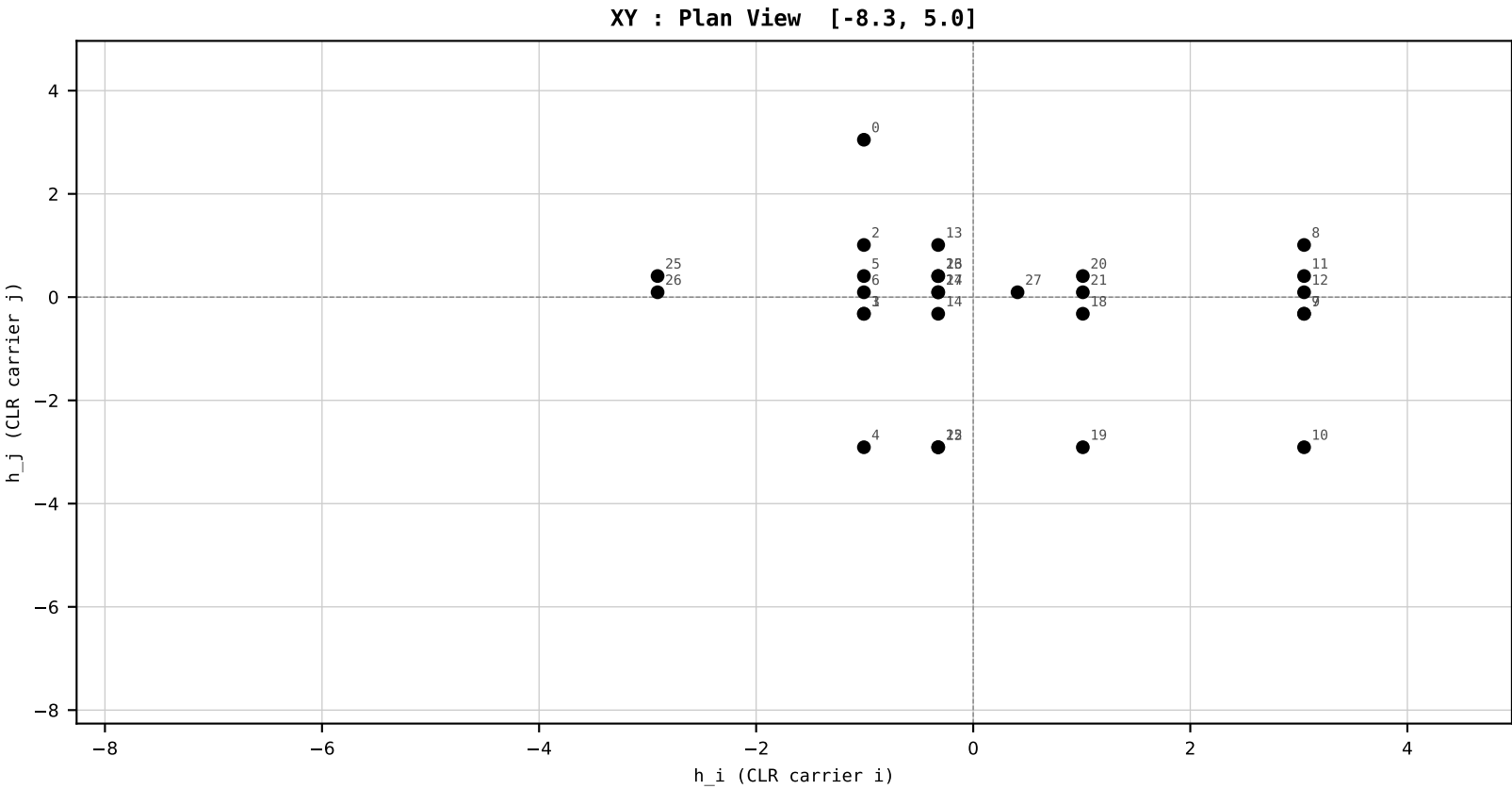
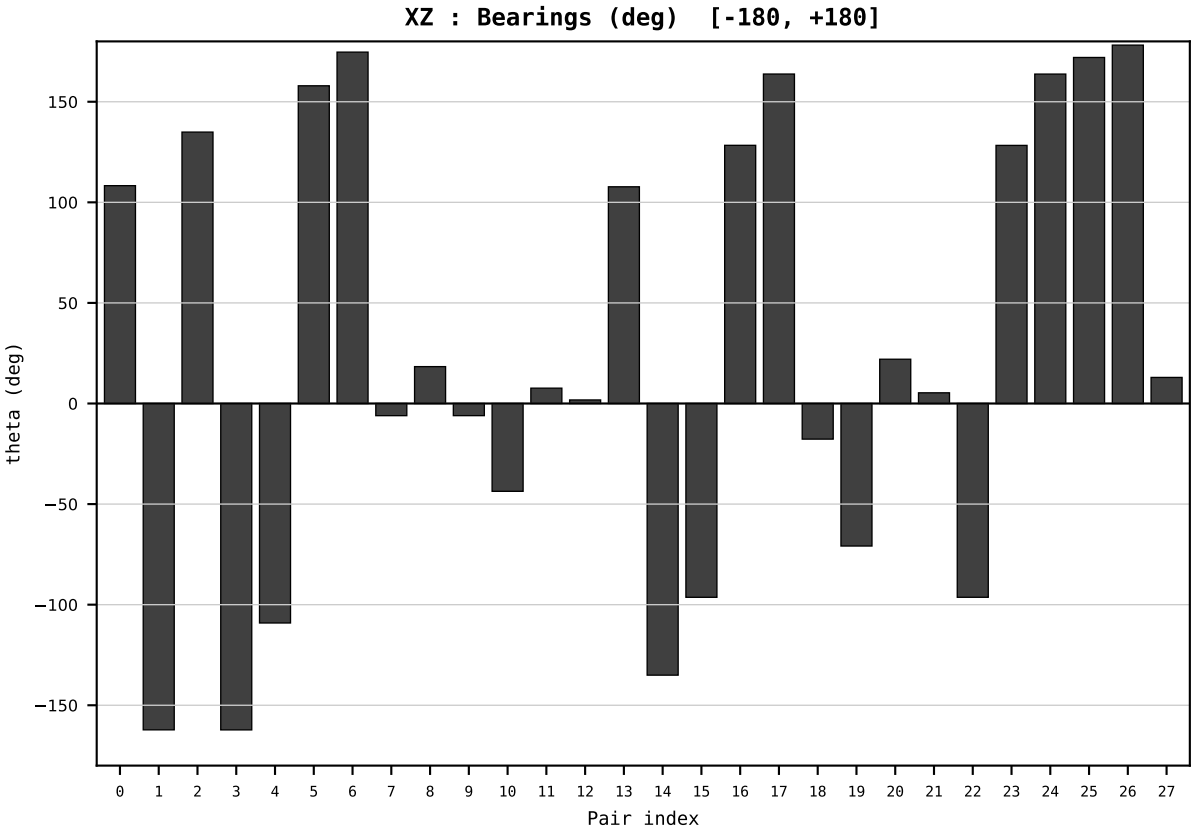
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2022
t : 22 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.527016
Ring = Hs-4
E_metric= 20.1694
kappa_HS= 386.33
omega = 5.9073 deg
d_A = 0.463267
Helm = Gas
Helm d = -0.326832
DR = 5.957 HLR
DR ratio= 386.3x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=23 | Year 2023 | D=8 N=26 pairs=28

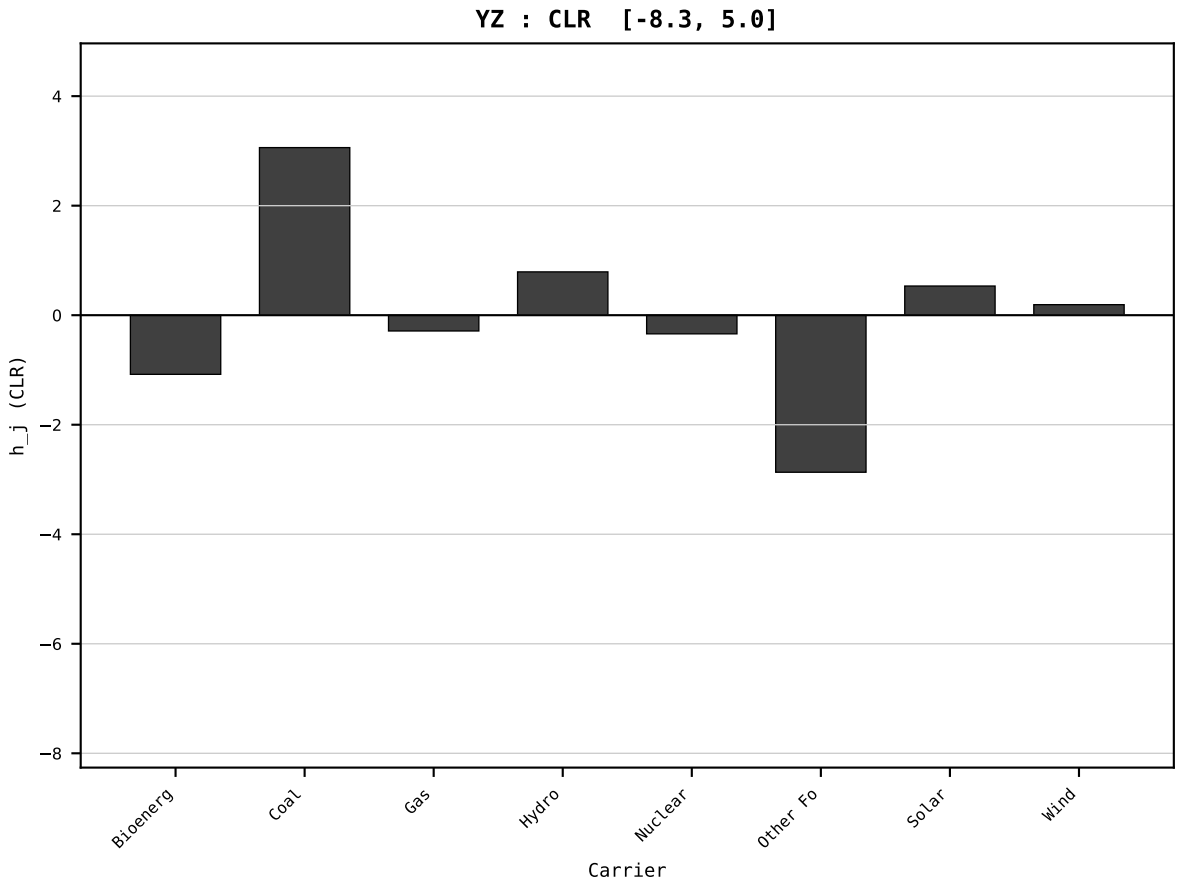
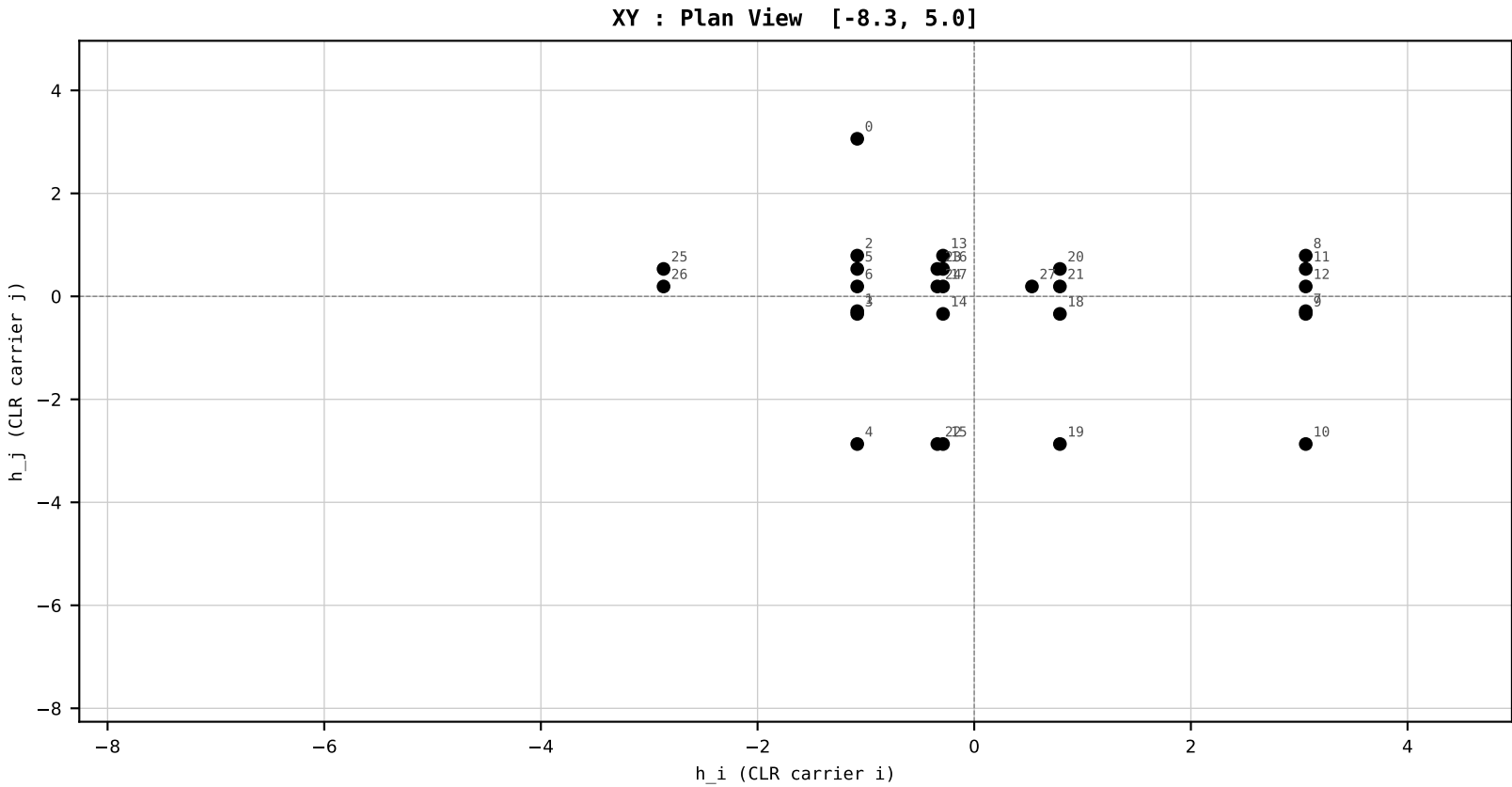
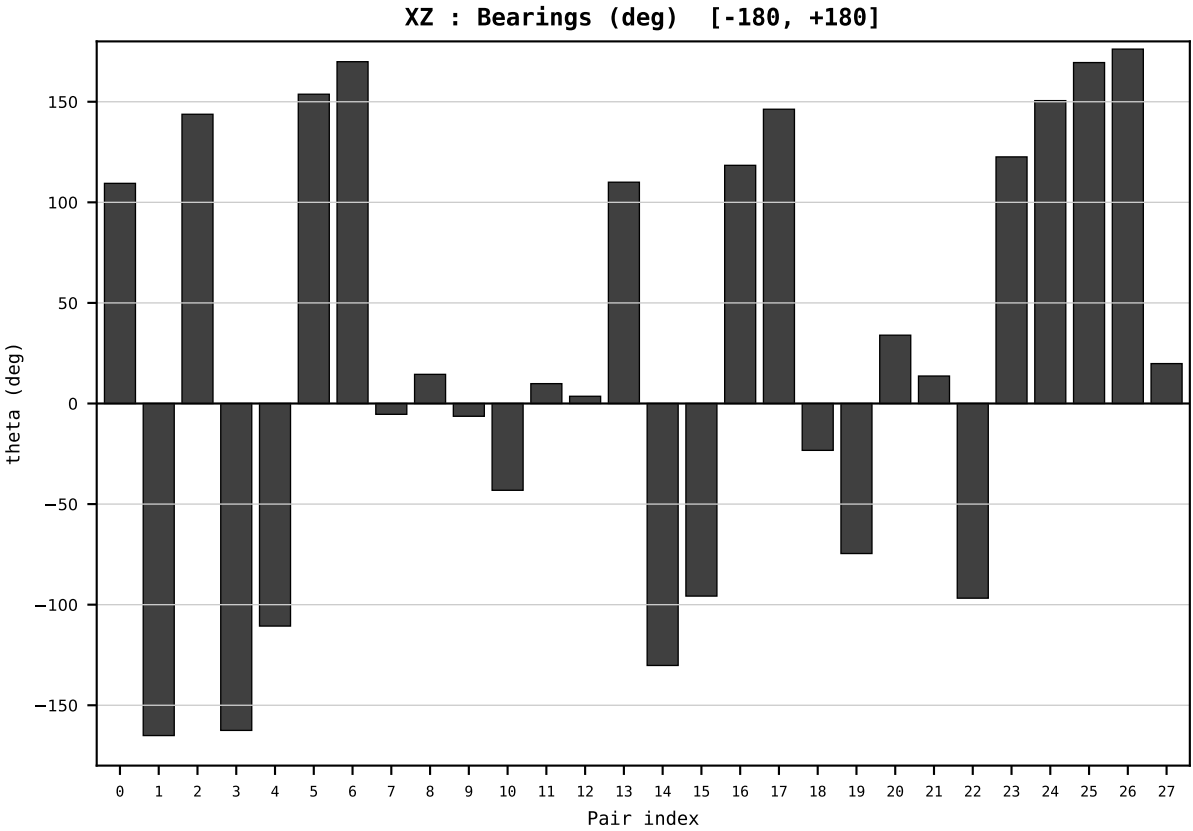
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2023
t : 23 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.533376
Ring = Hs-4
E_metric= 19.9012
kappa_HS= 375.44
omega = 3.6403 deg
d_A = 0.285909
Helm = Hydro
Helm d = -0.219097
DR = 5.928 HLR
DR ratio= 375.4x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=24 | Year 2024 | D=8 N=26 pairs=28

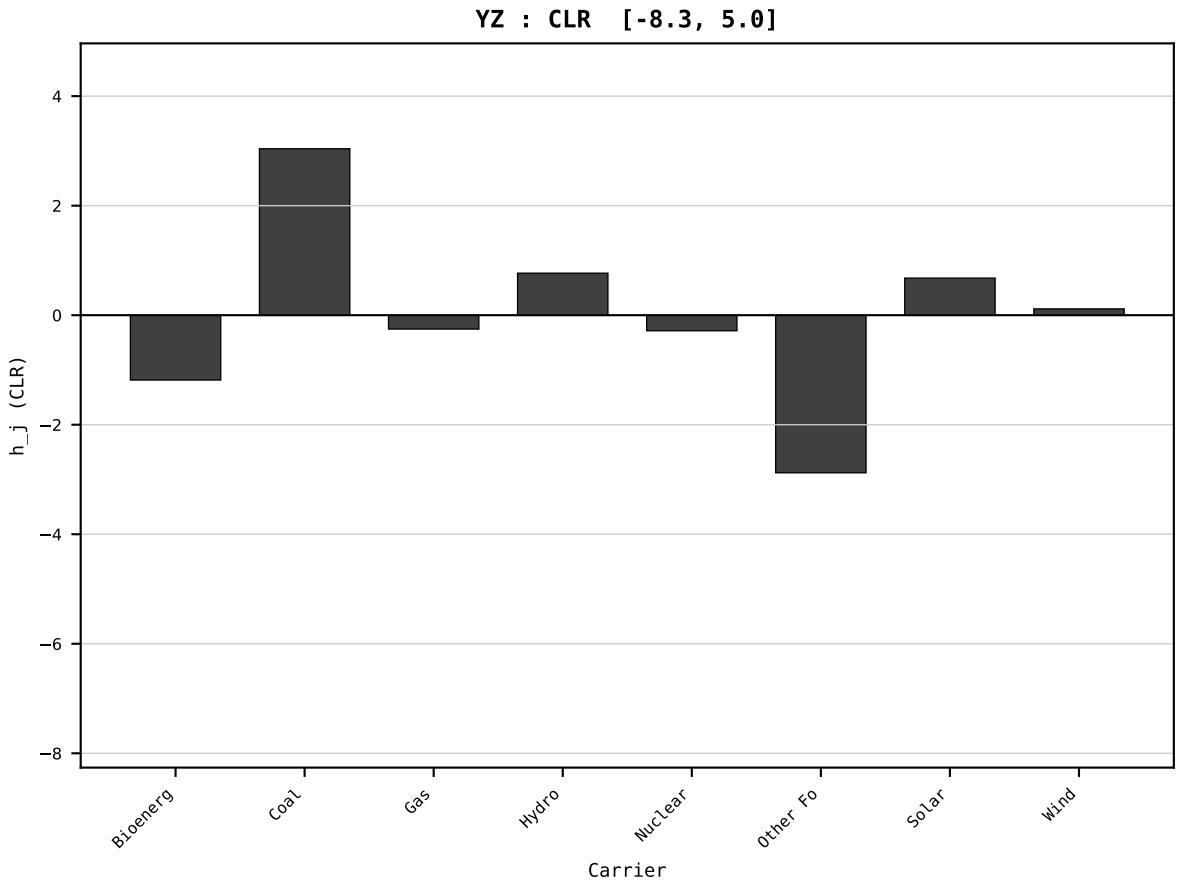
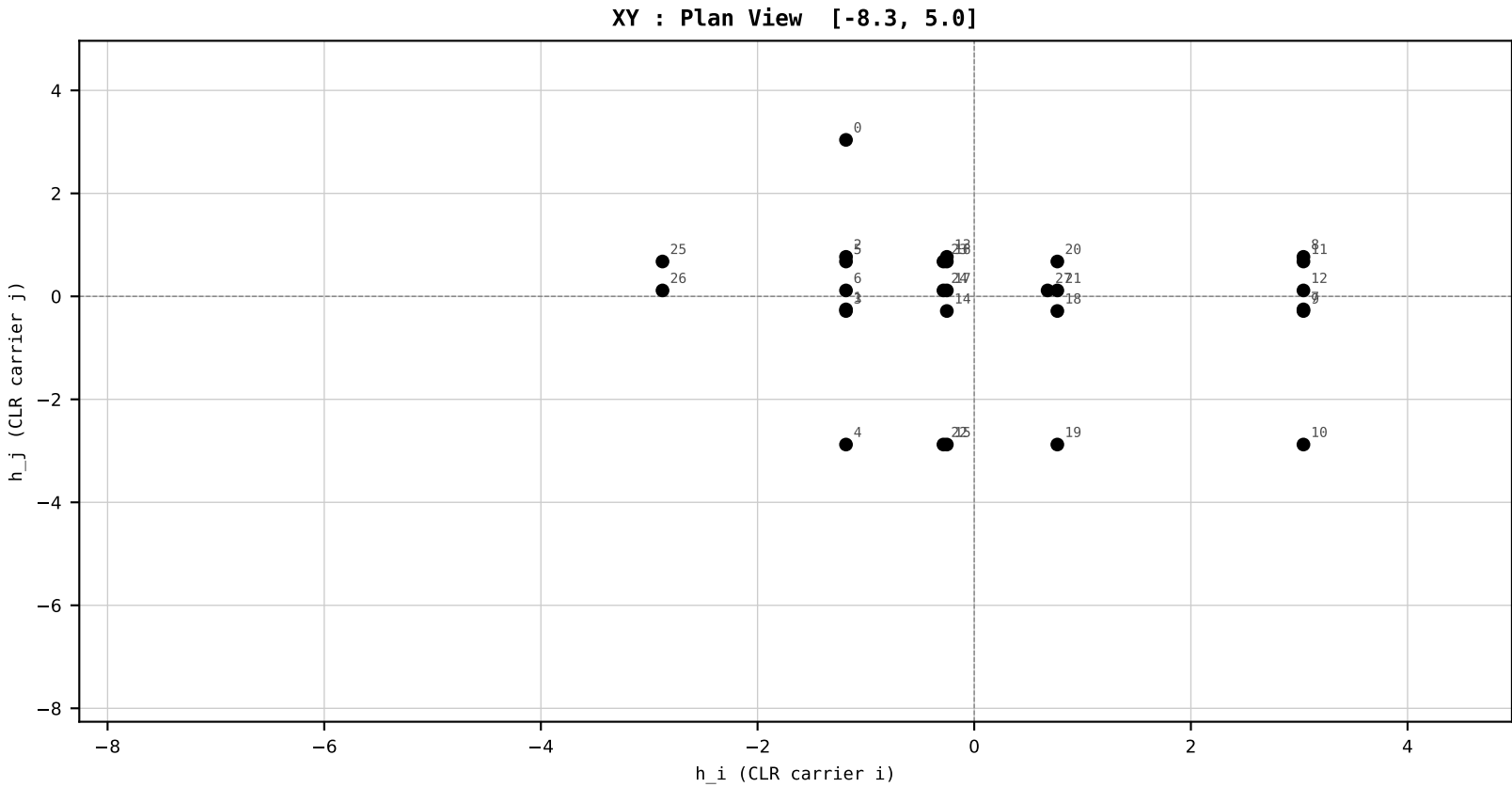
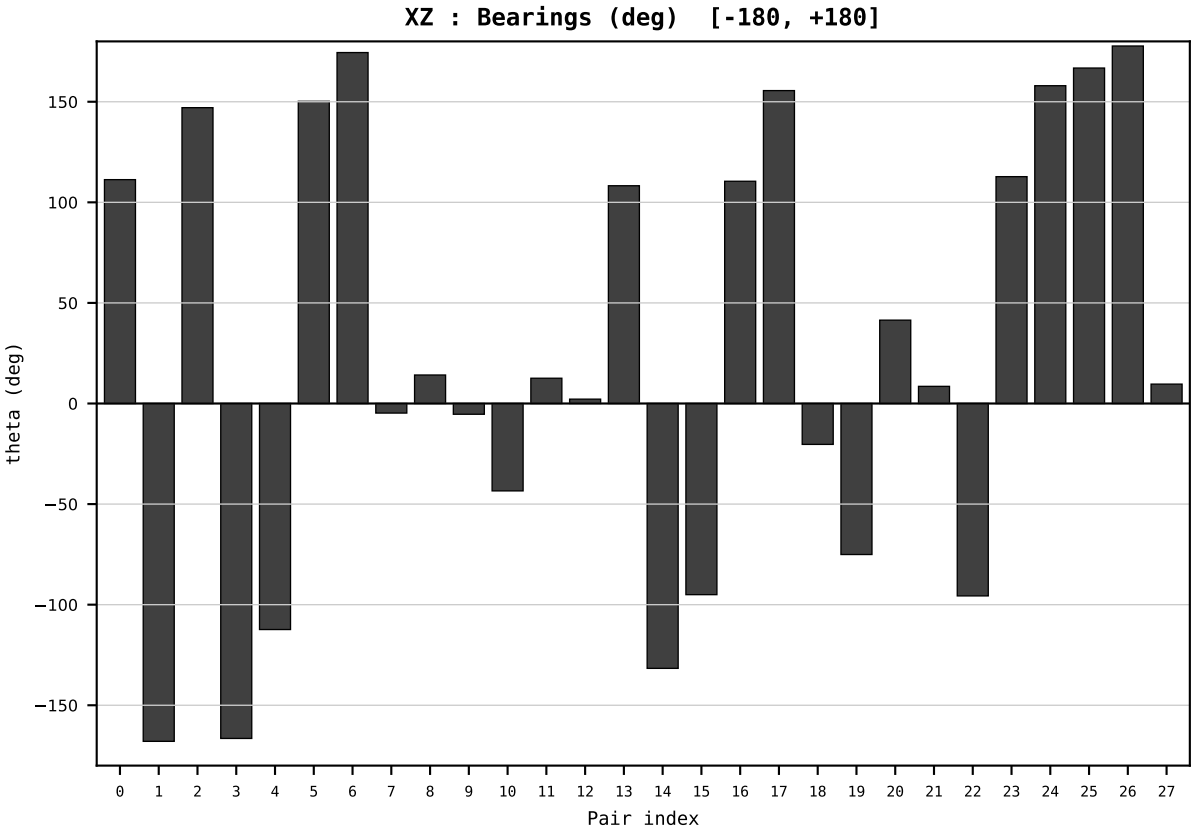
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

Dataset : ember_IND_India_generation_TWh.csv
Year : 2024
t : 24 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.523895
Ring = Hs-4
E_metric= 20.1222
kappa_HS= 371.12
omega = 2.6415 deg
d_A = 0.207695
Helm = Solar
Helm d = +0.144532
DR = 5.917 HLR
DR ratio= 371.1x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind



SECTION PLATE t=25 | Year 2025 | D=8 N=26 pairs=28

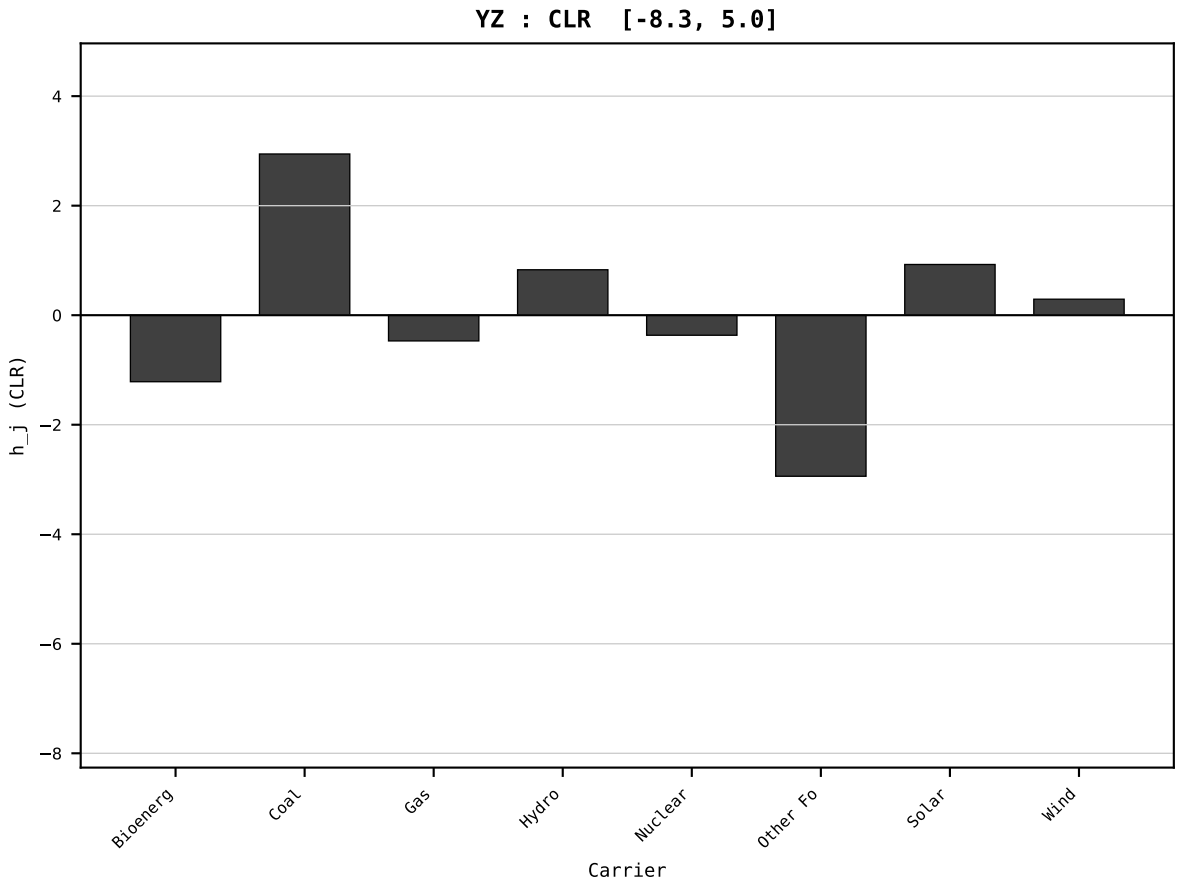
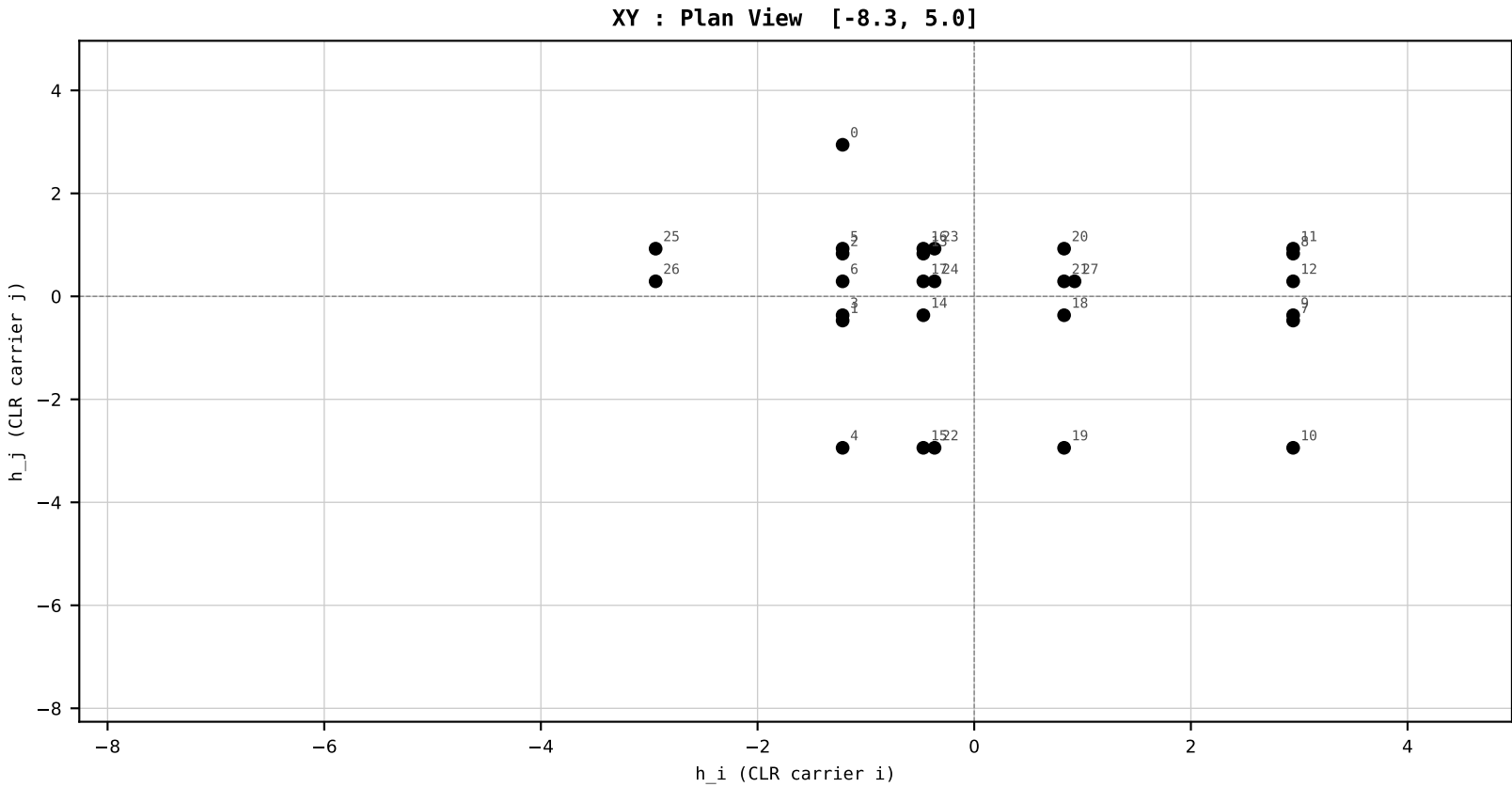
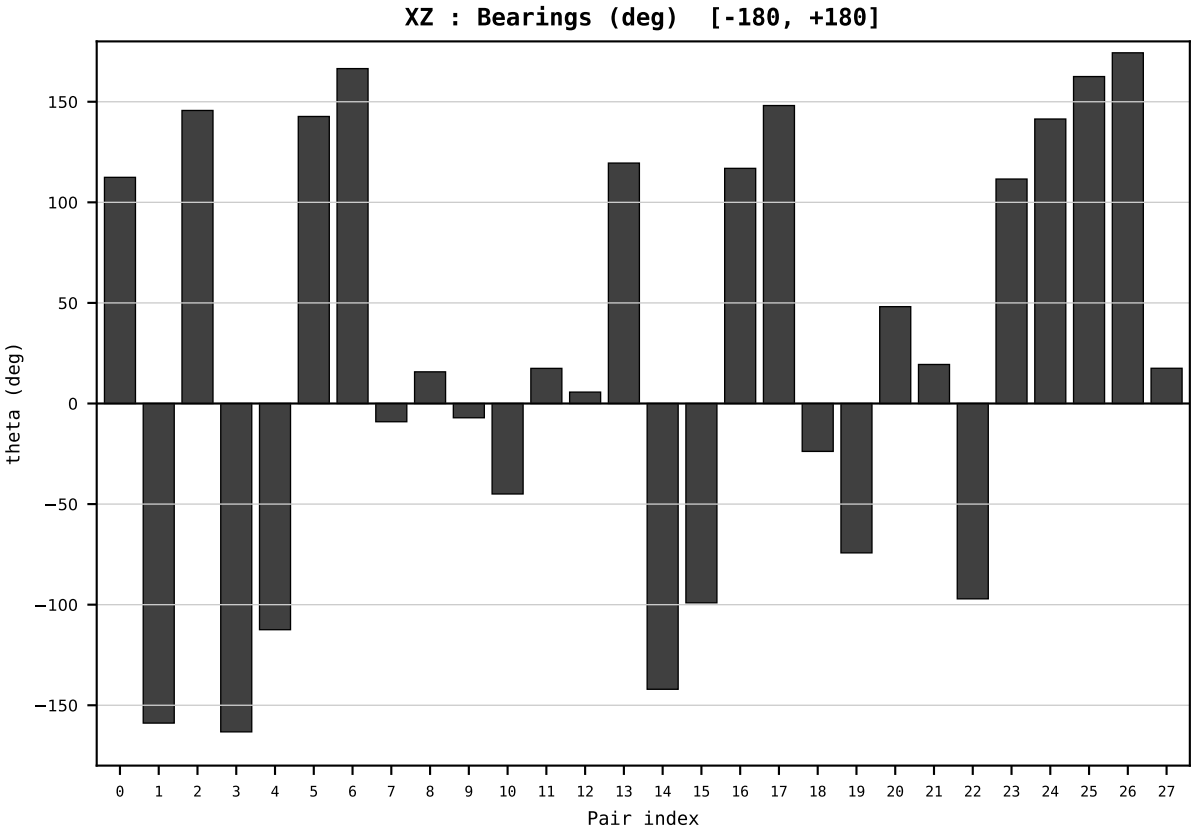
HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y→x=Y/T→h=CLR→section

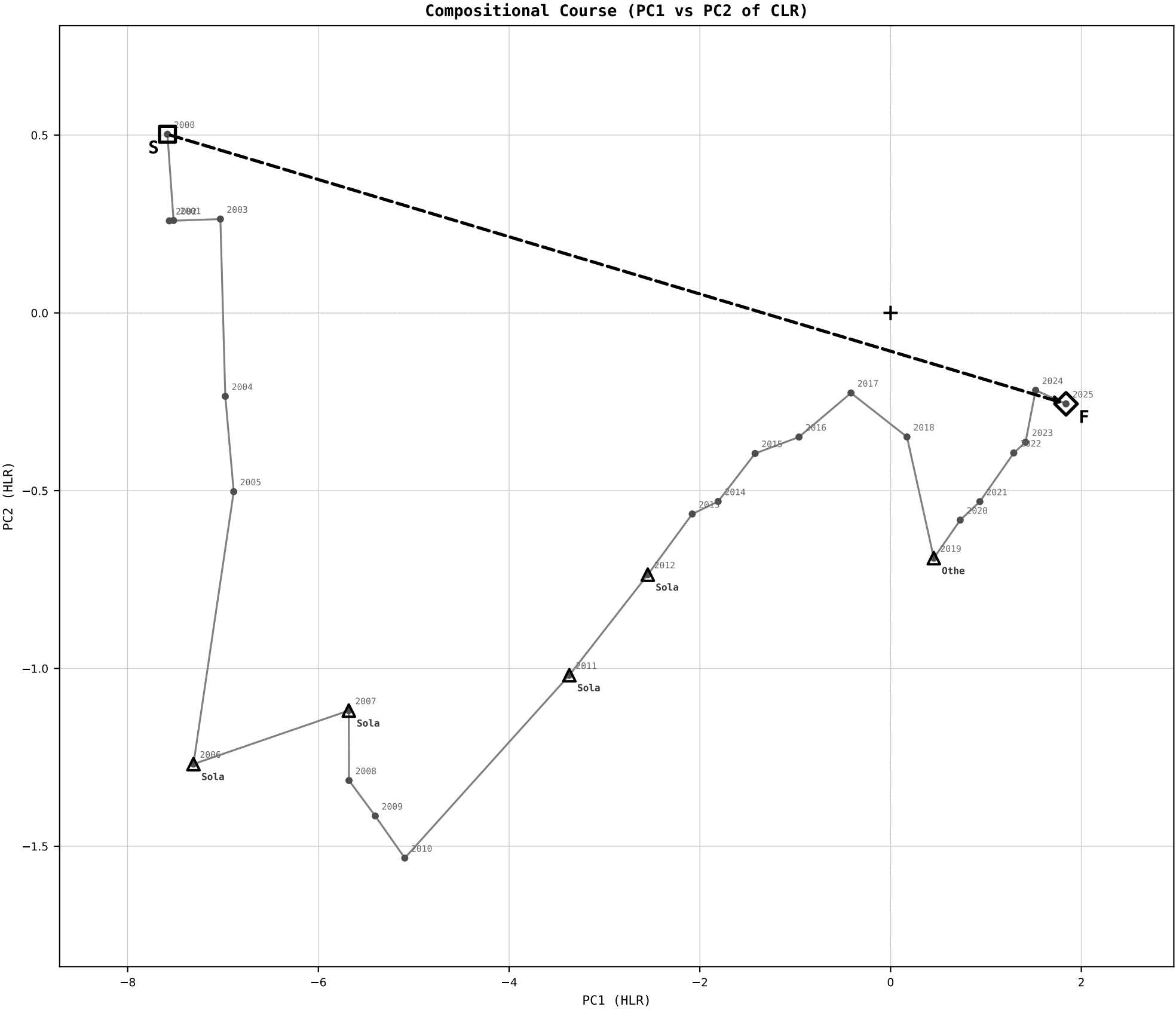
Dataset : ember_IND_India_generation_TWh.csv
Year : 2025
t : 25 / 25
D : 8 carriers
Pairs : 28 channels

Hs = 0.484928
Ring = Hs-3
E_metric= 20.7772
kappa_HS= 359.55
omega = 5.0650 deg
d_A = 0.406114
Helm = Solar
Helm d = +0.248797
DR = 5.885 HLR
DR ratio= 359.5x

XZ [-180,180]
CLR[-8.3,5.0]
Hs [0, 1]

PAIR INDEX:	
0=Bioenergy-Coal	14=Gas-Nuclear
1=Bioenergy-Gas	15=Gas-Other Fossil
2=Bioenergy-Hydro	16=Gas-Solar
3=Bioenergy-Nuclear	17=Gas-Wind
4=Bioenergy-Other Fossil	18=Hydro-Nuclear
5=Bioenergy-Solar	19=Hydro-Other Fossil
6=Bioenergy-Wind	20=Hydro-Solar
7=Coal-Gas	21=Hydro-Wind
8=Coal-Hydro	22=Nuclear-Other Fossil
9=Coal-Nuclear	23=Nuclear-Solar
10=Coal-Other Fossil	24=Nuclear-Wind
11=Coal-Solar	25=Other Fossil-Solar
12=Coal-Wind	26=Other Fossil-Wind
13=Gas-Hydro	27=Solar-Wind





COURSE METRICS

Scale Provenance:
Y_j → x_j=Y_j/T → h_j=CLR
→ PC projection
Unit: HLR (Higgins Log-Ratio)

Net distance = 9.4603 HLR
Path length = 14.6188 HLR
Directness = 0.6471

DR start = 10.572 HLR
ratio = 39023.0x
DR final = 5.885 HLR
ratio = 359.5x

E_{metric} start = 75.8817
E_{metric} final = 20.7772

PC1 loadings:
Solar +0.841
Other Fo -0.425
Gas -0.241

PC2 loadings:
Wind -0.574
Bioenerg -0.487
Other Fo +0.458

Top displacement events:
2011 Solar d=1.820
2007 Solar d=1.692
2006 Solar d=0.943
2012 Solar d=0.912
2019 Other Fo d=0.756

The instrument reads.
The expert decides.
The loop stays open.